Master Architecture of the DBHE-Driven Heating, Cooling and Dehumidification Plant

Streams, Junctions, Valve Network, Operating Modes,
Load Interface, and Verification Protocol

(Generated from model version 10.6, stamped 2026-08-01 09:35 BRT)

Craft & Hawkins Department of Petroleum Engineering
Louisiana State University

August 4, 2026

Abstract

This document specifies the *master architecture* of the surface plant driven by a closed-loop Deep Borehole Heat Exchanger formed from a repurposed legacy oil-and-gas well. Where the companion model report derives the governing equations of each component, this report defines the *topology*: which streams exist, where they join and split, which valve or damper controls each branch, and which subset of components is active in each operating mode. The architecture is deliberately a superset. It contains every component the plant could need across the full plane of sensible and latent loads, and a given operating mode is obtained by activating a subset of branches rather than by rebuilding the plant. Section 5 enumerates the resulting modes; Section 7 records which components are implemented in the code and which are pending.

Part II gives the mathematical model of every component, written against the source rather than from memory, together with a specification for the four components the architecture contains but the code does not yet.

The report is generated, not transcribed. Part I's tables and the version history are extracted at build time from the notebook, the frozen regression fixtures and the parametric sweep archives; the configuration constants quoted in Part II's prose are frozen in a fixture of their own and the build fails if any of them moves. The document is regenerated whenever the model version changes.

Contents

Part I

The plant: architecture, modes, and the resource that limits them

1 Scope, and how this document stays current

1.1 What is specified here

The plant comprises five fluid circuits: the geothermal water loop, a chilled water loop, a cooling-tower water loop, the process (building supply) air stream, and the scavenging (regeneration) air stream. The master architecture is the union of all flow paths among them. It is not a single machine but a family: each operating mode activates a subset of the branches, and the inactive branches are present but idle.

Two consequences follow, and they shape the whole study. First, the number of distinct plants is finite and small once the valves and dampers are treated as two-position devices rather than continuously modulating ones. Second, the question "which plant should be built" becomes "which subset of the master architecture serves the load quadrants that actually occur, using the water temperature the well can actually deliver over the design life". That is the question Sections 5 and 6 address.

1.2 Generation, not transcription

This report is emitted by `report/build_report.py` in the model repository. The build step reads:

- the notebook, for the version stamp and the configuration constants;
- `verification/reference/v*_baseline.json`, the frozen per-version results recorded by the regression harness;
- the parametric sweep archives, for the resource-feasibility tables.

When the model advances to the next version the report is rebuilt and republished, and the version history in Section 20 gains a row by itself.

Where the guarantee is weaker, and what protects it there. No number in Part I is typed by hand; a Part I statement that contradicts the code is a defect in the build script, not a stale paragraph. Part II cannot work that way. Its model sections quote constants inside prose and inside derivations — *the damper floor is* 0.05, *the coil is sized at* 3000 W/K *per* kg/s — and templating prose at that density would make it unreadable. Those numbers are typed, so they can rot.

The guard is a configuration fixture. `report/check_config.py` snapshots every scalar assignment in the configuration cell, and every subsequent build compares the live notebook against that snapshot. If a constant moves, the build stops and names the symbol, so the affected prose is re-read and the fixture re-frozen deliberately rather than by accident. This is the same discipline the model itself uses — freeze a reference, never overwrite it silently — applied to the document. It does not prove the prose is right; it guarantees that a change in the code cannot pass unnoticed.

2 Nomenclature convention

State labels follow a two- or three-letter convention. The first subscript letter identifies the stream: w for geothermal water, c for chilled water, p for process air, r for scavenging (regeneration) air, t for cooling-tower water. Subsequent letters identify the station. Valves on water circuits are $v_A \ldots v_H$; dampers on air circuits are $d_{pA} \ldots d_{pI}$ for the process stream and d_{rA}, d_{rB} for the scavenging stream.

Symbol	**Circuit**	**Station**
T_{in}	water	injection into the annulus
T_{ret}	water	production from the tubing
T_{wr}	water	after the v_F well-bypass rejoins the production stream
T_{wa}	water	accumulator outlet
T_{wg}	water	absorption-generator outlet
T_{wma}	water	generator-skip branch merged with the v_B accumulator bypass
T_{wmg}	water	T_{wg} tempered by T_{wma} through v_D
T_{wx}	water	regeneration heat-exchanger outlet
T_{wmx}	water	after the v_H remainder blends back; returns to the pump
T_{pc}, w_{pc}	process air	building comfort state (return)
T_{pmc}, w_{pmc}	process air	mixed state entering the supply fan
T_{pd}, w_{pd}	process air	desiccant-wheel process outlet
T_{pmo}, w_{pmo}	process air	after fork 1 remerges, entering fork 2
T_{pi}, w_{pi}	process air	building supply
T_{ea}, RH_{ea}	process air	building exhaust
T_{rm}, w_{rm}	scavenging air	mixed state entering the scavenging fan
T_{ri}, w_{ri}	scavenging air	regeneration-coil inlet
T_{ro}, w_{ro}	scavenging air	regeneration-coil outlet, wheel inlet
T_{rd}, w_{rd}	scavenging air	wheel regeneration outlet, vented

3 The five circuits

3.1 Geothermal water

The water circuit is a closed loop with four decision points. Reading from the pump:

1. v_E / v_F — **well bifurcation.** v_E sends flow down the annulus; v_F bypasses the well entirely and rejoins the production stream. The mixed state is T_{wr}. This decouples the loop flow from the well flow, so the surface plant can be run at a flow the subsurface would not tolerate, and vice versa.
2. v_A / v_B — **accumulator.** v_A charges or discharges the thermal store; v_B bypasses it. The accumulator is a buffer, not a topology choice: its state varies through the day within any mode.
3. v_C — **absorption generator.** The branch that skips the generator merges with the v_B bypass to form T_{wma}.
4. v_D**, then** v_G / v_H**.** v_D blends T_{wma} into the generator outlet T_{wg}, raising the regeneration-coil inlet to T_{wmg}. v_G passes that stream through the coil; v_H carries the remainder of T_{wma} around the coil to blend with the coil outlet T_{wx}, giving T_{wmx}.

The pairing of v_D and v_G deserves emphasis because the two are easily confused. v_D **raises** the coil inlet temperature: it can only blend *hotter* bypass water into a generator outlet that the chiller

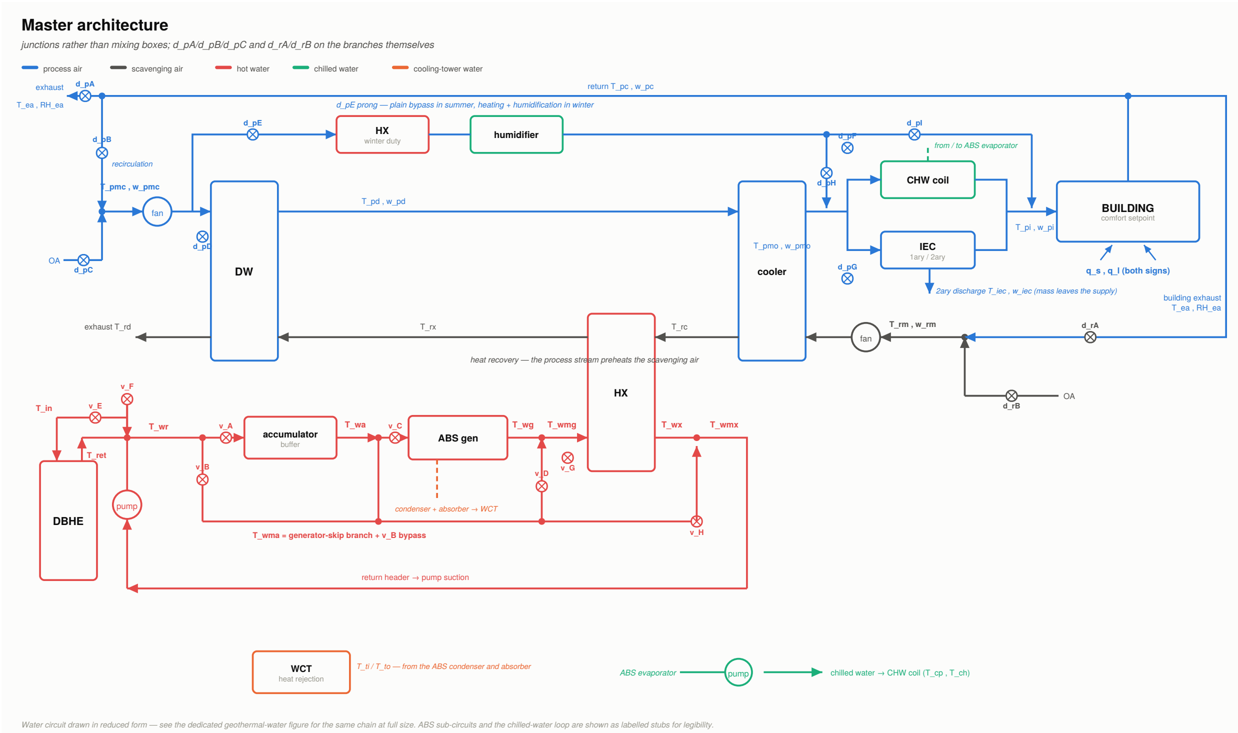

Figure 1: Master architecture. Every component the plant could need across the full load plane is present; an operating mode is a subset of active branches, not a different machine.

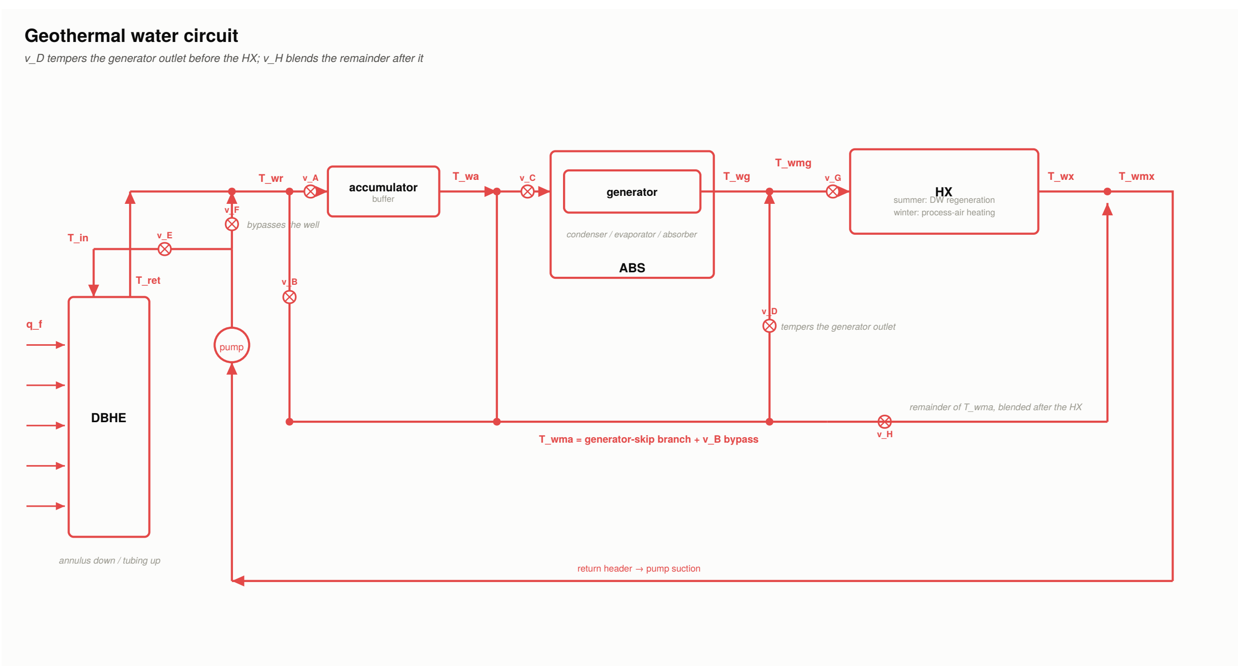

Figure 2: Geothermal water circuit. v_E/v_F decouple loop flow from well flow; v_A/v_B gate the accumulator; v_C gates the absorption generator; v_D raises the coil inlet, while v_G/v_H throttle the coil duty.

has cooled. v_G **throttles**: reducing the share of flow through the coil lowers the coil duty, hence the regeneration air temperature T_{ro}, hence the drying the wheel performs. When the wheel must be turned *down*, v_G is the control and v_D is not.

3.2 Process air

$$\text{mix} \;\rightarrow\; \text{fan} \;\rightarrow\; \underbrace{\left\{ d_{pD} \rightarrow \text{DW} \rightarrow \text{cooler} \;\middle|\; d_{pE} \rightarrow \text{HX}_{\text{winter}} \rightarrow \text{humidifier} \right\}}_{\text{fork 1}} \;\rightarrow\; T_{pmo} \;\rightarrow\; \underbrace{\left\{ d_{pF} \rightarrow \text{CHW} \;\middle|\; d_{pG} \rightarrow \text{IEC}\right.}_{\text{fork 2}}$$

The building return stream bifurcates at a node: d_{pA} exhausts, d_{pB} recirculates, and the recirculated air joins outdoor air admitted through d_{pC} at a mixing junction whose state is T_{pmc}, w_{pmc}.

Fork 1 selects between conditioning (wheel followed by the recuperative cooler) and the alternative prong that carries the winter heating coil and the humidifier. The d_{pE} prong terminates in two dampers: d_{pH} returns it to T_{pmo}, upstream of fork 2, which is the summer bypass; d_{pI} carries it past fork 2 directly into the supply duct, which is the winter path. Fork 2 is a genuine parallel pair — chilled-water coil *or* indirect evaporative cooler, never both — rejoining at the supply state T_{pi}, w_{pi}.

Fork 2 needs no third "neither" prong: with no chilled water circulating and no evaporative water supplied, air passes either branch at nothing but a pressure drop.

3.3 Scavenging air

$$\text{mix}\big(d_{rA}\,\text{building exhaust} + d_{rB}\,\text{outdoor air}\big) \rightarrow \text{fan} \rightarrow \text{cooler} \rightarrow \text{HX} \rightarrow \text{DW} \rightarrow \text{vented}$$

Two features of this circuit carry more weight than their simplicity suggests.

First, the scavenging stream passes through the **cooler before the regeneration coil**, so the process stream's rejected heat preheats it and the coil has less lifting to do. The cooler is therefore a recuperator between the two air streams, not a heat rejection device.

Second, the mixing junction draws on **building exhaust and outdoor air**, and never on the wheel's own regeneration outlet. Recirculating the wheel exhaust returns its desorbed moisture to its own inlet, which collapses the relative-humidity difference the wheel drives on. The code did exactly that until v10.5; correcting it nearly tripled moisture removal.

A physical bound constrains the mix. The building can only exhaust what it admits as fresh air, so the exhaust available to the scavenging loop is $\dot{m}_{\text{vent}} = 10.00\,\text{kg/s}$ against a scavenging demand of $\dot{m}_{\text{reg}} = 23.52\,\text{kg/s}$. The building-exhaust fraction is therefore capped at 0.425 and the balance made up with outdoor air.

3.4 Chilled water and heat rejection

The absorption chiller is a single indivisible subsystem with its chilled-water distribution: a generator fed by geothermal water, an evaporator serving the chilled-water coil in the process stream, and a condenser and absorber rejecting to a wet cooling tower. A chiller with nowhere to send chilled water is not a configuration, so **ABS and CHW are one switch, not two**.

4 The load interface

The plant sees the building only through a steady, well-mixed pair of balances. With $q_s > 0$ meaning cooling is required and $\dot{m}_w > 0$ meaning dehumidification is required,

$$q_s = \dot{m}_{\text{pro}}\, c_p\, (T_{pc} - T_{pi}), \tag{1}$$

$$\dot{m}_w = \dot{m}_{\text{pro}}\, (w_{pc} - w_{pi}). \tag{2}$$

Two equations, three unknowns $(\dot{m}_{\text{pro}}, T_{pi}, w_{pi})$. **Exactly one closure must be supplied**, and which one is a design decision rather than a physical fact:

- give $\dot{m}_{\text{pro}}$ — constant volume; the supply state follows;
- give $\Delta T_{\text{sup}} = T_{pc} - T_{pi}$ — variable volume; the flow follows.

Dividing (1) by (2) eliminates the flow:

$$\frac{T_{pc} - T_{pi}}{w_{pc} - w_{pi}} = \frac{q_s}{\dot{m}_w}\,\frac{1}{c_p}. \tag{3}$$

The load *ratio* alone fixes the direction of the room condition line on the psychrometric chart; the closure fixes how far along that line the supply state sits. This is the classical construction, and it is why the interface needs exactly one more piece of information and no more.

Latent load is carried as a moisture rate, not as a power. Expressing q_l in kilowatts requires choosing h_{fg}, and inverting (2) requires choosing it again. If the two choices differ — and values between 2450 and 2501 kJ/kg are all in common use — w_{pi} is wrong by about 2% with nothing on screen to show it. The interface therefore takes $\dot{m}_w$ in kg/s and offers a convenience wrapper that converts from kilowatts with an explicit h_{fg}.

The closure in the hatchery case is derived, not given. The supply flow is set by a design sensible load carried at a prescribed supply depression, floored by the forced ventilation requirement:

$$\dot{m}_{\text{pro}} = \max\!\left(\frac{Q_{s,\text{design}}}{c_p\, \Delta T_{\text{sup,design}}},\ \dot{m}_{\text{vent}}\right) = \max(33.60\,\text{kg/s},\ 10.00\,\text{kg/s}) = 33.60\,\text{kg/s}. \tag{4}$$

The depression rule binds. Its consequence reaches much further than fan sizing: a supply depression of only 3 K demands a large airflow, and a large airflow needs only a very small humidity difference to carry the same latent load. The wheel is not underperforming in this plant — it is barely being asked to work.

5 Operating modes

5.1 Naming

Load quadrants are named by the *processes they require*, from the sign pair $(\operatorname{sgn} q_s, \operatorname{sgn} q_l)$: cooling or heating, combined with dehumidification or humidification. Seasonal names were considered and rejected: they are climate-relative, since hot-and-dry is the only summer in a semi-arid region, and setpoint-relative, since the quadrant boundaries sit at the comfort state rather than at some neutral condition.

The sensible heat ratio cannot classify the quadrants on its own. Two negative loads give a positive ratio, indistinguishable from two positive ones. The sign pair is the primitive; the ratio describes position *within* a quadrant.

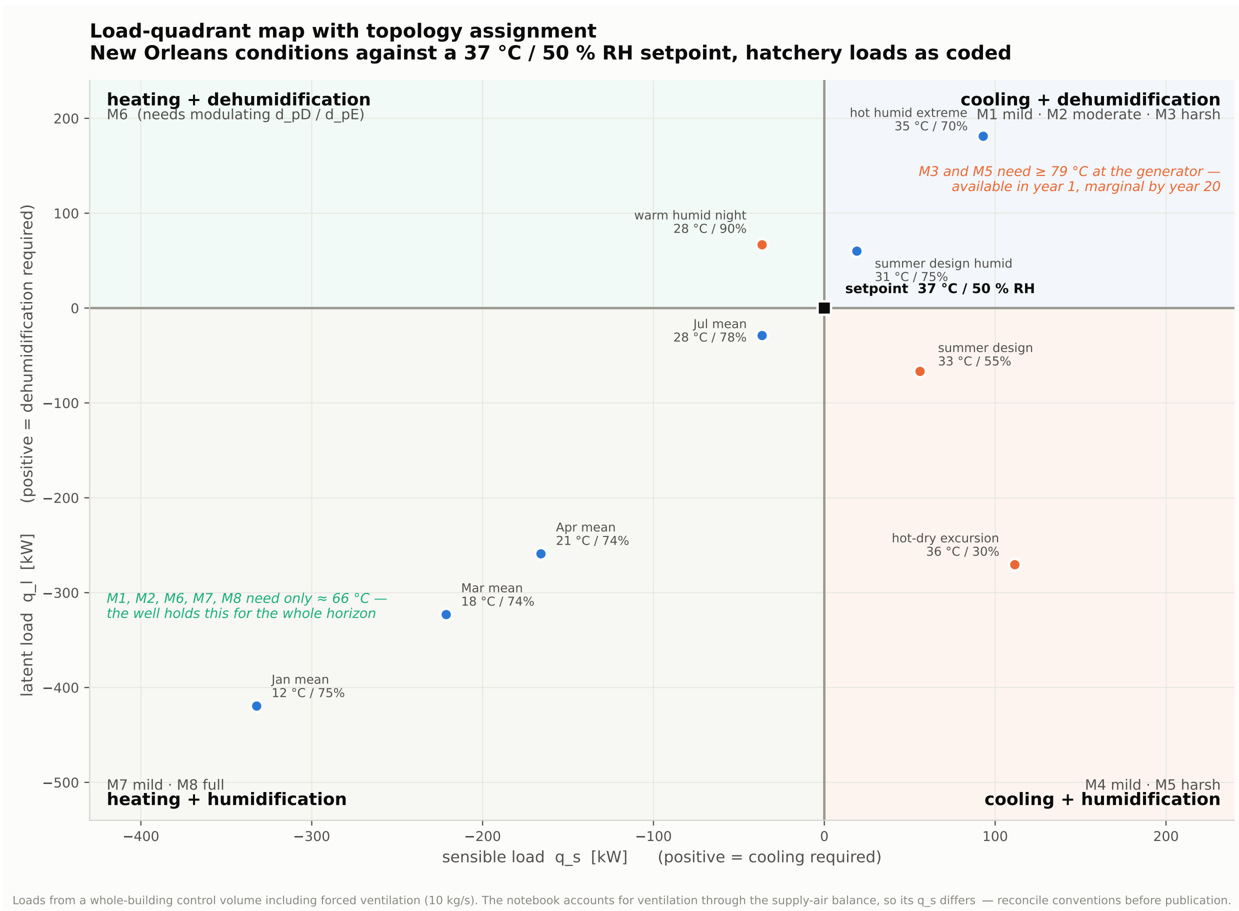

Figure 3: The load plane. Quadrants are named by the processes they require; the tier within a quadrant is set by load magnitude against what the well can deliver.

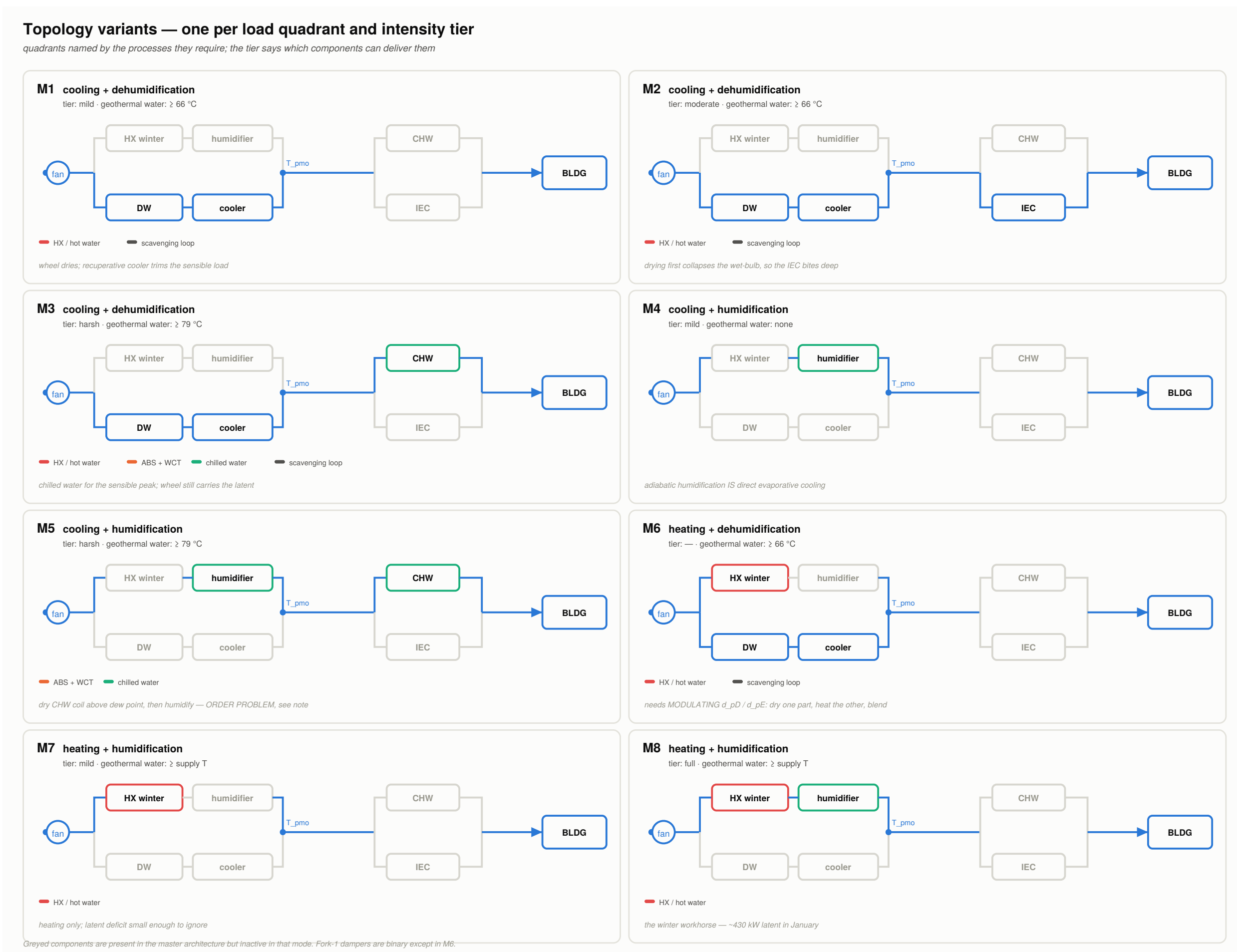

Figure 4: Topology variants. Each card shows the branches active in one mode; inactive branches remain installed but idle.

5.2 Two axes: quadrant and tier

The quadrant selects which processes are needed. The load *magnitude* selects which components can deliver them. The resulting mode set is:

Mode	Quadrant	Active components	Water required
M1	cooling + dehumidification	DW, cooler	$\geq 66\,°C$
M2	cooling + dehumidification	DW, cooler, IEC	$\geq 66\,°C$
M3	cooling + dehumidification	DW, cooler, ABS + CHW	$\geq 79\,°C$
M4	cooling + humidification	humidifier alone (adiabatic saturation *is* direct evaporative cooling)	none
M5	cooling + humidification	ABS + CHW, humidifier	$\geq 79\,°C$
M6	heating + dehumidification	DW, cooler and winter coil, *blended*	$\geq 66\,°C$
M7	heating + humidification	winter coil	$\geq$ supply T
M8	heating + humidification	winter coil, humidifier	$\geq$ supply T

Three of these carry conditions worth stating explicitly.

M6 is the only mode requiring modulating dampers. Dehumidification and heating sit on parallel prongs of fork 1, so the quadrant is reached by drying part of the flow, heating the rest, and blending at T_{pmo}. Every other mode is satisfied by two-position devices.

M5 has an unresolved ordering problem. The humidifier sits on the d_{pE} prong, upstream of the chilled-water coil in fork 2, so the air is humidified before it is cooled and may reach saturation. Cooling and then humidifying is the correct sequence, and the architecture does not currently provide it.

M7 and M8 route through prong D with the wheel stopped. Air passes a stationary wheel at nothing but a pressure drop, which turns the cooler into an exhaust-to-supply recuperator at no capital cost — see Section 6.3.

6 The resource constraint

6.1 Temperature: which tiers the well can serve

Only M3 and M5 require the 79 °C minimum at the absorption generator. Every other mode needs approximately 66 °C for wheel regeneration, which the well holds for the whole design life at almost any flow. The serviceable region of the load plane therefore *shrinks with reservoir depletion, and it shrinks from the cooling corner first.*

From the archived parametric sweep, year-20 conditions:

L [m]	BHT [°C]	q [m^3/h]	T_{ret} [°C]	T_{wa} [°C]	Q_{well} [kW]	generator viable
4000	144.0	3	79.8	71.4	130	no
4000	144.0	4	78.7	72.2	168	no
4000	144.0	5	76.4	71.4	197	no
4000	144.0	6	73.8	69.8	220	no
4000	144.0	8	69.1	66.3	251	no
4000	144.0	10	65.4	63.4	270	no
4500	159.5	3	86.5	77.0	149	no
4500	159.5	4	86.4	78.9	198	yes

L [m]	BHT [°C]	q [m^3/h]	T_{ret} [°C]	T_{wa} [°C]	Q_{well} [kW]	generator viable
4500	159.5	5	84.5	78.5	237	yes
4500	159.5	6	82.0	77.2	267	no
4500	159.5	8	77.0	73.7	309	no
4500	159.5	10	72.8	70.4	336	no
4677	165.0	3	88.8	78.8	156	yes
4677	165.0	4	89.1	81.2	208	yes
4677	165.0	5	87.3	81.1	251	yes
4677	165.0	6	84.9	79.8	283	yes
4677	165.0	8	79.8	76.3	331	no
4677	165.0	10	75.6	73.0	360	no

6.2 Capacity: the flow squeeze

Raising loop flow raises the generator duty but lowers the water temperature, and the two requirements are not satisfied at the same operating point. Using a design generator drop of 10 K and a maximum of 18 K, the coded sensible demand of 101 kW needs a generator duty of about 145 kW, that is roughly 7 m^3/h — but at 4500 m that flow does not reach 79 °C. The temperature-optimal flow starves the chiller; the capacity-optimal flow cools the water below its minimum. Satisfying both requires depth: approximately 5000 m to 5500 m at 6 m^3/h to 7 m^3/h.

6.3 Winter capacity, and why recovery dominates

Sizing the January duty over the whole outdoor-air stream, the enthalpy lift from a cold, dry ambient to the comfort state is roughly 60 kJ/kg, giving a total near 772 kW of which more than half is latent. Net of internal gains the plant duty is about 630 kW, against a well output between 198 kW and 336 kW: the well covers only 31–53%.

A dry recuperator at 70% effectiveness recovers about 227 kW, leaving 403 kW unserved. A *latent-capable* device at the same effectiveness recovers about 540 kW, leaving 90 kW — comfortably inside the well's capability. Because 58% of the winter duty is latent, only the second closes the gap.

This is the strongest argument in the architecture for routing M7 and M8 through prong D with the wheel halted: the cooler then acts as an exhaust-to-supply recuperator using hardware already installed. Whether the desiccant wheel itself, operated differently, can recover the latent half — a rotary wheel moving moisture from a warm humid exhaust to a cold dry supply *is* an enthalpy wheel — remains open.

7 Implementation status

Component	**Status**	**Note**
DBHE (GITT + coaxial BVP)	implemented	transient formation coupling, 20-year march
Thermal accumulator	implemented	lumped capacitance
Regeneration coil (HX)	implemented	effectiveness–NTU, water to air
Desiccant wheel	implemented	Jurinak F_1/F_2 with prescribed effectiveness
Recuperative cooler	implemented	process air to scavenging air

Component	Status	Note
Load interface	implemented	v10.6; generic $(q_s, \dot{m}_w)$ with explicit closure
Humidifier	pending	adiabatic saturation; unlocks M4
Indirect evaporative cooler	pending	effectiveness form; unlocks M2
Winter heating coil	pending	reuses the coil model on different streams; unlocks M6–M8
Absorption chiller + tower + CHW	pending	unlocks M3, M5

A known limitation of the wheel model. The Jurinak formulation carries no size: no diameter, no desiccant mass, no rotation speed, no transfer area. The outlet is placed a fixed fraction of the way from the process inlet toward the regeneration inlet in F_1/F_2 space, and those fractions are the two prescribed effectivenesses. It can be verified directly that the process outlet state is *independent of both mass flows*. Three consequences follow: the wheel cannot be resized within this model, it cannot be throttled by scavenging flow, and any claim about wheel sizing, rotation speed or part-load behaviour is outside what the formulation can represent. Deriving the effectivenesses from an NTU using the analogy method would remove these restrictions at essentially no computational cost, and is deferred.

Part II
The models: what each component actually computes

Part I described *which* components exist and *when* each is active. Part II gives the equations. Each section states the governing hypothesis, derives the result, and then says plainly what the assumption buys and what it costs — the ordering matters, because in almost every case the interesting engineering is in the cost. Sections 8 through 16 describe code that runs today and were written against the source, not from memory; Section 18 is a specification for the four components the architecture contains but the code does not yet.

8 Psychrometric foundations

Everything the plant does to air — heating it in the regeneration coil, drying it in the desiccant wheel, cooling it before it enters the hatchery — is bookkeeping on two numbers per stream. This section derives those numbers and the relations that move them around, exactly as implemented in the notebook (`dbhe_master.ipynb`, v10.6, psychrometric block of the desiccant-wheel cell). The equations are elementary and unforgiving: the whole latent story of this plant lives in the third decimal place of a humidity ratio.

8.1 Why the humidity ratio, and why "per kilogram of dry air"

Moist air is a mixture of dry air and water vapour. As a stream travels around either loop, water is added (desorption in the wheel, egg metabolism, infiltration) and removed (adsorption in the wheel), so the *total* moist-air mass flow is not conserved across a component. The dry-air mass flow is: nothing here creates or destroys dry air. That single fact fixes the convention. Define the humidity ratio

$$w \;\equiv\; \frac{m_\mathrm{v}}{m_\mathrm{a}} \;=\; \frac{\text{mass of water vapour}}{\text{mass of dry air}} \qquad [\mathrm{kg/kg}], \tag{5}$$

and carry every extensive property *per kilogram of dry air*. A water balance on any component is then literally "$\dot{m}w$ in equals $\dot{m}w$ out", with the same constant $\dot{m}$ on both sides: the bookkeeping is linear and cannot drift. Had we used relative humidity, or moist-air mass, every balance would carry a denominator that changes as the process runs, and small errors would compound around the loop. This is why the notebook passes (T, w) pairs — never (T, RH) — between components; relative humidity is computed only where a human or a warning message has to read it.

8.2 Humidity ratio from partial pressures

Treat dry air and vapour as ideal gases sharing a volume V at temperature T, with partial pressures summing to the total by Dalton's law, $p_\mathrm{atm} = p_\mathrm{a} + p_\mathrm{v}$. Then, using $R_i = R_u/M_i$,

$$w \;=\; \frac{m_\mathrm{v}}{m_\mathrm{a}} \;=\; \frac{p_\mathrm{v} V/(R_\mathrm{v} T)}{p_\mathrm{a} V/(R_\mathrm{a} T)} \;=\; \frac{p_\mathrm{v}}{p_\mathrm{a}}\,\frac{M_\mathrm{v}}{M_\mathrm{a}}. \tag{6}$$

With $M_\mathrm{v} = 18.015\,\mathrm{g/mol}$ and $M_\mathrm{a} = 28.966\,\mathrm{g/mol}$ the molar-mass ratio is 0.6219, which is the constant hard-coded (as 0.622) in every psychrometric routine of the notebook:

$$\boxed{w \;=\; 0.622\,\frac{p_\mathrm{v}}{p_\mathrm{atm} - p_\mathrm{v}}} \tag{7}$$

Two things to notice. First, w depends on the *total* pressure — which is why $p_\mathrm{atm} = 101\,325\,\mathrm{Pa}$ is a declared configuration constant and every psychrometric call is routed through that one symbol: at another altitude the same T and RH give a different w. Second, the relation is non-linear in p_v, because the denominator shrinks as the air gets wetter; at the humidities this plant works in ($20\,\mathrm{g/kg}$ to $22\,\mathrm{g/kg}$) that curvature is a few percent, not negligible.

8.3 Saturation pressure and relative humidity

Relative humidity is the ratio of the actual vapour partial pressure to the saturation pressure at the *same* temperature,

$$\mathrm{RH} \;\equiv\; \frac{p_\mathrm{v}}{p_\mathrm{sat}(T)}, \qquad \text{equivalently} \qquad p_\mathrm{v} \;=\; \mathrm{RH}\cdot p_\mathrm{sat}(T). \tag{8}$$

The code does not call a steam table for p_sat. It uses a single-line Magnus-type correlation, with T in °C and p_sat in Pa:

$$\boxed{p_\mathrm{sat}(T) \;=\; 610.94\,\exp\!\left(\frac{17.625\,T}{T + 243.04}\right)} \tag{9}$$

These are the Alduchov–Eskridge coefficients for saturation over liquid water, accurate to better than 0.4 % over −40 °C to 50 °C. That covers the process loop comfortably, but the regeneration air runs at 60 °C to 100 °C, outside the fitted range, and (9) is still evaluated there by the wheel's relative-humidity diagnostic. The cost is confined to that diagnostic: p_sat never enters a mass or energy balance, it only converts w to RH for display and regime checking.

Substituting (8) and (9) into (7) gives the routine `w_from_RH`(T, RH) used throughout:

$$w(T, \mathrm{RH}) \;=\; 0.622\,\frac{\mathrm{RH}\,p_{\mathrm{sat}}(T)}{p_{\mathrm{atm}} - \mathrm{RH}\,p_{\mathrm{sat}}(T)}. \tag{10}$$

The implementation clips RH to $[0, 1]$ first, so a caller cannot request a supersaturated state through this door — the state simply saturates. That is a guard, not physics: it hides an inconsistent request rather than reporting it.

Inverting (7) for p_{v} and dividing by p_{sat} gives the reverse map:

$$\boxed{\mathrm{RH}(T, w) \;=\; \frac{1}{p_{\mathrm{sat}}(T)}\,\frac{w\,p_{\mathrm{atm}}}{0.622 + w}} \tag{11}$$

There is no single named `RH_from_w` helper in v10.6; this same expression is written out three times (in the wheel regime diagnostic, in the results dashboard, and inline in the load interface), algebraically identical in all three.

The consequence that drives this project. p_{sat} grows nearly exponentially with T, so a fixed relative humidity at a high temperature is a very large absolute moisture content. The hatchery setpoint of 37 °C at 50 % RH is not a dry condition: from (9), $p_{\mathrm{sat}}(37) = 6271\,\mathrm{Pa}$, so $p_{\mathrm{v}} = 3136\,\mathrm{Pa}$ and $w = 19.86\,\mathrm{g/kg}$. Whether the plant must dehumidify or humidify is settled entirely by the sign of $w_{\mathrm{oa}} - w_{\mathrm{pc}}$ — a comparison of humidity ratios, never of relative humidities.

8.4 Moist-air enthalpy as coded

Take dry air at 0 °C and saturated liquid water at 0 °C as reference states. The enthalpy per kilogram of dry air is the dry-air sensible term plus the vapour term (latent heat at the reference, plus the vapour's own sensible heat):

$$h(T, w) \;=\; \underbrace{c_{p,\mathrm{a}}\,T}_{\text{dry air}} \;+\; w\;\underbrace{\big(h_{\mathrm{fg},0} + c_{p,\mathrm{v}}\,T\big)}_{\text{vapour}}, \tag{12}$$

and the notebook fixes the three constants at $c_{p,\mathrm{a}} = 1.006\,\mathrm{kJ/(kg\,K)}$, $h_{\mathrm{fg},0} = 2501\,\mathrm{kJ/kg}$, $c_{p,\mathrm{v}} = 1.86\,\mathrm{kJ/(kg\,K)}$:

$$\boxed{h(T, w) \;=\; 1.006\,T + w\,\big(2501 + 1.86\,T\big) \qquad [\mathrm{kJ/kg\ dry\ air}]} \tag{13}$$

Both specific heats are constants — no temperature or pressure dependence. Over 0 °C to 100 °C the true $c_{p,\mathrm{a}}$ moves by well under 1 %, so the cost is small, and as the next subsection shows it is what makes mixing exactly linear. Because mixing delivers (h, w) rather than (T, w) we need the inverse, and solving (13) for T is a one-line algebraic step precisely because the constants are constant:

$$h = 1.006\,T + 2501\,w + 1.86\,w\,T \quad\Longrightarrow\quad \boxed{T(h, w) \;=\; \frac{h - 2501\,w}{1.006 + 1.86\,w}} \tag{14}$$

This is `T_from_hw`. Note what the denominator is: the moist-air specific heat per kilogram of dry air,

$$c_{p,\mathrm{m}}(w) \;=\; 1.006 + 1.86\,w \qquad [\mathrm{kJ/(kg\,K)}]. \tag{15}$$

Equation (15) reappears in that exact form in the air-to-air cooler (at the stream's own w), in the building load calculation (at the mean $\frac{1}{2}(w_{\mathrm{pd}} + w_{\mathrm{pc}})$), and in the generic load interface (at the room humidity). The regeneration coil is the exception: it uses the constant dry-air value 1006 J/(kg K) on the air side and drops the $1.86\,w$ term entirely. At the regeneration inlet humidity, about 20 g/kg, that understates the air-side capacity rate — and hence the coil duty — by roughly 3.7 %. It is a bounded inconsistency, not a justified simplification.

A second note: (13) and the building latent load use $h_{\mathrm{fg},0} = 2501\,\mathrm{kJ/kg}$, while the load generators and the load interface convert moisture rate to latent power with a separate configuration constant $h_{\mathrm{fg}} = 2.5\,\mathrm{MJ/kg}$. They differ by 0.04 % and no result depends on it, but a reader comparing q_l against an enthalpy difference will find a residual that is arithmetic, not physics.

8.5 The adiabatic mixing rule

Every junction in both air loops is this one derivation, so it is worth doing carefully once. Two streams with dry-air mass flows $\dot m_1$ and $\dot m_2$, at states (T_1, w_1) and (T_2, w_2), mix adiabatically at constant pressure into one stream $\dot m_3$ at (T_3, w_3). Three balances close it. *Dry air*, which nothing creates or destroys:

$$\dot m_3 \;=\; \dot m_1 + \dot m_2. \tag{16}$$

Water, since stream i carries $\dot m_i w_i$ of vapour:

$$\dot m_1 w_1 + \dot m_2 w_2 = \dot m_3 w_3 \quad\Longrightarrow\quad w_3 = \frac{\dot m_1 w_1 + \dot m_2 w_2}{\dot m_1 + \dot m_2}. \tag{17}$$

Energy: with no work and no heat crossing the boundary, enthalpy is conserved:

$$\dot m_1 h_1 + \dot m_2 h_2 = \dot m_3 h_3 \quad\Longrightarrow\quad h_3 = \frac{\dot m_1 h_1 + \dot m_2 h_2}{\dot m_1 + \dot m_2}. \tag{18}$$

The mixed temperature then follows from (14), $T_3 = T(h_3, w_3)$. Mixing is done in (w, h) and converted back to T at the end, never the other way round.

Why the temperature is not mass-averaged. It is tempting to write $T_3 = (\dot m_1 T_1 + \dot m_2 T_2)/\dot m_3$. In general this is wrong, and the reason is instructive. Substituting (12) into (18),

$$c_{p,\mathrm{a}} T_3 + h_{\mathrm{fg},0} w_3 + c_{p,\mathrm{v}} w_3 T_3 \;=\; \frac{\dot m_1 \big(c_{p,\mathrm{a}} T_1 + h_{\mathrm{fg},0} w_1 + c_{p,\mathrm{v}} w_1 T_1\big) + \dot m_2(\cdots)}{\dot m_3},$$

the terms linear in T and in w separate cleanly and by themselves would give a mass-averaged T_3. The cross term does not separate, because $\overline{wT} \neq \bar w\,\bar T$. Physically, the moist-air specific heat depends on humidity, so a wetter stream carries more energy per kelvin and the mixed temperature is weighted by $\dot m_i c_{p,\mathrm{m},i}$, not by $\dot m_i$. Under the code's constant-c_p form, however, that weighting can be written out in closed form; using (15) and w_3 from (17),

$$T_3 \;=\; \frac{\dot m_1 c_{p,\mathrm{m}}(w_1)\, T_1 + \dot m_2 c_{p,\mathrm{m}}(w_2)\, T_2}{(\dot m_1 + \dot m_2)\, c_{p,\mathrm{m}}(w_3)}. \tag{19}$$

So the honest statement is not that temperature is never averaged, but this: *under the constant-c_p enthalpy of* (13) *the mixed temperature is exactly the $\dot m_i c_{p,\mathrm{m},i}$-weighted average of the inlet temperatures*, degenerating to the plain mass average only when $w_1 = w_2$. Mixing is linear in (w, h) and non-linear in (w, T). Had either specific heat been temperature-dependent, even (19) would not close in one step and the junction would need iteration; the constant-c_p assumption buys an exact, non-iterative mixing node. The error from naive mass-averaging scales as $1.86\,(w_1 - w_2)$ against 1.006, and at the scavenging mixer, where exhaust near 19.9 g/kg meets outdoor air near 21.4 g/kg, it is hundredths of a kelvin — a property of this operating point, not a licence. On a dry day it grows quickly.

Fractional form as it appears in the code. At both mixing nodes the streams are fractions of one loop flow summing to unity. Writing d_A for the fresh-air fraction and $d_\mathrm{B} = 1 - d_\mathrm{A}$ for the recirculated one, (17) and (18) collapse to convex combinations with $\dot{m}_3 = \dot{m}$:

$$w_3 = d_\mathrm{B} w_1 + d_\mathrm{A} w_2, \qquad h_3 = d_\mathrm{B} h_1 + d_\mathrm{A} h_2, \qquad T_3 = T(h_3, w_3). \tag{20}$$

This is literally the body of `process_inlet_state` (recirculated room return mixed with outdoor makeup, weights d_pB and d_pA) and of the scavenging node inside `regen_air_pass` (building exhaust at the room state mixed with outdoor air, weights d_rA and d_rB). Both then add a fan temperature rise, set to 0 K in the shipped configuration, so both fans are currently isenthalpic and (20) is the entire node.

8.6 Sensible processes: what a coil does on the chart

A coil that heats or cools air without condensation transfers energy but no mass:

$$w_\mathrm{out} \;=\; w_\mathrm{in} \qquad \text{(sensible process)}, \tag{21}$$

and the state moves *horizontally* on a chart with T on the abscissa and w on the ordinate. Two components are sensible by construction here: the regeneration coil, whose ε–NTU routine returns only an air outlet temperature, and the air-to-air cooler, which does the same.

Heating at constant w collapses the relative humidity: in (11) the numerator is fixed while $p_\mathrm{sat}(T)$ climbs steeply. This is the entire purpose of the regeneration coil — it does not dry the air, it makes the air *thirsty*, and that collapsed relative humidity is what pulls water out of the sorbent. The wheel model says so explicitly: its diagnostic warns that the wheel dries the process stream only while the process-side relative humidity exceeds the regeneration-side value, and reverses to *humidifying* once that gap closes.

Cooling at constant w does the opposite: RH rises, and (21) holds only while the outlet stays above the stream's dew point. *The cooler routine does not test this.* It back-solves a conductance to hit a requested outlet temperature and returns $w_\mathrm{out} = w_\mathrm{in}$ regardless, capping at a maximum UA if the target is unreachable. The only supersaturation check anywhere is in the load interface, which evaluates (11) on the *required* supply state and warns if it exceeds unity. That is a check on demand, one step downstream, and it will not fire when the cooler is merely asked for an outlet close to saturation. Any run with a low supply temperature should be audited by evaluating (11) at the cooler outlet.

8.7 Wet-bulb temperature

The notebook contains *no* wet-bulb routine, and no dew-point routine either. This is worth stating plainly, because a reader from a conventional HVAC background will look for one. Nothing in the solution path needs it: cooling here is dry-coil sensible, dehumidification is by sorption rather than condensation, and the state is carried as (T, w) throughout.

Where a wet-bulb variable would appear classically, the wheel model uses Jurinak's characteristic potentials instead. F_1 is very nearly constant along a line of constant enthalpy — the notebook's own self-check shows this by evaluating it at three temperatures along $h = 85\,\mathrm{kJ/kg}$ — and constant enthalpy is, to good approximation, constant thermodynamic wet bulb; F_2 plays the same role for relative humidity. The wheel therefore works in an enthalpy-like and an RH-like coordinate, carrying what a wet-bulb/dry-bulb pair would carry, in the variables its effectiveness is actually fitted in. Should a wet-bulb temperature ever be wanted for reporting, it is the root of the adiabatic-saturation equation solved with the same p_sat; that routine would have to be written, and no result here depends on one.

8.8 Worked example: the shipped design point

Table 5 evaluates the two anchor states of the model with the configuration constants as shipped: $p_{\text{atm}} = 101\,325\,\text{Pa}$, ambient 31 °C at 75 % RH (Gulf design day), comfort 37 °C at 50 % RH (hatchery setter).

Table 5: Anchor states from the shipped configuration, evaluated with Eqs. (9), (10) and (13).

State	T [°C]	RH [–]	w [g/kg]	h [kJ/kg]
Outdoor air (design day)	31.0	0.75	21.36	85.8
Comfort (hatchery setter)	37.0	0.50	19.86	88.3

Check the outdoor value by hand: $p_{\text{sat}}(31) = 610.94 \exp(17.625 \times 31/274.04) = 4486\,\text{Pa}$; $p_{\text{v}} = 0.75 \times 4486 = 3365\,\text{Pa}$; $w = 0.622 \times 3365/(101325 - 3365) = 0.021\,36\,\text{kg/kg}$; $h = 1.006(31) + 0.02136\,(2501 + 1.86 \times 31) = 31.2 + 54.7 = 85.8\,\text{kJ/kg}$.

Now read the table the way the model does. The outdoor air is 6 K *cooler* than the room and its relative humidity is 25 percentage points higher, yet it carries only 1.5 g/kg more water: ventilation imposes a real but modest dehumidification duty while contributing sensible *cooling*, an unusual combination that shapes the whole control problem, since the dominant sensible load comes from the eggs and the envelope. Note also that the two enthalpies differ by only 2.5 kJ/kg across that 6 K gap — almost all of the sensible difference is repaid by the latent one. Reading enthalpy alone would suggest these are nearly the same air. They are not, and that is exactly why w is carried separately.

9 The DBHE subsurface model

The Deep Borehole Heat Exchanger is a closed coaxial loop built inside a repurposed well. Water is injected at the surface down the *annulus* (between the production casing and a central tubing string), is warmed by the surrounding rock as it descends, turns at the bottom, and returns up the *tubing bore*. Nothing is produced from the formation; only heat crosses the borehole wall.

Two sub-models are coupled. The *formation* model is transient and lives on a time scale of months to decades: the rock around the well cools progressively as heat is withdrawn. The *fluid* model is quasi-steady and lives on a time scale of minutes: the water traverses the well and leaves long before the rock notices. The whole subsurface exposes exactly two numbers to the rest of the plant — the produced temperature T_{ret} and the wall heat flux that drives the formation forward one step.

9.1 Formation: radial conduction and its two hypotheses

Around each axial cell the rock is treated as a homogeneous, radially symmetric conducting medium with no source:

$$\frac{1}{r}\frac{\partial}{\partial r}\left(r\,\frac{\partial \theta}{\partial r}\right) = \frac{1}{\alpha_e}\frac{\partial \theta}{\partial t}, \qquad r \in [r_{ce},\, R_{\text{far}}], \tag{22}$$

where $\theta(r, z, t) = T - T_\infty(z)$ is the temperature *excess* over the undisturbed geothermal profile

$$T_\infty(z) = T_{\text{surf}} + G_g z, \tag{23}$$

with $T_{\text{surf}} = 20\,°\text{C}$ and $G_g = 0.031\,\text{K/m}$ in the baseline configuration — so the undisturbed bottom-hole temperature at the demonstration depth $L = 4500\,\text{m}$ is 159.5 °C. Working in θ rather than T is not cosmetic: it makes the initial condition identically zero and the far-field condition homogeneous, which is what allows the transform of Section 9.2 to be linear and history-free.

The inner boundary at the borehole wall $r = r_{ce} = 0.107\,95\,\mathrm{m}$ (the 8.5 inch hole radius) carries a prescribed heat flux; the outer boundary carries a prescribed temperature. Two hypotheses are buried here and both deserve to be stated with their failure modes.

Hypothesis 1: axial conduction in the rock is negligible. Equation (22) has no $\partial^2\theta/\partial z^2$ term, so the N_Z axial cells are thermally independent of one another: each one solves its own one-dimensional radial problem and they communicate only through the fluid. The justification is a ratio of driving gradients. With $N_Z = 50$ cells over 4500 m the cell height is 90 m, and the undisturbed temperature difference between neighbouring cells is only $G_g \times 90\,\mathrm{m} = 2.79\,\mathrm{K}$. The radial problem, by contrast, is driven by tens of kelvin developed over a diffusion length $\delta \sim \sqrt{\alpha_e t}$ of a few metres. The axial conductive flux $k_e\,\Delta T_z/\Delta z$ is therefore smaller than the radial flux $k_e\,\Delta T_r/\delta$ by two to three orders of magnitude.

Where it fails. The argument assumes *smooth* axial variation. It breaks wherever the axial field develops a genuine kink — at the very top, where cold injected water meets warm near-surface rock over a short interval, and at the bottom turn — and it breaks if the well is refined to cells of order the radial diffusion length (metres), at which point the neglected term is no longer small relative to the retained one. At $N_Z = 50$ this is comfortable; the hypothesis should be re-examined before pushing N_Z past a few hundred.

Hypothesis 2: the domain is truncated at a finite radius R_{far}, held at the undisturbed temperature. The rock is unbounded; the model is not. The code truncates at

$$R_{\mathrm{far}} = \max\!\left(200\,\mathrm{m},\ 3.5\sqrt{4\alpha_e\, t_{\mathrm{total}}}\right), \tag{24}$$

i.e. at several diffusion lengths of the *whole* simulated horizon. For the baseline $\alpha_e = 1.05\times10^{-6}\,\mathrm{m^2/s}$ and $t_{\mathrm{total}} = 20\,\mathrm{yr}$ the diffusion length is $\sqrt{4\alpha_e t_{\mathrm{total}}} = 51.5\,\mathrm{m}$ and $3.5 \times 51.5\,\mathrm{m} = 180\,\mathrm{m}$, so it is the 200 m *floor* that is active and $R_{\mathrm{far}} = 200\,\mathrm{m}$.

The outer condition imposed there is $\theta(R_{\mathrm{far}}, t) = 0$: a Dirichlet condition pinning the far field to the undisturbed geothermal temperature. This is worth emphasising because it is easy to write "adiabatic outer boundary" by reflex, and the two choices are not interchangeable. Under a zero-flux outer boundary a continuously extracting well has *no* steady state — the enclosed rock simply cools without bound — whereas under the Dirichlet condition the wall-to-far-field problem possesses the finite steady resistance

$$R_{ss} = \frac{\ln(R_{\mathrm{far}}/r_{ce})}{2\pi k_e}, \tag{25}$$

which the tail correction of Section 9.4 depends on entirely. The code's characteristic equation (below) imposes the Dirichlet form, and Eq. (25) is used consistently with it. With $k_e = 2.2\,\mathrm{W/(m\,K)}$, $R_{ss} = 0.544\,\mathrm{K\,m/W}$.

What it costs. The truncation turns a slowly-diverging unbounded problem into a bounded one that eventually reaches steady state. Provided the disturbance never actually reaches R_{far}, the two are indistinguishable, and Eq. (24) is precisely the condition guaranteeing that. Extend the horizon or raise the flux without re-deriving R_{far} and the model starts drawing heat from an artificial infinite reservoir at the boundary, *over*-predicting sustainable output. The safeguard is that R_{far} is computed from t_{total}.

9.2 The generalized integral transform

9.2.1 Eigenvalue problem

Separating Eq. (22) gives the Sturm–Liouville problem

$$\frac{1}{r}\frac{\mathrm{d}}{\mathrm{d}r}\left(r\frac{\mathrm{d}\psi}{\mathrm{d}r}\right)+\beta^2\psi=0,\qquad \left.\frac{\mathrm{d}\psi}{\mathrm{d}r}\right|_{r_{ce}}=0,\qquad \psi(R_{\mathrm{far}})=0, \tag{26}$$

whose solutions are combinations of the zeroth-order Bessel functions J_0, Y_0. The code builds the eigenfunction so that the inner Neumann condition is satisfied identically:

$$\psi_n(r)=Y_1(\beta_n r_{ce})\,J_0(\beta_n r)-J_1(\beta_n r_{ce})\,Y_0(\beta_n r), \tag{27}$$

since $\mathrm{d}J_0(\beta r)/\mathrm{d}r=-\beta J_1(\beta r)$ makes $\psi_n'(r_{ce})\propto Y_1J_1-J_1Y_1=0$ automatically. The eigenvalues are then the roots of the remaining (outer) condition,

$$\psi(\beta,R_{\mathrm{far}})=Y_1(\beta r_{ce})\,J_0(\beta R_{\mathrm{far}})-J_1(\beta r_{ce})\,Y_0(\beta R_{\mathrm{far}})=0. \tag{28}$$

The roots are found by brute force — a scan of 800 000 points over $\beta\in(0,200]$ 1/m for sign changes, each bracket refined by Brent's method to a tolerance of 10^{-14} — and the first $N=60$ are kept. Brute force is the right choice: the roots are computed once and reused for all 29 220 time steps, so robustness matters far more than speed. At the baseline geometry the retained band runs from $\beta_1=0.0120$/m to $\beta_{60}=0.939$/m — from a mode whose time constant $1/(\alpha_e\beta_1^2)$ is of order two centuries down to one of order 300 h.

9.2.2 Transform pair and the norm

The ψ_n are orthogonal under the weight r inherited from the cylindrical Laplacian. Defining the norm

$$\mathcal{N}_n=\int_{r_{ce}}^{R_{\mathrm{far}}} r\,\psi_n^2(r)\,\mathrm{d}r, \tag{29}$$

the transform pair is

$$\bar{\theta}_n(t)=\int_{r_{ce}}^{R_{\mathrm{far}}} r\,\psi_n(r)\,\theta(r,t)\,\mathrm{d}r\qquad(\text{transform}),\qquad \theta(r,t)=\sum_{n=1}^{N}\frac{\psi_n(r)}{\mathcal{N}_n}\,\bar{\theta}_n(t)\qquad(\text{inverse}). \tag{30}$$

Only the wall value is ever needed downstream, so the code stores the *inversion weights* $\Omega_n=\psi_n(r_{ce})/\mathcal{N}_n$ once and reconstructs the wall temperature as a single dot product.

The norms are evaluated by the trapezoidal rule on a radial grid that adapts to the mode being integrated: at least 6000 points, and at least 20 points per half-wavelength π/β_n. Resolving each oscillation is the whole issue — the integrand $r\psi_n^2$ oscillates n times across the domain and is weighted towards large r.

The non-positive-norm guard. Mathematically $\mathcal{N}_n>0$ always: it is the integral of a non-negative quantity. The code nevertheless checks it and warns, naming the offending mode indices, whenever $\mathcal{N}_n\le 0$. The paranoia is not about the mathematics but about the quadrature: a non-positive norm can only mean the radial grid has aliased ψ_n and returned a meaningless number. Because $\mathcal{N}_n$ sits in the *denominator* of every inversion weight, such a failure does not produce a mildly wrong answer but a diverging one, so the warning names the two correct remedies — raise the grid resolution, or reduce the number of modes to the band the grid can support. Silence here would let a discretisation failure masquerade as a physical result.

9.2.3 The transformed equations

Multiply Eq. (22) by $r\psi_n$, integrate over $[r_{ce}, R_{\text{far}}]$, and integrate the radial operator by parts twice. Orthogonality annihilates every cross term and leaves

$$\frac{\mathrm{d}\bar{\theta}_n}{\mathrm{d}t} = -\alpha_e\beta_n^2\,\bar{\theta}_n + \alpha_e\Big[r\big(\psi_n\theta' - \psi_n'\theta\big)\Big]_{r_{ce}}^{R_{\text{far}}}. \tag{31}$$

The boundary term is where the physics enters, and both of our boundary conditions were chosen so that it collapses. At R_{far} both ψ_n and θ vanish, so the upper limit contributes nothing. At r_{ce}, $\psi_n'(r_{ce}) = 0$ kills the second piece, and Fourier's law $-k_e\theta'(r_{ce}) = q'' = q'/(2\pi r_{ce})$ converts the first, leaving

$$\boxed{\frac{\mathrm{d}\bar{\theta}_n}{\mathrm{d}t} + \alpha_e\beta_n^2\,\bar{\theta}_n = \alpha_e\beta_n^2\,F_n\,q', \qquad F_n = \frac{\psi_n(r_{ce})}{2\pi k_e\,\beta_n^2}} \tag{32}$$

where q' is the heat rate per unit borehole length delivered *into* the rock at the wall (so $q' < 0$ during extraction) and F_n is the code's precomputed forcing coefficient. Writing the right-hand side as $\alpha_e\beta_n^2 F_n q'$ rather than as a bare $F_n q'$ is deliberate: it makes F_n the *steady gain* of the mode, $\bar{\theta}_n^\infty = F_n q'$, which is what the tail correction of Section 9.4 needs.

This decoupling is the entire point of the transform. A partial differential equation in (r, t) has become N scalar ordinary differential equations that do not talk to each other.

9.3 Exact time march of the transformed modes

Over one step Δt the flux is held piecewise constant at its value q'^k from the fluid solve. Equation (32) is then a linear first-order ODE with constant coefficients and integrates *exactly*:

$$\boxed{\bar{\theta}_n^{k+1} = \bar{\theta}_n^k\,d_n + F_n\,(1 - d_n)\,q'^k, \qquad d_n = \exp\!\big(-\alpha_e\beta_n^2\Delta t\big)} \tag{33}$$

with $\Delta t = 0.25\,\text{d} = 6\,\text{h}$ in the baseline. There is no time-step stability limit and no truncation error in time: whatever error exists comes from holding q' constant across the step, not from the integrator. The decay factors d_n are computed once at initialisation.

Two properties are worth naming. First, the cost per step is $O(N)$ per axial cell and *no history is stored*: the state is the N numbers $\bar{\theta}_n$, the same size after twenty years as after one step. This is the genuine advantage of the transform over a convolution formulation (g-function or infinite cylindrical source), whose history buffer grows with simulated time. It is *not* that the transform avoids iteration — the coupled fluid–formation problem is solved by fixed point either way. Second, $d_n \to 1$ for slow modes and $d_n \to 0$ for fast ones, so Eq. (33) degrades gracefully: fast modes relax immediately to their quasi-steady value $F_n q'$, slow modes accumulate.

9.4 The analytical tail correction

9.4.1 Why a truncated series is not merely "a bit less accurate"

The high-order eigenfunctions are the ones with short radial wavelength, and short radial wavelength is what resolves the steep temperature gradient hugging the borehole wall at early time. Truncating at $N = 60$ therefore does not blur the solution uniformly; it removes the near-wellbore structure specifically. The symptom is systematic: the retained modes reproduce too *little* resistance between wall and far field, so the model under-predicts how far the wall temperature must drop to drive a given flux, and gets the first weeks to months badly wrong.

9.4.2 Deriving the deficit

The deficit can be computed exactly, because both the true and the truncated steady limits are known in closed form.

The true steady resistance is Eq. (25), the classical logarithmic result for flux at r_{ce} and fixed temperature at R_{far}. What do the N retained modes give in the same limit? Setting $\mathrm{d}\bar{\theta}_n/\mathrm{d}t = 0$ in Eq. (32) gives $\bar{\theta}_n^{\infty} = F_n q'$; inverting at the wall with Eq. (30) and dividing by q' gives the partial resistance

$$R_{\text{partial}} = \frac{1}{q'} \sum_{n=1}^{N} \frac{\psi_n(r_{ce})}{\mathcal{N}_n} F_n q' = \frac{1}{2\pi k_e} \sum_{n=1}^{N} \frac{\psi_n^2(r_{ce})}{\beta_n^2 \mathcal{N}_n}. \tag{34}$$

The identity being exploited is that the *complete* series reproduces the logarithm exactly,

$$\sum_{n=1}^{\infty} \frac{\psi_n^2(r_{ce})}{\beta_n^2 \mathcal{N}_n} = \ln(R_{\text{far}}/r_{ce})\,, \tag{35}$$

so the truncation error is not estimated — it is the closed-form remainder

$$\boxed{R_{\text{tail}} = R_{ss} - R_{\text{partial}} = \frac{\ln(R_{\text{far}}/r_{ce})}{2\pi k_e} - \frac{1}{2\pi k_e} \sum_{n=1}^{N} \frac{\psi_n^2(r_{ce})}{\beta_n^2 \mathcal{N}_n}} \tag{36}$$

This quantity is not a rounding error. Because the series converges only as $1/n^2$ under constant-flux forcing, evaluating Eqs. (25) and (34) at the baseline configuration ($N = 60$, $R_{\text{far}} = 200\,\text{m}$, $r_{ce} = 0.107\,95\,\text{m}$) leaves R_{tail} at roughly a third of R_{ss}. A model that ignored it would be wrong about the borehole-wall temperature by a first-order amount, not a refinement.

9.4.3 Carrying the deficit as one extra state

The missing resistance is given its own lumped mode, advanced with exactly the same exponential recursion but with the decay rate of the first *omitted* eigenvalue. The code estimates that eigenvalue by the asymptotic spacing of the Bessel roots, $\beta_{N+1} \approx \beta_N + \pi/R_{\text{far}}$:

$$\bar{\theta}_{\text{tail}}^{k+1} = \bar{\theta}_{\text{tail}}^{k}\, d_{\text{tail}} + R_{\text{tail}}\,(1 - d_{\text{tail}})\, q'^{\,k}, \qquad d_{\text{tail}} = \exp\!\left[-\alpha_e\, (\beta_N + \pi/R_{\text{far}})^2\, \Delta t\right]. \tag{37}$$

Note that R_{tail} plays the role of F_n here: it *is* the steady gain of the lumped remainder, which is why Eq. (32) was normalised as it was. Assigning the whole remainder a single time constant is an approximation — the omitted modes span a range of rates — but it is the conservative one, since the slowest omitted mode governs when the lumped state has finished responding.

The borehole-wall temperature is then reconstructed cell by cell as

$$T_{bw}(z) = T_\infty(z) + \sum_{n=1}^{N} \bar{\theta}_n\, \Omega_n + \bar{\theta}_{\text{tail}}, \qquad \Omega_n = \frac{\psi_n(r_{ce})}{\mathcal{N}_n}. \tag{38}$$

9.4.4 The optional second-set correction

The notebook also contains a finer variant of the same idea: instead of lumping the remainder into one state, compute a second block of eigenvalues (modes $N + 1$ through N_B, with $N_B = 1200$) explicitly, and represent their contribution by the quasi-steady resistance

$$C_B = \frac{1}{2\pi k_e} \sum_{n=N+1}^{N_B} \frac{\psi_n^2(r_{ce})}{\beta_n^2 \mathcal{N}_n}. \tag{39}$$

This is **disabled in the baseline** (`USE_QS_CORRECTION = False`, $C_B = 0$), so the results reported here use the single-state analytical tail of Eq. (37). The machinery is retained because it provides the natural convergence check on the tail approximation: as N_B grows, C_B must approach R_{tail}.

9.5 Fluid: the coaxial boundary-value problem

9.5.1 Quasi-steady hypothesis

The fluid is treated as steady *within* each time step. The separation of scales is wide: the water transits the borehole in minutes to hours, the time step is 6 h, and the formation responds over months to decades. The cost is that transients *inside* the well — the thermal front travelling down the annulus after a step change in injection temperature, and the inertia of the water column and steel — are not represented. The model must therefore not be asked about the first few hours after a start-up or a flow change; it is a tool for seasonal and multi-year behaviour, and honest only at that scale.

9.5.2 The two coupled energy balances

Two counter-current streams exchange heat with each other through the tubing wall — the parasitic "short circuit" by which the hot returning water reheats, or is cooled by, the descending water — and the annulus additionally exchanges with the borehole wall. Steady balances on differential elements, nondimensionalised with $\zeta = z/L$ and

$$\theta_j = \frac{T_j - T_{\text{in}}}{\Delta T_{sc}}, \qquad \Delta T_{sc}(t) = \max\Big(\max_z T_{bw}(z) - T_{\text{in}},\ 1\,\text{K}\Big), \tag{40}$$

give

$$\frac{\mathrm{d}\theta_a}{\mathrm{d}\zeta} = \text{NTU}_{bw}\,(\theta_{bw} - \theta_a) - \text{NTU}_{at}\,(\theta_a - \theta_t)\,, \tag{41}$$

$$\frac{\mathrm{d}\theta_t}{\mathrm{d}\zeta} = -\,\text{NTU}_{at}\,(\theta_a - \theta_t)\,. \tag{42}$$

The minus sign on the tubing equation is the counter-flow: ζ increases downward for both equations, but the tubing stream travels upward, so the derivative along its own direction of travel is $-\mathrm{d}/\mathrm{d}\zeta$.

The scale ΔT_{sc} is *recomputed every step* from the current wall field, not fixed once. The 1 K floor prevents division by zero when the well has been driven to thermal equilibrium with the injection temperature.

9.5.3 The NTU groupings

Both groups are conductance per unit length times length divided by stream capacity rate, evaluated at the *well* flow $\dot{m}_{\text{well}}$ (not the loop flow — see Section 9.8):

$$\text{NTU}_{at} = \frac{G^{\text{eff}}_{a,t}\,L}{\dot{m}_{\text{well}}c_{p,w}}, \qquad \text{NTU}_{bw} = \frac{G^{\text{eff}}_{a,ce}\,L}{\dot{m}_{\text{well}}c_{p,w}}. \tag{43}$$

Each conductance is a series chain of resistances per unit length, built on log-mean (thermodynamic centroid) radii $\bar{r} = (r_o - r_i)/\ln(r_o/r_i)$ so that each solid wall can be split into an inner and an outer half without introducing an error:

$$\frac{1}{G_{a,t}^{\text{eff}}} = \underbrace{\frac{1}{2\pi r_{t,o} h_a} + \frac{\ln(r_{t,o}/\bar{r}_t)}{2\pi k_t}}_{\text{annulus film + outer half of tubing}} + \underbrace{\frac{\ln(\bar{r}_t/r_{t,i})}{2\pi k_t} + \frac{1}{2\pi r_{t,i} h_{tb}}}_{\text{inner half of tubing + bore film}}, \tag{44}$$

$$\frac{1}{G_{a,ce}^{\text{eff}}} = \underbrace{\frac{1}{2\pi r_{ca,i} h_a} + \frac{\ln(\bar{r}_{ca}/r_{ca,i})}{2\pi k_{ca}}}_{\text{annulus film + inner half of casing}} + \underbrace{\frac{\ln(r_{ca,o}/\bar{r}_{ca})}{2\pi k_{ca}} + \frac{\ln(\bar{r}_{ce}/r_{ca,o})}{2\pi k_{ce}}}_{\text{outer half of casing + inner half of cement}} + \underbrace{\frac{\ln(r_{ce}/\bar{r}_{ce})}{2\pi k_{ce}}}_{\text{outer half of cement}}. \tag{45}$$

The steel conductivities ($k_{ca} = k_{t,\text{bare}} = 45\,\text{W/(m K)}$) make the wall terms negligible in Eq. (44) for a *bare* tubing, which is exactly the problem: the short circuit is then strong. Vacuum insulated tubing, $k_t = 0.02\,\text{W/(m K)}$, raises the tubing resistance by more than three orders of magnitude and is what suppresses it. The model carries both cases explicitly and labels the state "Bare" or "Insulated" from the value of k_t.

Water properties (ρ, c_p, μ, k_f, Pr) come from IAPWS-IF97 at the operating pressure $P_{\text{op}} = 50\,\text{bar}$, evaluated once per well state at a representative mean temperature. A constant-property shortcut was explicitly rejected in the code: a fixed 70 °C estimate misstates μ and Pr by about −38% at 43 °C and +90% at 130 °C, and both feed the film coefficients directly.

9.6 Transfer-matrix solution of the two-point BVP

Equations (41)–(42) are linear with coefficients that are constant within each depth cell (the wall temperature θ_{bw} is piecewise constant, one value per cell). Writing $\boldsymbol{\theta} = (\theta_a, \theta_t)^{\mathsf{T}}$,

$$\frac{\mathrm{d}\boldsymbol{\theta}}{\mathrm{d}\zeta} = \mathbf{A}\boldsymbol{\theta} + \mathbf{b}_k, \qquad \mathbf{A} = \begin{pmatrix} -(\text{NTU}_{at} + \text{NTU}_{bw}) & +\text{NTU}_{at} \\ -\text{NTU}_{at} & +\text{NTU}_{at} \end{pmatrix}, \qquad \mathbf{b}_k = \begin{pmatrix} \text{NTU}_{bw}\,\theta_{bw,k} \\ 0 \end{pmatrix}, \tag{46}$$

and the exact propagator across one cell of thickness $\Delta\zeta = 1/N_Z$ is

$$\boldsymbol{\theta}_{k+1} = \boldsymbol{\Phi}\,\boldsymbol{\theta}_k + \boldsymbol{\Psi}\,\mathbf{b}_k, \qquad \boldsymbol{\Phi} = e^{\mathbf{A}\Delta\zeta}, \qquad \boldsymbol{\Psi} = \int_0^{\Delta\zeta} e^{\mathbf{A}s}\,\mathrm{d}s = \mathbf{A}^{-1}(\boldsymbol{\Phi} - \mathbf{I}). \tag{47}$$

Both $\boldsymbol{\Phi}$ and $\boldsymbol{\Psi}$ are built once per flow state and reused down the whole well. Because $\det\mathbf{A} = -\text{NTU}_{at}\,\text{NTU}_{bw}$, the closed form for $\boldsymbol{\Psi}$ requires both groups strictly positive; a perfectly insulating tubing ($k_t \to 0$, so $\text{NTU}_{at} \to 0$) would make $\mathbf{A}$ singular. The vacuum-insulated case is small but finite, so this is a limit to be aware of rather than a present defect.

Closing the boundary conditions. The problem is a two-point BVP: the annulus condition is known at the surface ($\theta_a(0) = 0$, injection at T_{in}) but the tubing condition is known only at the bottom (the two streams join, $\theta_a(1) = \theta_t(1)$). The code exploits linearity instead of shooting iteratively. It marches two independent solutions from the surface,

$$\mathbf{y}_0^P = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \text{ (particular, with the } \mathbf{b}_k \text{ forcing)}, \qquad \mathbf{y}_0^H = \begin{pmatrix} 0 \\ 1 \end{pmatrix} \text{ (homogeneous, unforced)}, \tag{48}$$

both of which already satisfy $\theta_a(0) = 0$, and forms $\boldsymbol{\theta} = s\,\mathbf{y}^H + \mathbf{y}^P$. The single unknown s is the surface tubing temperature, and the bottom joining condition fixes it in one division:

$$s = \frac{y_t^P(1) - y_a^P(1)}{y_a^H(1) - y_t^H(1)}. \tag{49}$$

No iteration, no stability limit, no step-size restriction — the axial discretisation error comes only from treating θ_{bw} as piecewise constant, since the ODE itself is integrated exactly within each cell. The produced temperature is read off directly:

$$T_{\text{ret}} = T_{\text{in}} + \theta_t(0)\,\Delta T_{sc}. \tag{50}$$

9.7 Closing the loop: flux back to the formation

The fluid solve returns the annulus temperature at each cell centre, obtained by averaging the two cell-face values, $T_{a,cc,k} = T_{\text{in}} + \frac{1}{2}(\theta_{a,k} + \theta_{a,k+1})\Delta T_{sc}$. That temperature sets the wall flux that drives Eq. (33):

$$\boxed{q'_k = G_{\text{eff}}\,(T_{a,cc,k} - T_{bw,k})\,, \qquad G_{\text{eff}} = \frac{\text{NTU}_{bw}\,\dot{m}_{\text{well}}\,c_{p,w}}{L} \;=\; G^{\text{eff}}_{a,ce}} \tag{51}$$

per unit borehole length. The last equality is a consistency identity, not a new model: substituting Eq. (43) recovers the conductance chain of Eq. (45) exactly. The same conductance therefore appears on both sides of the coupling, which is what guarantees the energy the fluid gains is the energy the rock loses. During extraction $T_{a,cc} < T_{bw}$, so $q' < 0$, and Eq. (33) drives $\bar{\theta}_n$ negative and T_{bw} below T_∞: the formation depletes.

The order within a step is *evaluate, then advance*: the fluid is solved against the wall field left by the previous step, and only after the surface loop has converged is the formation marched with the resulting flux. Relative to the formation's own time constants a 6 h lag is negligible — far smaller than what the quasi-steady fluid hypothesis already concedes.

Retirement of the operating fraction f_{op}. Earlier versions derated the forcing by a duty fraction, $q' = f_{\text{op}}\,G_{\text{eff}}(T_{a,cc} - T_{bw})$, to represent a load running only part of the day. That factor has been **removed** from Eq. (51). In the coupled model the subsurface runs *continuously* and the duty cycle is expressed where it physically lives: in the surface valve schedule, principally the well-flow fraction $w_{v\text{E}}$, and in the thermal accumulator. Derating the flux conflated two distinct questions — how hard the well is run, and how much load is served — which made the predicted formation depletion genuinely ambiguous. The consequence must be stated plainly: the formation now sees the full, un-derated flux, so for a given well flow the drawdown is faster than under the derated formulation, and any earlier write-up carrying f_{op} inside the transform forcing requires amendment. The symbol survives in the configuration at $f_{\text{op}} = 1.00$ only so that legacy references do not break; it multiplies nothing.

9.8 Film coefficients, the Reynolds clamp, and the validity flag

Both film coefficients h_a and h_{tb} in Eqs. (44)–(45) come from the Gnielinski correlation with the Petukhov friction factor,

$$f_D = (0.790\ln\text{Re} - 1.64)^{-2}\,, \qquad \text{Nu} = \frac{(f_D/8)\,(\text{Re} - 1000)\,\text{Pr}}{1 + 12.7\sqrt{f_D/8}\left(\text{Pr}^{2/3} - 1\right)}, \qquad h = \frac{\text{Nu}\,k_f}{D_h}, \tag{52}$$

evaluated on the hydraulic diameters $D_{h,a} = 2(r_{ca,i} - r_{t,o})$ for the annulus and $D_{h,tb} = 2r_{t,i}$ for the tubing bore. The two velocities are tied by mass conservation, $v_{tb} = v_a A_a / A_{tb}$.

Equation (52) is valid for Re $\gtrsim$ 3000; below that it is not merely inaccurate but pathological, since the factor (Re − 1000) drives Nu to zero at Re = 1000 and negative below it. The well flow is $w_{v\text{E}}\dot{m}$, and $w_{v\text{E}}$ is a control that the surface plant may throttle, so low Reynolds numbers are reachable in normal operation. The code handles this in two separate places, and the distinction matters.

1. *Numerically*, the correlation is **clamped**: the Reynolds number fed to Eq. (52) is $\max(\mathrm{Re}, 3100)$. Below the clamp the film coefficient simply freezes at its value at $\mathrm{Re} = 3100$. This keeps h finite and positive so the conductance chain never produces a negative or singular NTU, and it deliberately does *not* switch to a laminar correlation mid-march.

2. *Diagnostically*, the true unclamped Reynolds numbers of both streams are recomputed every time the flow is set, and a one-shot message is printed naming $w_{v\mathrm{E}}$ and both values whenever either falls below 3000. The flag fires once per well state, so a long throttled run reports the condition without flooding the log.

Two facts make the clamp acceptable rather than fatal. First, the regime where the correlation is weakest is the regime where the answer barely depends on it: at low $w_{v\mathrm{E}}$ the capacity rate $\dot{m}_{\mathrm{well}} c_{p,w}$ is small, NTU_{bw} is correspondingly large, the annulus water nearly equilibrates with the wall regardless of h_a, and T_{ret} becomes insensitive to it. Second, the control carries a floor $w_{v\mathrm{E}} \geq w_{v\mathrm{E,min}} = 0.05$, keeping a trickle circulating rather than shutting down; a true shutdown would require a different physical model (conduction into a stagnant column), and switching models mid-march would introduce a discontinuity the tail correction of Section 9.4 has not been validated against. One continuous model with an honest warning is the more defensible choice — but the warning must be read, because a run spending most of its time below $\mathrm{Re} = 3000$ is reporting film coefficients the correlation does not support.

9.9 Summary of the subsurface parameters

Two derived quantities close the section, because both are load-bearing and neither is a free parameter. The truncation radius follows from the horizon by Eq. (24) and lands on its own floor at 200 m; the number of time steps follows from the horizon and Δt and is 29 220 for the twenty-year baseline. Every other number in Table 6 is a declared input, and the two outputs the rest of the plant consumes are Eq. (50) and Eq. (51).

10 The surface water circuit

The water circuit sits between the wellhead and the regeneration coil: a pump that fixes the loop mass flow $\dot{m}$, a set of splits and mixing junctions that decide *how much* water sees the well, the store and the coil, and a lumped accumulator that moves heat in time rather than only in space. The modelling hypothesis is deliberately blunt — incompressible water, one constant $c_{p,w}$ evaluated at a representative loop temperature, adiabatic instantaneous junctions, pipe runs with neither capacitance nor heat loss. The cost is stated at the end of this section; the benefit is that every junction collapses from an enthalpy balance to a flow-weighted average of temperatures, which is what makes the coupled fixed point of Section 11 cheap enough to run 29 220 time steps.

10.1 Topology

Starting at the pump discharge, at T_{wi}: at the **E/F bifurcation** a fraction $w_{v\mathrm{E}}$ of the loop flow goes *down the well* and the remainder $w_{v\mathrm{F}} = 1 - w_{v\mathrm{E}}$ *bypasses* it at T_{wi}. Branch E is produced at T_{ret} and re-mixes with branch F to give T_{wr}. At the next fork, $w_{v\mathrm{A}}$ enters the **accumulator** and leaves at T_{wa} while $w_{v\mathrm{B}} = 1 - w_{v\mathrm{A}}$ heads straight for the coil; the accumulator outlet is then split

Table 6: Baseline subsurface configuration, as declared in the model.

Symbol	Quantity	Value
Geometry (legacy S. Louisiana well: 2-7/8 in tubing, 7 in casing, 8.5 in hole)		
$r_{t,i}$	tubing inner radius	0.030 99 m
$r_{t,o}$	tubing outer radius	0.036 52 m
$r_{ca,i}$	casing inner radius	0.079 75 m
$r_{ca,o}$	casing outer radius	0.088 90 m
r_{ce}	cement outer radius (hole)	0.107 95 m
Materials and formation		
k_{ca}	casing steel conductivity	45.0 W/(m K)
k_{ce}	cement conductivity	0.95 W/(m K)
k_t	tubing: bare steel / VIT	45.0 / 0.02 W/(m K)
k_e	formation conductivity	2.2 W/(m K)
α_e	formation diffusivity	1.05×10^{-6} m^2/s
T_{surf}	mean surface temperature	20.0 °C
G_g	geothermal gradient	0.031 K/m
Operation and numerics		
T_{in}	injection temperature	25.0 °C
P_{op}	wellbore operating pressure	50 bar
L	depth (demonstration case)	4500 m
N_Z	axial cells in the fluid BVP	50
N	retained eigenvalues (Set A)	60
Δt	time step	0.25 d = 6 h
R_{far}	transform truncation radius	200 m
$w_{v\text{E,min}}$	floor on well-flow fraction	0.05
$T_{\text{cement}}^{\text{max}}$	cement reliability limit	165.0 °C

again, $w_{v\mathrm{C}}$ of it returning to the coil inlet and $w_{v\mathrm{D}} = 1 - w_{v\mathrm{C}}$ bypassing the coil. The coil inlet blend is T_{wm} and its water outlet T_{wh}; mixing that with branch D gives the loop return T_{wo}, and the pump closes the loop back to T_{wi}.

Table 7 collects the split fractions. In the baseline $w_{v\mathrm{E}} = 1$, so the F bypass is closed and the well carries the full loop flow; the branch exists because a well-flow schedule is the natural optimisation variable, not because the baseline uses it.

Table 7: Valve split fractions of the water circuit (Section K of the configuration cell). The complements $w_{v\mathrm{B}}, w_{v\mathrm{D}}, w_{v\mathrm{F}}$ are not free parameters: each is 1 minus its partner.

Symbol	Meaning	Baseline	Complement
$w_{v\mathrm{A}}$	loop flow into the accumulator	0.30	$w_{v\mathrm{B}} = 0.70$
$w_{v\mathrm{C}}$	accumulator outlet returned to the coil	0.50	$w_{v\mathrm{D}} = 0.50$
$w_{v\mathrm{E}}$	loop flow sent down the well	1.00	$w_{v\mathrm{F}} = 0$
$w_{v\mathrm{E,min}}$	floor imposed on the well flow	0.05	—

10.2 The E/F bifurcation, and why branch F is not just a mixing node

Write an enthalpy balance on the junction where the produced stream $w_{v\mathrm{E}}\dot{m}$ at T_{ret} meets the bypass $w_{v\mathrm{F}}\dot{m}$ at T_{wi}. With $c_{p,w}$ constant across the junction the specific heats cancel and only the flow weights survive:

$$\boxed{T_{\mathrm{wr}} = w_{v\mathrm{E}}\, T_{\mathrm{ret}} + (1 - w_{v\mathrm{E}})\, T_{\mathrm{wi}}} \tag{53}$$

Setting $w_{v\mathrm{E}} = 1$ returns $T_{\mathrm{wr}} = T_{\mathrm{ret}}$, so the bifurcation is backward compatible with the plain single-well loop.

Equation (53) looks like a trivial dilution, and if the well were a fixed temperature source it would be. It is not. The DBHE model is *parametrised by the flow through it*: the well is re-evaluated at $w_{v\mathrm{E}}\dot{m}$, which sets the annulus and tubing velocities, hence the Reynolds numbers, hence the Gnielinski film coefficients, hence the transfer units NTU_{at}, NTU_{bw} and the effective wall conductance G_{eff} that drives the formation. Closing branch F therefore does two opposing things at once: it sends more water down the hole, *increasing* the extracted duty, and it shortens the residence time while changing the internal conductances, *decreasing* T_{ret}. Branch F is a temperature-versus-duty knob on the subsurface, not a mixing valve.

This is why the well model is split in two. The expensive, flow-independent objects — the GITT eigenmodes, the tail resistance, the persistent wall-temperature state — are built once at initialisation; only the cheap flow-dependent terms are refreshed each step, which makes a time-varying $w_{v\mathrm{E}}$ affordable.

A consistency caveat. The code refreshes the well terms at $\max(w_{v\mathrm{E}}, w_{v\mathrm{E,min}})$ with $w_{v\mathrm{E,min}} = 0.05$ — a scheduled shutdown leaves a trickle circulating rather than a stagnant column — while the junction Eq. (53) uses the raw $w_{v\mathrm{E}}$. For any $w_{v\mathrm{E}} \geq w_{v\mathrm{E,min}}$, including the baseline, the two coincide; below the floor they do not, and a future scheduling study must either clamp the schedule or apply the floor in both places.

10.3 Mixing junctions and the mass-conservation check

Two streams reach the coil inlet: the direct branch B, carrying $w_{v\mathrm{B}}\dot{m}$ at T_{wr}, and the recycled branch C, carrying $w_{v\mathrm{C}}w_{v\mathrm{A}}\dot{m}$ at the accumulator outlet T_{wa}. The fraction of the loop flow that actually passes through the coil is therefore

$$\phi_{\text{coil}} = w_{v\text{B}} + w_{v\text{C}}\, w_{v\text{A}} = (1 - w_{v\text{A}}) + w_{v\text{C}}\, w_{v\text{A}}, \tag{54}$$

and the same enthalpy-balance argument that gave Eq. (53) gives the coil inlet and the loop return:

$$T_{\text{wm}} = \frac{w_{v\text{B}}\, T_{\text{wr}} + w_{v\text{C}} w_{v\text{A}}\, T_{\text{wa}}}{\phi_{\text{coil}}}, \tag{55}$$

$$T_{\text{wo}} = \phi_{\text{coil}}\, T_{\text{wh}} + w_{v\text{D}} w_{v\text{A}}\, T_{\text{wa}}. \tag{56}$$

Equation (55) is normalised by ϕ_{coil} because it is an average over the coil stream only; Eq. (56) is not, because it is an average over the whole loop. That last statement holds only if the weights close, so check it rather than assert it. Substituting $w_{v\text{B}} = 1 - w_{v\text{A}}$ and $w_{v\text{D}} = 1 - w_{v\text{C}}$,

$$\phi_{\text{coil}} + w_{v\text{D}} w_{v\text{A}} = (1 - w_{v\text{A}}) + w_{v\text{C}} w_{v\text{A}} + (1 - w_{v\text{C}})\, w_{v\text{A}} = 1. \tag{57}$$

Equation (56) is thus a proper convex combination for *any* valve setting, and the circuit conserves mass identically. This is not decoration: because the loop is closed and iterated to a fixed point, a weight set that did not close would inject a spurious enthalpy source that the iteration would happily converge to, and the error would look like a physical result.

The coil throttle is currently a constant. It is tempting to read ϕ_{coil} as the plant's heat-delivery actuator, and eventually it will be. In the code of record it is not: ϕ_{coil} is evaluated *once*, before the time loop, from the fixed constants of Table 7, so with $w_{v\text{A}} = 0.30$ and $w_{v\text{C}} = 0.50$,

$$\phi_{\text{coil}} = 0.70 + 0.50 \times 0.30 = 0.85, \qquad w_{v\text{D}} w_{v\text{A}} = 0.15, \tag{58}$$

for the whole 20-year run. The humidity actuator in the present model is the regeneration vent damper, not the water throttle. Making ϕ_{coil} a controlled variable — throttling water away from the coil when the wheel needs less regeneration and banking the surplus in the accumulator — is the intended next step, and Eqs. (54)–(57) already admit a time-varying $w_{v\text{A}}, w_{v\text{C}}$ without breaking the mass balance.

10.4 The thermal accumulator

The store — a packed sand or concrete box, or a water tank — is one lumped capacitance: a single temperature T_{acc}, no internal gradients, no stratification. Its heat capacity is

$$C_{\text{acc}} = \rho_{\text{acc}}\, c_{p,\text{acc}}\, V_{\text{acc}} = 1600 \times 830 \times 2000 \approx 2.66 \times 10^9 \text{ J/K}. \tag{59}$$

Water passing through it at $w_{v\text{A}}\dot{m}$ does not reach T_{acc}; the approach is set by one heat-exchange effectiveness ε_{acc}, defined as always by actual over maximum possible temperature change, so the exit temperature is

$$T_{\text{wa}} = T_{\text{wr}} - \varepsilon_{\text{acc}} \left(T_{\text{wr}} - T_{\text{acc}}\right), \qquad \varepsilon_{\text{acc}} = 0.70. \tag{60}$$

As $\varepsilon_{\text{acc}} \to 1$ the water leaves at the store temperature; as $\varepsilon_{\text{acc}} \to 0$ the store is invisible. With no through-flow the code returns $T_{\text{wa}} = T_{\text{acc}}$, keeping Eq. (55) finite in the degenerate case.

An energy balance on the store — what the water gives up minus what leaks to the surroundings — is the governing ODE:

$$\boxed{C_{\text{acc}} \frac{\mathrm{d}T_{\text{acc}}}{\mathrm{d}t} = \underbrace{\varepsilon_{\text{acc}}\, w_{v\text{A}} \dot{m}\, c_{p,w} \left(T_{\text{wr}} - T_{\text{acc}}\right)}_{\dot{Q}_{\text{charge}}} - \underbrace{(UA)_{\text{acc}} \left(T_{\text{acc}} - T_{\text{amb}}\right)}_{\dot{Q}_{\text{loss}}}} \tag{61}$$

The charge term is exactly the enthalpy drop of the through-flow implied by Eq. (60), since

$$w_{v\mathrm{A}}\dot{m}\,c_{p,w}\,(T_{\mathrm{wr}}-T_{\mathrm{wa}})=\varepsilon_{\mathrm{acc}}\,w_{v\mathrm{A}}\dot{m}\,c_{p,w}\,(T_{\mathrm{wr}}-T_{\mathrm{acc}})\,.$$

The store and the water agree on how much heat changed hands, so the model is energy consistent at this junction — not automatic when an effectiveness and a capacitance are written by different hands.

The code integrates Eq. (61) with a forward Euler step, committed once per time step after the fixed point has converged:

$$T_{\mathrm{acc}}^{k+1}=T_{\mathrm{acc}}^{k}+\frac{\Delta t}{C_{\mathrm{acc}}}\Big[\varepsilon_{\mathrm{acc}}\,w_{v\mathrm{A}}\dot{m}\,c_{p,w}\big(T_{\mathrm{wr}}^{k}-T_{\mathrm{acc}}^{k}\big)-(UA)_{\mathrm{acc}}\big(T_{\mathrm{acc}}^{k}-T_{\mathrm{amb}}\big)\Big],\tag{62}$$

with $\Delta t = 21\,600\,\mathrm{s}$ (a quarter day). Explicit Euler on a linear relaxation is stable only for $\Delta t \le C_{\mathrm{acc}}/(\varepsilon_{\mathrm{acc}} w_{v\mathrm{A}}\dot{m}\,c_{p,w}+(UA)_{\mathrm{acc}})$, which a store of $2.66\times10^{9}\,\mathrm{J/K}$ driven by a few kilograms per second satisfies by three orders of magnitude. First-order accuracy is adequate because the store's own time constants are months while the forcing is resolved four times a day.

The sign of $\dot{Q}_{\mathrm{charge}}$ takes care of itself: when $T_{\mathrm{wr}} > T_{\mathrm{acc}}$ the store charges, and when the formation has been drawn down over several years and T_{wr} falls below the store, the same term goes negative and the store discharges into the loop. That reversal is the entire purpose of the component, and it needs no switch.

The loss term is not a detail. It is easy to treat $(UA)_{\mathrm{acc}}$ as a rounding error, because for a single overnight cycle it is one. With $(UA)_{\mathrm{acc}} = 150\,\mathrm{W/K}$ the store's loss time constant is

$$\tau_{\mathrm{loss}}=\frac{C_{\mathrm{acc}}}{(UA)_{\mathrm{acc}}}=\frac{2.66\times10^{9}\,\mathrm{J/K}}{150\,\mathrm{W/K}}\approx1.77\times10^{7}\,\mathrm{s}\approx205\,\mathrm{d},\tag{63}$$

so over one night an excess above ambient decays by about 0.5 % — negligible, as expected. But the store is not proposed as an overnight buffer; it is proposed as the thing that carries heat from a fresh year into a depleted one. On that horizon the same number reads very differently: an unforced excess retains only $\exp(-365/205)\approx 17\,\%$ of itself after a year and loses about 14 % in a single month. Whether "charge now, recover later" is engineering or wishful thinking is decided entirely by $(UA)_{\mathrm{acc}}$ integrated over the storage duration, and Eq. (63) deserves a sensitivity study before any claim about seasonal storage is made. Note also that the sink temperature is bound to the outdoor air, $T_{\mathrm{amb}} = T_{\mathrm{oa}}$, so under weather-file forcing the loss will track the season automatically — largest in summer, which is precisely when the store is being charged.

10.5 Frictional pressure drop and pump power

The coaxial well is two passages in series: down the annulus, up the tubing. Each contributes a Darcy–Weisbach loss over the full depth L, and the model sums them:

$$\Delta p=\sum_{i\in\{a,t\}} f_i\,\frac{L}{D_{h,i}}\,\frac{\rho\,v_i^2}{2},\qquad \mathrm{Re}_i=\frac{\rho\,|v_i|\,D_{h,i}}{\mu}.\tag{64}$$

The friction factor is a two-branch correlation:

$$f_i=\begin{cases}\dfrac{64}{\mathrm{Re}_i}, & \mathrm{Re}_i<2300\quad\text{(fully developed laminar)},\\[2ex](0.790\ln\mathrm{Re}_i-1.64)^{-2}, & \mathrm{Re}_i\ge2300\quad\text{(smooth-pipe Petukhov fit)}.\end{cases}\tag{65}$$

The laminar branch is exact for a circular pipe; the turbulent branch is Petukhov's explicit fit, chosen so that hydraulics and heat transfer stay consistent — it is the same f the Gnielinski correlation uses for the film coefficients, so a single friction factor serves both.

What the switch costs. The two branches do not meet. At $\mathrm{Re} = 2300$ the laminar expression gives $f = 0.0278$ and the Petukhov fit gives $f = 0.0499$ — a factor of 1.8 jump, with no blending applied — and Petukhov's fit, correlated for $\mathrm{Re} \gtrsim 10^4$, is being extrapolated down to 2300. In the design regime both passages are comfortably turbulent and neither issue bites; but a low-flow schedule, exactly what a time-varying $w_{v\mathrm{E}}$ produces, will walk the annulus through the discontinuity and Δp will show a step that is numerical, not physical.

Hydraulic power divided by the wire-to-water efficiency gives the pump consumption, and the dissipated power is returned to the water as a temperature rise on the loop return:

$$P_{\mathrm{pump}} = \frac{\Delta p\, \dot{V}}{\eta_{\mathrm{pump}}}, \qquad \eta_{\mathrm{pump}} = 0.65, \qquad T_{\mathrm{wi}} = T_{\mathrm{wo}} + \frac{P_{\mathrm{pump}}}{\dot{m}\, c_{p,w}}. \tag{66}$$

Equation (66) closes the loop, and with it the water circuit.

Scope note. Three simplifications live in Eqs. (64)–(66) and should be named rather than buried. First, friction is evaluated on the *well* stream only: surface piping, fittings, and the throttling loss required to push $w_{v\mathrm{E}}\dot{m}$ through a deep, high-resistance well while a near-zero-resistance bypass sits in parallel, are not counted. Second, $\dot{V}$ in Eq. (66) is the *well* volumetric flow while the temperature rise is spread over the *loop* flow $\dot{m}$; these coincide only at $w_{v\mathrm{E}} = 1$, the baseline, so any partial-flow study inherits an attribution error. Third, the pump's thermal effect is small next to the coil duty; the term is retained so that the loop energy balance closes exactly, not because it matters numerically.

11 The regeneration coil

The coil is the plant's water-to-air interface: geothermal water inside the tubes, regeneration air across the fins, no phase change on either side. Because the air is heated and nothing condenses, the process is purely sensible and the humidity ratio passes through unchanged, $w_{\mathrm{ro}} = w_{\mathrm{ri}}$. Everything the coil does is captured by one duty and two outlet temperatures.

11.1 Why effectiveness–NTU and not LMTD

Both methods describe the same exchanger; the choice is decided by what is known and by how often the calculation must run. This is a *rating* problem: the inlet temperatures, the capacity rates and UA are known, and the outlets are wanted. The LMTD method builds the duty from a log-mean temperature difference computed from the *outlets* — the very quantities being sought — so it must be iterated, and each iteration takes a logarithm of a difference that passes through zero when the streams pinch. The ε–NTU method inverts that dependency: effectiveness is a closed-form function of NTU and the capacity ratio alone, both known before the solve, so the coil is evaluated once and exactly.

That matters here more than usual, because the coil does not sit on its own. It is evaluated inside the coupled fixed point on $(T_{\mathrm{wi}}, T_{\mathrm{ri}}, w_{\mathrm{ri}})$, which runs up to 40 iterations; that fixed point is itself re-run inside the humidity back-solve, which allows up to 12 outer iterations plus two anchoring solves; and the whole thing repeats for 29 220 time steps. A coil needing its own inner iteration would put a third loop inside the second and a non-smooth convergence criterion inside a relaxed fixed point — a reliable way to stall the outer solve.

11.2 Capacity rates and the effectiveness relation

The hot side is the coil water stream, the cold side the regeneration air:

$$C_h = \phi_{\text{coil}}\, \dot{m}\, c_{p,w}, \qquad C_c = \dot{m}_{\text{reg}}\, c_{p,a}, \qquad c_{p,a} = 1006\,\text{J/(kg K)}. \tag{67}$$

From these follow the three quantities the method needs:

$$C_{\min} = \min(C_h, C_c), \quad C_{\max} = \max(C_h, C_c), \quad C_r = \frac{C_{\min}}{C_{\max}}, \quad \text{NTU} = \frac{UA_{\text{coil}}}{C_{\min}}. \tag{68}$$

$C_{\min}$ is the stream that runs out of capacity first, and it alone sets the thermodynamic ceiling on the duty: no exchanger, however large, can transfer more than $C_{\min}(T_{\text{wm}} - T_{\text{ri}})$, since doing so would drive that stream past the other's inlet temperature. C_r measures how badly the streams are matched; NTU measures how much exchanger has been bought relative to the stream that limits it.

The coil is a finned-tube crossflow unit with *both fluids unmixed* — water confined to tubes, air to fin passages, so neither stream can even out its own transverse temperature profile. This is the arrangement the code selects by default and the one used throughout the facility model. Its standard closed-form approximation is

$$\boxed{\varepsilon = 1 - \exp\left\{ \frac{\text{NTU}^{0.22}}{C_r} \left[\exp\left(-C_r\, \text{NTU}^{0.78}\right) - 1\right] \right\}} \tag{69}$$

This is a correlation, not an exact solution — the exact unmixed–unmixed crossflow result is an infinite series — accurate to about 1 % over the usual design range. A counterflow branch also exists in the code for comparison, but it is not the default and the facility driver never selects it; quoting counterflow effectiveness here would overstate the coil, since counterflow is the best possible arrangement and unmixed crossflow is not.

11.3 Duty and outlets

Effectiveness is actual duty over the thermodynamic maximum, so the duty and both outlets follow immediately from an energy balance on each stream:

$$Q_{\text{coil}} = \varepsilon\, C_{\min}\, (T_{\text{wm}} - T_{\text{ri}}), \qquad T_{\text{ro}} = T_{\text{ri}} + \frac{Q_{\text{coil}}}{C_c}, \qquad T_{\text{wh}} = T_{\text{wm}} - \frac{Q_{\text{coil}}}{C_h}. \tag{70}$$

T_{ro} is the regeneration temperature delivered to the desiccant wheel; T_{wh} returns to the mixing node of Eq. (56). Since $\varepsilon < 1$ strictly, $T_{\text{ro}} < T_{\text{wm}}$ always: the coil approaches the water inlet temperature but never reaches it, and no amount of UA changes that once C_c is not the limiting stream.

Equation (70) carries no sign guard. If the regeneration air ever arrives hotter than the coil water, Q_{coil} goes negative and the coil cools the air — physically correct for a sensible exchanger, and deliberately left in, but it makes a negative Q_{coil} a diagnostic worth reading rather than an error.

One approximation worth naming. The cold-side capacity rate uses the dry-air specific heat $c_{p,a} = 1006\,\mathrm{J/(kg\,K)}$ rather than the moist-air value $c_{p,a} + 1860\,w$. At a humidity ratio of $20\,\mathrm{g/kg}$ this understates C_c by about 3.7 %, which propagates into NTU, ε and T_{ro}. The load side of the model does carry the moist-air correction, so the inconsistency is confined to the coil and is small; it is recorded here so that it is not rediscovered as a bug.

11.4 The temperature–throughput trade-off

This is the coil's controlling design tension and it constrains the whole plant, so it is worth deriving rather than asserting. Take the air side first, at a fixed water stream. If the air flow is small, $C_c \ll C_h$, then $C_{\min} = C_c$ and $\mathrm{NTU} = UA/C_c$ is large: the air leaves close to the water inlet, $T_{\mathrm{ro}} \to T_{\mathrm{wm}}$, but the duty $Q = \varepsilon C_c(T_{\mathrm{wm}} - T_{\mathrm{ri}})$ is small because C_c is. *Hot air, little heat.* If the air flow is large, $C_c \gg C_h$, then $C_{\min} = C_h$, the duty saturates at $\varepsilon C_h(T_{\mathrm{wm}} - T_{\mathrm{ri}})$ — the water is now the bottleneck — and the temperature rise Q/C_c tends to zero. *Much air, cool air.* The wheel needs a minimum T_{ro} to regenerate at all; the plant needs enough Q to cover the latent load. The two requirements pull in opposite directions and the air-side design sits on that frontier.

The water side needs a more careful statement than it is usually given. Taken in isolation, at *fixed* T_{wm}, raising the water flow raises C_h and raises both Q and T_{ro}; there is no trade-off internal to the coil, only diminishing returns as the duty saturates at the air-limited value $\varepsilon C_c(T_{\mathrm{wm}} - T_{\mathrm{ri}})$. The trade-off appears only once the coil is closed on the well. T_{wm} is not independent: through Eqs. (53) and (55) it is inherited from T_{ret}, and the DBHE delivers less temperature the harder it is pumped. Raising the loop flow therefore raises the well's duty and lowers T_{ret}, and the coil passes that on faithfully: *more water raises the duty and lowers the temperature the wheel sees.* The mechanism is the well, not the exchanger.

The same mechanism explains why one well starves a large plant. Scaling the hatchery scales $\dot m_{\mathrm{reg}}$ and hence C_c, but the water stream is set by a single borehole. Once $C_h < C_c$ the coil is water-limited, the duty is pinned at $\varepsilon C_h(T_{\mathrm{wm}} - T_{\mathrm{ri}})$, and the delivered temperature rise Q/C_c collapses towards zero as C_c grows. No amount of coil substitutes for a second well.

11.5 Sizing rule for the conductance

Fixing UA_{coil} as an absolute constant would silently mis-size the coil whenever the plant is scaled — a ten-million-egg hatchery served by an exchanger sized for one million. The model therefore ties the conductance to the stream it serves:

$$UA_{\mathrm{coil}} = (ua)_{\mathrm{coil}}\,\dot m_{\mathrm{reg}}, \qquad (ua)_{\mathrm{coil}} = 3000\,\mathrm{W/K\ per\ kg/s}, \tag{71}$$

with $\dot m_{\mathrm{reg}}$ itself derived from the process air flow. The choice of 3000 is not arbitrary: it fixes the number of transfer units against the air stream at

$$\mathrm{NTU}\big|_{C_{\min}=C_c} = \frac{(ua)_{\mathrm{coil}}\,\dot m_{\mathrm{reg}}}{\dot m_{\mathrm{reg}}\,c_{p,a}} = \frac{3000}{1006} \approx 2.98, \tag{72}$$

independent of plant size — a conventional finned-coil design point, giving $\varepsilon \approx 0.95$ in the small-C_r limit. The scaling is therefore self-consistent: enlarge the plant and the coil grows with it, keeping the air-side design point fixed. What does *not* stay fixed is C_r, which is exactly the starvation mechanism of Section 11.4.

What Eq. (71) **hides.** The conductance is physically three resistances in series,

$$\frac{1}{UA_{\text{coil}}} = \frac{1}{h_w A_w} + R_{\text{wall}} + \frac{1}{\eta_o h_a A_a}, \tag{73}$$

with h_w from a Gnielinski or Dittus–Boelter correlation on the tube side, h_a from a finned-surface Colburn-j correlation on the air side, and η_o the overall fin efficiency. Because $A_a \gg A_w$ the air-side term dominates — another way of saying that the physical size and cost of the coil live on the air side. Equation (73) is what one would use to design and price an actual coil; Eq. (71) is what the system-level model needs, and the two should be reconciled before any hardware quotation is taken seriously.

12 The desiccant wheel

The wheel is the component that makes this plant a *dehumidification* plant rather than a heating plant, and it is where the geothermal heat is finally spent. A rotary desiccant wheel is a porous matrix coated with a sorbent — silica gel here — rotating slowly between two counter-flowing air streams. In the *process* sector it adsorbs moisture from the air destined for the building; in the *regeneration* sector, bathed in the hot air leaving the coil, it desorbs that moisture and is readied for the next revolution. Everything the well does upstream exists to keep that regeneration air hot enough.

12.1 Notation

Four air states matter. The third column gives the name carried in the notebook, so that report and code can be read side by side.

Station	Symbol	Notebook name	Origin
Process inlet	$(T_{\text{pi}}, w_{\text{pi}})$	`T_p, w_p`	room return + fresh makeup
Process outlet	$(T_{\text{pd}}, w_{\text{pd}})$	`T_po, w_po`	dried air, feeds the cooler
Regeneration inlet	$(T_{\text{ro}}, w_{\text{ri}})$	`T_reg, w_reg`	coil outlet (w unchanged: the coil is sensible)
Regeneration outlet	$(T_{\text{rd}}, w_{\text{rd}})$	`T_rd, w_rd`	wheel exhaust, vented

Dry-air mass flows are $\dot{m}_{\text{pro}}$ (process) and $\dot{m}_{\text{reg}}$ (regeneration), and in the code $\dot{m}_{\text{reg}} = 0.70\,\dot{m}_{\text{pro}}$ through the configuration parameter `reg_pro_ratio`. Humidity ratios w are in kg/kg of dry air, and moist-air enthalpy is taken from the same psychrometric block used everywhere else in the model,

$$h(T, w) = 1.006\,T + w\,(2501.0 + 1.86\,T) \qquad [\text{kJ/kg}], \quad T \text{ in } ^\circ\text{C}, \tag{74}$$

with the inverse $T(h, w) = (h - 2501.0\,w)/(1.006 + 1.86\,w)$.

12.2 Why a rotating matrix is hard

Inside the matrix two conservation laws hold at once and are strongly coupled. Writing z for distance along the flow, τ for time in the rotation, w and h for the air humidity ratio and enthalpy, and W and H for the *sorbent* water loading and enthalpy,

$$\text{moisture:} \qquad \frac{\partial w}{\partial z} + \frac{\partial W}{\partial \tau} = 0, \tag{75}$$

$$\text{energy:} \qquad \frac{\partial h}{\partial z} + \frac{\partial H}{\partial \tau} = 0. \tag{76}$$

The coupling runs through the sorption equilibrium. The loading $W = W(T, w)$ follows the isotherm, so it depends on *both* air properties, and the sorbent enthalpy $H = H(T, W)$ carries the heat of adsorption released when water binds to the gel. The physical loop is short and vicious: adsorbing water releases heat, the heat raises the local temperature, the higher temperature shifts the equilibrium, and the shifted equilibrium changes how much more water can be adsorbed. Temperature and humidity cannot be solved one at a time.

Equations (75)–(76) form a coupled non-linear hyperbolic system. Solving them directly means a spatially and temporally resolved simulation of the matrix, repeated inside every step of a twenty-year march that already carries a fixed-point loop around the well — unaffordable, and in any case dependent on matrix data (channel geometry, sorbent mass, isotherm) that we do not have for a specific commercial wheel.

12.3 The Banks/Close non-linear analogy

The route out is the non-linear analogy method of Close and Banks, developed for combined heat-and-mass regenerators and reduced to a usable form for silica gel by Jurinak (1982). The idea is a change of variables. One seeks two *characteristic potentials* $F_1(T, w)$ and $F_2(T, w)$ such that the coupled system becomes two *independent* wave equations, each potential being conserved along its own characteristic and travelling at its own speed γ_i:

$$\frac{\partial F_i}{\partial z} + \frac{1}{\gamma_i}\frac{\partial F_i}{\partial \tau} = 0, \qquad i = 1, 2. \tag{77}$$

What this buys is decoupling. In (F_1, F_2) coordinates the entangled heat-and-mass problem becomes two separate single-variable transfer problems, each characterised by one effectiveness exactly as a sensible heat exchanger is characterised by one ε. The cost is that the potentials are not universal: they are tied to the sorbent's isotherm, so the fits below are silica-gel fits and nothing else.

12.4 The two potentials, exactly as coded

The notebook codes Jurinak's silica-gel potentials as

$$\boxed{F_1(T, w) = -\frac{2865.0}{T^{1.490}} + 4.344\, w^{0.8624}} \tag{78}$$

$$\boxed{F_2(T, w) = \frac{T^{1.490}}{6360.0} - 1.127\, w^{0.07969}} \tag{79}$$

which are, character for character, the two lines

```
def F1(T_K, w):  return -2865.0*T_K**(-1.490) + 4.344*w**0.8624
def F2(T_K, w):  return  T_K**(1.490)/6360.0 - 1.127*w**0.07969
```

of cell 12 of the model of record.

Units warning — read this before using the formulas. Equations (78)–(79) are *dimensional* fits, not dimensionless groups. T **must** be in kelvin and w **must** be in kg/kg of dry air, never g/kg. Feeding °C or g/kg produces numbers, no warning, and a silently wrong wheel; with w in g/kg the first term of F_2 is unchanged while the second grows by a factor $1000^{0.07969} \approx 1.73$, which is enough to invert the direction of the answer. The notebook respects this by converting at the single entry point of the wheel routine, `Tp_K, Tr_K = T_p+273.15, T_reg+273.15`, and by carrying w in kg/kg everywhere in the psychrometric block. Every temperature that crosses the wheel boundary in either direction is in °C; kelvin exists only inside.

12.5 What the potentials mean physically

The two potentials have an interpretation that is also a way to test them.

Lines of constant F_1 are very nearly lines of constant *enthalpy*. F_1 is the fast potential, associated with the combined heat-and-mass wave that sweeps quickly through the matrix; it is very nearly the adiabatic-saturation invariant of the air, which is why holding F_1 fixed is almost the same as holding h fixed.

Lines of constant F_2 are very nearly lines of constant *relative humidity*. F_2 is the slow potential, associated with the sorption wave, and it is the moisture-capacity coordinate: since $\partial F_2/\partial w < 0$, drying the air *raises* F_2, and dragging the process state toward the hot, low-RH regeneration state is precisely dragging it up in F_2.

The self-check, and how the sign in F_2 was settled. Published transcriptions of Eq. (79) disagree on the sign of the second term, so the notebook resolves it numerically and re-runs the check on every execution. It evaluates F_1 along a constant-enthalpy line ($h = 85\,\mathrm{kJ/kg}$) and F_2 along a constant relative humidity line (RH = 0.40):

T [°C]	F_1 at $h = 85\,\mathrm{kJ/kg}$	F_2 at RH = 0.40 (coded, minus)	F_2 at RH = 0.40 (plus)
25	−0.41814	−0.00135	+1.53061
45	−0.41621	+0.00376	+1.68083
55	−0.41908	+0.00770	+1.75639
spread	0.00287	0.00905	0.22578

F_1 moves by 0.69% of its own magnitude across a 30 K span, so the iso-enthalpy reading holds well. For F_2 the test is comparative rather than absolute: the coded minus sign gives an iso-RH line *twenty-five times flatter* than the plus sign would (0.00905 against 0.22578), which settles the transcription without appeal to any of the conflicting sources. This check is not decoration — it is the only defence the model has against a sign error that would otherwise be invisible in the plant results.

12.6 The effectiveness method

Because the transformed problem (77) is decoupled, each potential behaves like a single-stream transfer problem and can be assigned an effectiveness. Define, exactly as ε is defined for a sensible exchanger — actual change over the maximum change available —

$$\eta_{Fi} \equiv \frac{F_i^{\mathrm{pd}} - F_i^{\mathrm{pi}}}{F_i^{\mathrm{ri}} - F_i^{\mathrm{pi}}}, \qquad i = 1, 2, \tag{80}$$

where $F_i^{\mathrm{ri}} \equiv F_i(T_{\mathrm{ro}}, w_{\mathrm{ri}})$ is the potential of the *regeneration inlet*: the maximum change available to the process stream is the full excursion from its own inlet to the state on the other side of the matrix. Rearranged, this is the working form coded in the notebook: the process state is dragged from its own inlet towards the regeneration inlet, in (F_1, F_2) space, by the prescribed fractions,

$$\boxed{F_i^{\text{pd}} = F_i(T_{\text{pi}}, w_{\text{pi}}) + \eta_{Fi}\Big[F_i(T_{\text{ro}}, w_{\text{ri}}) - F_i(T_{\text{pi}}, w_{\text{pi}})\Big], \qquad i = 1, 2.} \tag{81}$$

The geometric picture is worth holding on to: the outlet sits a prescribed fraction of the way along the segment joining the two inlets — but that segment is straight in (F_1, F_2) coordinates, not on the psychrometric chart, and it is the mapping back that makes the process path curve.

The values, and why they are asymmetric. The configuration cell prescribes

$$\eta_{F1} = 0.10, \qquad \eta_{F2} = 0.70. \tag{82}$$

The asymmetry is the design intent of a dehumidification wheel. A large η_{F2} means the wheel transfers most of the available sorption (that is, relative-humidity) potential — that is its job. A small η_{F1} means it transfers little of the enthalpy potential, because transferring enthalpy from the regeneration side would simply carry the regeneration heat into the process stream, which the downstream cooler would then have to remove again. A good wheel has η_{F2} high and η_{F1} low.

Standing flag. These two numbers are literature-typical placeholders for silica gel, and the configuration cell says so in a comment (`<- FIT to wheel data`). They are constants only at fixed rotation speed and balanced flow. Every quantitative statement in this report that depends on the wheel inherits their uncertainty, and they must be fitted to the catalogue or test data of a chosen wheel before any of it is claimed as a prediction.

12.7 Recovering the state: the 2-D Newton inversion

Equation (81) delivers the outlet *potentials*, but the rest of the plant needs the outlet *state* $(T_{\text{pd}}, w_{\text{pd}})$. Since Eqs. (78)–(79) are non-linear in both arguments and not analytically invertible, the code solves the 2×2 system

$$\mathbf{r}(T, w) = \begin{pmatrix} F_1(T, w) - F_1^{\text{pd}} \\ F_2(T, w) - F_2^{\text{pd}} \end{pmatrix} = \mathbf{0} \tag{83}$$

by Newton's method, with the Jacobian obtained by differentiating the fits directly:

$$\mathbf{J} = \begin{pmatrix} \dfrac{\partial F_1}{\partial T} & \dfrac{\partial F_1}{\partial w} \\ \dfrac{\partial F_2}{\partial T} & \dfrac{\partial F_2}{\partial w} \end{pmatrix} = \begin{pmatrix} +2865.0 \times 1.490\, T^{-2.490} & 4.344 \times 0.8624\, w^{-0.1376} \\ \dfrac{1.490\, T^{0.490}}{6360.0} & -\, 1.127 \times 0.07969\, w^{-0.92031} \end{pmatrix}. \tag{84}$$

Writing $\mathbf{J} = \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right)$ with $\det = ad - bc$, the update $\Delta\mathbf{x} = -\mathbf{J}^{-1}\mathbf{r}$ uses the closed-form 2×2 inverse $\mathbf{J}^{-1} = \frac{1}{\det}\left(\begin{smallmatrix} d & -b \\ -c & a \end{smallmatrix}\right)$, so that

$$\Delta T = -\frac{d\, r_1 - b\, r_2}{\det}, \qquad \Delta w = -\frac{-c\, r_1 + a\, r_2}{\det}, \tag{85}$$

which is line for line what `_invert_F1F2` evaluates. The Jacobian signs are the ones already discussed: $\partial F_1/\partial T > 0$ and $\partial F_2/\partial w < 0$.

The iteration starts from the process inlet itself, $(T, w) = (T_{\text{pi}}, w_{\text{pi}})$ in kelvin, which is always a good guess because the outlet is only a partial drag away from it. After each update w is floored, $w \leftarrow \max(w, 10^{-7})$, to protect the fractional powers $w^{-0.1376}$ and $w^{-0.92031}$ from a zero or negative argument. The routine converts back on exit and returns T in °C. Convergence is quadratic; the shipped self-check converges in five iterations.

What happens if it does not converge — and this matters. The loop runs a fixed `for _ in range(60)`. Its exit test, $|r_1| < 10^{-9}$ and $|r_2| < 10^{-9}$, is applied to the residual computed *before* the current update, so on success the routine performs one harmless extra step. On failure there is no branch at all: after sixty iterations the routine simply `return`s the last iterate, with no convergence flag, no exception, and no warning. A non-converged inversion is therefore indistinguishable, downstream, from a converged one. The only mode that raises rather than lies is a singular Jacobian, $\det = 0$, which produces a `ZeroDivisionError` on native floats. Adding a convergence flag that propagates into the facility diagnostics is a small change and should be made before the wheel is exercised far outside the conditions tested here.

12.8 The regeneration outlet: conservation, not correlation

The effectiveness method gives only the process outlet. The regeneration outlet comes from overall balances on the wheel — which is exactly why the process side must be closed before the regeneration side can be, and that ordering propagates outward into the structure of the whole air-loop solution.

Treat the wheel as **adiabatic** and at **cyclic steady state**: the sorbent stores nothing net over a full revolution, and nothing is lost to the surroundings. Then whatever moisture and enthalpy leave one stream must enter the other. With $\Delta w_\mathrm{p} \equiv w_\mathrm{pi} - w_\mathrm{pd}$ the moisture removed from the process air (the quantity `dW_process` returned by the routine),

$$\dot{m}_\mathrm{pro}\,\Delta w_\mathrm{p} = \dot{m}_\mathrm{reg}\,(w_\mathrm{rd} - w_\mathrm{ri}) \quad\Longrightarrow\quad \boxed{w_\mathrm{rd} = w_\mathrm{ri} + \frac{\dot{m}_\mathrm{pro}}{\dot{m}_\mathrm{reg}}\,\Delta w_\mathrm{p}} \tag{86}$$

$$\dot{m}_\mathrm{pro}\,\big(h_\mathrm{pi} - h_\mathrm{pd}\big) = \dot{m}_\mathrm{reg}\,\big(h_\mathrm{rd} - h_\mathrm{ro}\big) \quad\Longrightarrow\quad \boxed{h_\mathrm{rd} = h_\mathrm{ro} + \frac{\dot{m}_\mathrm{pro}}{\dot{m}_\mathrm{reg}}\,\big(h_\mathrm{pi} - h_\mathrm{pd}\big)} \tag{87}$$

and finally $T_\mathrm{rd} = T(h_\mathrm{rd}, w_\mathrm{rd})$ from the inverse of Eq. (74).

Reading the signs. When the wheel dries, $\Delta w_\mathrm{p} > 0$, so Eq. (86) gives $w_\mathrm{rd} > w_\mathrm{ri}$: the regeneration air picks the moisture up, which is what desorption means. At the same time η_{F1} is small but positive, so the process air gains a little enthalpy, $h_\mathrm{pd} > h_\mathrm{pi}$, and Eq. (87) then gives $h_\mathrm{rd} < h_\mathrm{ro}$: the regeneration air *cools* as it desorbs. Both directions are correct, and neither was imposed — they fall out of the balances once the process side is known.

Where conservation alone is not enough. Equations (86)–(87) carry no bound. They will return a regeneration outlet at any temperature the arithmetic demands, including states below the process inlet wet-bulb, if the flow ratio is pushed. At $\dot{m}_\mathrm{reg}/\dot{m}_\mathrm{pro} = 0.30$ with the self-check inlets, the balance returns $T_\mathrm{rd} = -3.2\,^\circ\mathrm{C}$ — arithmetically consistent and physically impossible. Nothing in the wheel routine flags this, because nothing in the wheel routine knows about transfer limits. It is a reason to keep $\dot{m}_\mathrm{reg}/\dot{m}_\mathrm{pro}$ near its configured value of 0.70 unless the outlet state is inspected.

12.9 Why the flow ratio is a lever on one side only

The ratio $\dot{m}_\mathrm{pro}/\dot{m}_\mathrm{reg}$ appears in Eqs. (86)–(87) and nowhere else. On the *regeneration* side it is a real lever: a small regeneration flow concentrates the transferred moisture and enthalpy into a small stream, giving a large rise in w_rd and a large drop in T_rd, while a large flow dilutes both. Combined with the coil's temperature-versus-throughput trade-off, $\dot{m}_\mathrm{reg}/\dot{m}_\mathrm{pro}$ is one of the more consequential design choices in the plant.

On the *process* side it is not a lever at all. Equations (81)–(85) contain no mass flow of any kind. The process outlet $(T_{\mathrm{pd}}, w_{\mathrm{pd}})$ depends only on the two inlet states and on η_{F1}, η_{F2}. Doubling both air flows, or halving the regeneration flow alone, leaves the process outlet bit-for-bit identical; only the regeneration outlet moves. Numerically, with the self-check inlets, $\dot{m}_{\mathrm{reg}}/\dot{m}_{\mathrm{pro}} \in \{0.30, 0.70, 2.00\}$ all return $T_{\mathrm{pd}} = 57.86\,°\mathrm{C}$ and $w_{\mathrm{pd}} = 12.33\,\mathrm{g/kg}$, while T_{rd} moves from -3.2 to 43.4 to 67.0 °C. Any conclusion drawn from the model about process-side flow effects is an artefact of something else in the plant, not of the wheel.

12.10 The central consequence: latent becomes sensible

Because $\eta_{F1} \ll 1$, the process path is nearly iso-enthalpic — F_1 barely moves, and constant F_1 is very nearly constant h. Along a constant-enthalpy line, Eq. (74) forces a reduction in w to be paid for in T:

$$h = \text{const} = 1.006\,T + w\,(2501 + 1.86\,T) \quad \Longrightarrow \quad \frac{\mathrm{d}T}{\mathrm{d}w} \approx -\frac{2501}{1.006 + 1.86\,w} < 0. \tag{88}$$

Numerically, removing 1 g/kg at constant enthalpy heats the air by about $2501/1.04 \approx 2.4\,\mathrm{K}$.

This is not a defect of the wheel; it is thermodynamics. The latent heat released when vapour is adsorbed does not vanish — it reappears as sensible heat in the very same air stream, plus whatever extra enthalpy the small η_{F1} drag imports from the hot regeneration side. **A desiccant wheel is a latent-to-sensible converter.** It follows that the air-to-air cooler downstream of the wheel is not an optional refinement or a polish stage: without it the wheel delivers air that is drier than the room but far hotter, and the building cannot be held at comfort. The wheel creates the sensible load that the cooler — and, in the modes that use it, the chiller — exists to reject.

12.11 The regime diagnostic: when the wheel runs backwards

A desiccant wheel dries the process stream only while the regeneration air is drier *in relative-humidity terms* than the process air. Below that crossover the drag in Eq. (81) points the wrong way, the wheel humidifies the process air, and Δw_{p} simply goes negative. Nothing breaks: the fixed-point loop still converges, the humidity controller still reports its target met, and the reversal is invisible in the plant results.

The notebook therefore instruments it. A module-level dictionary `DW_REGIME` counts every wheel evaluation and every reversed one, records the worst (most negative) Δw_{p}, and raises a single `UserWarning` on the first reversal quoting both inlet states, both relative humidities, and the RH gap that must stay positive; at the end of a run `dw_regime_report()` prints the reversed fraction and the worst excursion. This is a reporting device, not a correction: the model still returns the reversed state and the plant still runs on it. It exists so that a reversal is never silent.

12.12 Worked example: the shipped self-check

The last lines of cell 12 exercise the wheel at a Gulf-Coast-like process inlet, 32 °C at RH = 0.68, against a regeneration inlet at 80 °C carrying the same absolute humidity, at the configured flow ratio $\dot{m}_{\mathrm{reg}}/\dot{m}_{\mathrm{pro}} = 0.70$.

Stream	T [°C]	w [g/kg]	h [kJ/kg]	RH [%]	
Process inlet (pi)	32.0	20.48	84.6	68.0	room return + makeup
Process outlet (pd)	57.9	12.33	90.4	10.9	**dried** 8.15 g/kg, **heated** 25.9 K
Regen inlet (ro)	80.0	20.48	134.7	6.7	coil outlet
Regen outlet (rd)	43.4	32.11	126.5	—	**cooler and wetter**

Check the moisture balance, Eq. (86):

$$w_{\mathrm{rd}} = 20.48 + \frac{1}{0.70}\,(20.48 - 12.33) = 20.48 + 1.4286 \times 8.15 = 32.11\,\mathrm{g/kg}. \quad \checkmark$$

Two readings. The process enthalpy rise is only $90.4 - 84.6 = 5.7\,\mathrm{kJ/kg}$ out of an available $134.7 - 84.6 = 50.1\,\mathrm{kJ/kg}$ — the $\eta_{F1} = 0.10$ drag doing exactly what it was set to do. And the temperature rise of 25.9 K exceeds the pure iso-enthalpic estimate $2.4 \times 8.15 \approx 19.6\,\mathrm{K}$ by just that imported enthalpy. The regeneration stream mirrors it, leaving 36.6 K colder and 11.6 g/kg wetter than it entered.

12.13 Limitations: the formulation carries no size

This subsection is the one to read before quoting any wheel result.

There is no wheel in the model. Equations (81)–(85) contain no diameter, no depth, no desiccant mass, no channel geometry, no transfer area, no rotation speed and no NTU. The entire hardware description of the wheel is the pair of numbers $(\eta_{F1}, \eta_{F2}) = (0.10, 0.70)$. Everything the model knows about the component is compressed into those two constants, and they were prescribed, not derived. Three consequences follow, and none is subtle.

The process outlet is independent of both mass flows. Shown numerically in Section 12.9: the flows enter only the regeneration balances. In a real wheel the process outlet depends strongly on face velocity, because face velocity sets residence time and therefore the number of transfer units. The model cannot see this.

The wheel cannot be resized. There is no parameter to change that makes the wheel bigger or smaller. Scaling the plant — changing `N_eggs`, and with it $\dot{m}_{\mathrm{pro}}$ — rescales every duty in the facility but leaves the wheel's performance per kilogram of air exactly where it was. A wheel sized for the design flow and a wheel half that size are the same object inside this model.

Part-load and rotation-speed behaviour are outside the model's vocabulary. Wheel rotation speed is the primary control of a real desiccant wheel: too slow and the matrix saturates before it returns to regeneration, too fast and it carries hot regeneration air into the process stream as parasitic heat, and the optimum shifts with load. None of this exists here. Turning down the plant changes the inlet states, and the wheel responds to those, but it responds with the same two effectiveness values it uses at full load. Any statement in this report about wheel sizing, rotation speed, face-velocity selection or part-load degradation is outside what the model can represent, and should be read as an assumption or as a placeholder, never as a result.

The remedy, and why it is deferred. The analogy method itself supplies the missing link. Once the problem is decoupled into the two waves of Eq. (77), each wave admits a regenerator-style effectiveness correlation of the form $\eta_{Fi} = \eta_{Fi}(\mathrm{NTU}_i, C^*_{r,i})$, where the number of transfer units follows from the transfer coefficient and the transfer area, $\mathrm{NTU}_i \sim UA_i/(\dot{m}\,c_p)_i$, and the matrix capacity ratio follows from the desiccant mass and the rotation speed. Supplying wheel geometry, sorbent mass and rotation speed would then *produce* η_{F1} and η_{F2} instead of taking them as input, and every limitation listed above would lift at once: face-velocity dependence, resizing and part-load behaviour would all follow.

That extension is deferred. It needs geometric and sorbent data for a specific commercial wheel, which we do not have, and fitting it to a catalogue is a separate exercise from the plant-level architecture this report documents. Until it is done, η_{F1} and η_{F2} remain what the configuration cell calls them: numbers to be fitted to wheel data.

13 The scavenging air loop

The wheel of Section 12 only moves moisture; something must carry that moisture out of the building. That is the job of the scavenging air stream: it is heated by the geothermal coil, passed through the wheel's regeneration sector where it strips the sorbent, and then thrown away. The word *loop* survives from an earlier topology and is now slightly generous, as Section 13.6 explains, but the stream is still the one place where the whole plant's latent duty leaves the site.

13.1 Stations and the code names that carry them

One pass of the stream is the function `regen_air_pass`. Four states matter, and the notebook names them as follows.

Station	Symbol	Notebook name	Set by
Scavenging inlet	$(T_{\mathrm{ri}}, w_{\mathrm{ri}})$	`T_ri, w_ri`	the mixing node, downstream of the fan
Coil outlet	$(T_{\mathrm{ro}}, w_{\mathrm{ri}})$	`T_ro`	the geothermal coil (sensible: w unchanged)
Wheel exhaust	$(T_{\mathrm{rd}}, w_{\mathrm{rd}})$	`T_rd, w_rd`	the wheel regeneration balance; **vented**
Mixed state	$(T_{\mathrm{rm}}, w_{\mathrm{rm}})$	`T_rm, w_rm`	building exhaust + outdoor air

Two further states enter from outside the stream: the outdoor air $(T_{\mathrm{oa}}, w_{\mathrm{oa}})$, and the building exhaust, which is the room state $(T_{\mathrm{pc}}, w_{\mathrm{pc}})$ because the building is held at comfort and does not float.

13.2 The pass, station by station

Coil. The scavenging air enters the geothermal coil at $(T_{\mathrm{ri}}, w_{\mathrm{ri}})$ and leaves at T_{ro}, computed by the ε–NTU relation of Eq. (70) with the conductance UA_{coil} of Eq. (71). The coil is a liquid-to-air exchanger with a dry surface, so it is purely sensible:

$$T_{\mathrm{ro}} = T_{\mathrm{ri}} + \frac{Q_{\mathrm{coil}}}{\dot{m}_{\mathrm{reg}}\, c_{p,a}}, \qquad w_{\mathrm{ro}} = w_{\mathrm{ri}}. \tag{89}$$

The code never forms w_{ro} at all; it passes `w_ri` straight into the wheel, which *is* Eq. (89) written as an identity rather than an assignment.

Wheel. The regeneration sector is entered at $(T_{\mathrm{ro}}, w_{\mathrm{ri}})$ against the process stream $(T_{\mathrm{pi}}, w_{\mathrm{pi}})$; the outlets follow from Eqs. (80), (86) and (87). The wheel delivers two things at once: the dried process air $(T_{\mathrm{pd}}, w_{\mathrm{pd}})$, which the building will use, and the wet exhaust $(T_{\mathrm{rd}}, w_{\mathrm{rd}})$, which the plant must lose.

Exhaust, mixing node, fan. The wheel exhaust is vented in its entirety: in v10.6 the pair $(T_{\mathrm{rd}}, w_{\mathrm{rd}})$ is a *reported diagnostic and nothing else*, read by no downstream equation. That is the single most important structural fact about this section, and Section 13.5 explains why it was made so. A fresh scavenging inlet is then assembled from two reservoirs — building exhaust and outdoor air — and pushed through the fan.

13.3 The mixing node

Let $d_{r\mathrm{A}}$ be the fraction of the scavenging mass flow drawn from the *building exhaust*, and $d_{r\mathrm{B}} = 1 - d_{r\mathrm{A}}$ the fraction drawn from *outdoor air*. Applying the fractional mixing rule of Eq. (20) to the two conserved quantities — water and enthalpy — and then recovering the temperature with Eq. (14):

$$w_{\mathrm{rm}} = d_{r\mathrm{A}}\, w_{\mathrm{pc}} + d_{r\mathrm{B}}\, w_{\mathrm{oa}}, \tag{90}$$

$$h_{\mathrm{rm}} = d_{r\mathrm{A}}\, h(T_{\mathrm{pc}}, w_{\mathrm{pc}}) + d_{r\mathrm{B}}\, h(T_{\mathrm{oa}}, w_{\mathrm{oa}}), \tag{91}$$

$$T_{\mathrm{rm}} = \frac{h_{\mathrm{rm}} - 2501.0\, w_{\mathrm{rm}}}{1.006 + 1.86\, w_{\mathrm{rm}}}. \tag{92}$$

The fan is treated as isenthalpic, with the temperature-rise hook retained at zero so that fan heat can be added later without renaming a state:

$$T_{\mathrm{ri}} = T_{\mathrm{rm}} + \Delta T_{\mathrm{fan}}, \qquad w_{\mathrm{ri}} = w_{\mathrm{rm}}, \qquad \Delta T_{\mathrm{fan}} = 0. \tag{93}$$

Both balances must use the same damper position. Equations (90) and (91) share one pair of weights, and the code asserts it: `assert abs((d_rA_use + d_rB_use) - 1.0) < 1e-12`. The assertion looks pedantic; it is not. A mixing junction produces a state that lies *on the straight line* joining its two inlet states on the psychrometric chart, and it lies there precisely because water and enthalpy are apportioned with the same two numbers. Use one damper position in the moisture balance and a different one in the energy balance and the result is off the mixing line: a state that no mixing process can produce, because mass and energy have been created in different amounts.

An earlier version did exactly that, with weights that summed to 1.07. The error is not academic. Reconstructing it at the design point with weights (0.30, 0.77) instead of (0.30, 0.70): the mixed humidity rises from 20.91 g/kg to 22.41 g/kg and the mixed enthalpy from 86.5 to 92.6 kJ/kg, which lands the scavenging inlet about 2 K too hot and 1.5 g/kg too wet — seven per cent of an air stream conjured out of nothing, feeding a wheel whose whole output is a difference of humidities. Errors of this kind do not announce themselves: the fixed point still converges, the controller still reports success, and the plant appears slightly better than it is.

13.4 The physical cap on the exhaust fraction

The building cannot exhaust more than it admits. Its fresh-air intake is the forced ventilation $\dot{m}_{\mathrm{vent}}$ set by the eggs' oxygen requirement ($0.030\,\mathrm{m^3/h}$ per egg, Section S of the configuration), and everything else in the process stream is recirculated and never leaves the room. The exhaust available to the scavenging stream is therefore $\dot{m}_{\mathrm{vent}}$, and the largest exhaust fraction physically obtainable is

$$\boxed{d_{r\mathrm{A}}^{\mathrm{cap}} = \min\left(1,\ \frac{\dot{m}_{\mathrm{vent}}}{\dot{m}_{\mathrm{reg}}}\right)} \tag{94}$$

with $d_{r\mathrm{A}}^{\mathrm{use}} = \min(d_{r\mathrm{A}}, d_{r\mathrm{A}}^{\mathrm{cap}})$ and the balance made up with outdoor air. The code applies the cap inside `regen_air_pass` and warns once per run when the requested position exceeds it.

At the shipped configuration $\dot{m}_{\text{vent}} = 10.0\,\text{kg/s}$ and $\dot{m}_{\text{reg}} = 0.70\,\dot{m}_{\text{pro}} = 23.52\,\text{kg/s}$, so $d_{r\text{A}}^{\text{cap}} = 0.425$. The nominal damper range $[0.05, 1]$ is thus a fiction above 0.425: the scavenging stream is more than twice the building's exhaust, and no damper setting can make the wheel breathe more room air than the room produces. This has a direct consequence for the controller (Section 15.3) and it is worth stating as a design observation in its own right: *the authority of this actuator is set by the ratio of two flows that were sized for entirely unrelated reasons* — egg respiration on one side, the wheel's regeneration ratio on the other.

13.5 What changed in v10.5, and why it had to

Until v10.4 the node of Section 13.3 mixed the *wheel's own exhaust* $(T_{\text{rd}}, w_{\text{rd}})$ with outdoor makeup: a genuinely recirculating loop, in which a fraction $d_{r\text{A}}$ was vented and the rest returned to the coil. That arrangement has a closed-form steady state worth deriving, because it shows exactly how it fails.

Write the moisture the scavenging stream picks up in one pass, from Eq. (86),

$$\Delta w_{\text{dw}} \equiv \frac{\dot{m}_{\text{pro}}}{\dot{m}_{\text{reg}}}\,(w_{\text{pi}} - w_{\text{pd}}), \qquad \text{so} \qquad w_{\text{rd}} = w_{\text{ri}} + \Delta w_{\text{dw}}. \tag{95}$$

In the old topology the mixed humidity was $w_{\text{rm}} = d_{r\text{B}}\,w_{\text{rd}} + d_{r\text{A}}\,w_{\text{oa}}$, and at steady state $w_{\text{ri}} = w_{\text{rm}}$, so

$$\begin{aligned} w_{\text{ri}} &= d_{r\text{B}}\,(w_{\text{ri}} + \Delta w_{\text{dw}}) + d_{r\text{A}}\,w_{\text{oa}} \\ w_{\text{ri}}\,(1 - d_{r\text{B}}) &= d_{r\text{B}}\,\Delta w_{\text{dw}} + d_{r\text{A}}\,w_{\text{oa}}, \end{aligned}$$

and since $1 - d_{r\text{B}} = d_{r\text{A}}$,

$$\boxed{w_{\text{ri}} = w_{\text{oa}} + \frac{d_{r\text{B}}}{d_{r\text{A}}}\,\Delta w_{\text{dw}}} \tag{96}$$

The moisture the wheel had just removed was being handed back to it, amplified by the recirculation-to-vent ratio. Put the design numbers in. With $\Delta w_{\text{dw}} = (33.6/23.52) \times 2.34\,\text{g/kg}$ (the year-1 value; it relaxes to $1.85\,\text{g/kg}$ by year 20) $= 3.34\,\text{g/kg}$ and the damper at its floor $d_{r\text{A}} = 0.05$, Eq. (96) gives $w_{\text{ri}} = 85\,\text{g/kg}$ — against a saturation humidity of $83\,\text{g/kg}$ at the coil outlet temperature of $49.4\,°\text{C}$. The regeneration air arrived at the wheel supersaturated.

That is the failure, and it is a failure of the *right* variable. A desiccant wheel is not driven by humidity ratio but by the difference in relative humidity between its two faces: the F_2 potential of Eq. (79) is a relative-humidity-like coordinate, and the wheel dries the process air only while the regeneration side sits at lower RH than the process side. The old loop drove that difference to zero and beyond. The comparison at the design point is stark:

Regeneration inlet at $T_{\text{ro}} = 49.4\,°\text{C}$	w_{ri} [g/kg]	RH	Wheel
v10.6, exhaust + outdoor mixing ($d_{r\text{A}} = 0.05$)	21.3	28 %	dries
Pre-v10.5, own-exhaust recirculation ($d_{r\text{A}} = 0.05$)	84.9	102 %	reverses
Pre-v10.5, own-exhaust recirculation ($d_{r\text{A}} = 0.30$)	29.2	38 %	dries weakly

The process face sits at RH $\approx 56\,\%$, so the current topology opens a gap of some 28 percentage points, whereas the old one closed it entirely at the very damper positions the controller wanted to use. The wheel's regime diagnostic (Section 12.11) was written precisely to catch the resulting reversal.

Since v10.5 the node draws from two reservoirs that the wheel does not contaminate: room exhaust, which at comfort is drier in relative terms than outdoor air, and outdoor air itself. The wheel's own product is vented and never returns. The cost of the change is honest and should be stated: the plant now heats air it has never heated before on every pass, so the coil duty is higher than a recirculating loop would demand, and the exhaust enthalpy is thrown away entirely. What is bought is a wheel that is not fighting itself.

13.6 Does this stream still need a fixed point?

The question is worth asking explicitly, because the code still solves $(T_{\mathrm{ri}}, w_{\mathrm{ri}})$ as unknowns. Trace the dependency in the old topology: $T_{\mathrm{ri}} \to T_{\mathrm{ro}}$ (coil) $\to (T_{\mathrm{rd}}, w_{\mathrm{rd}})$ (wheel) $\to (T_{\mathrm{rm}}, w_{\mathrm{rm}})$ (mix) $\to (T_{\mathrm{ri}}, w_{\mathrm{ri}})$ (fan). The chain closed on itself, the inlet appeared on both sides, and a genuine two-variable fixed point had to be solved.

Now trace it in v10.6. Equations (90)–(93) depend on $(T_{\mathrm{pc}}, w_{\mathrm{pc}})$, on $(T_{\mathrm{oa}}, w_{\mathrm{oa}})$ and on $d_{r\mathrm{A}}$ — all prescribed — and on *nothing produced by the wheel or the coil.* **Venting the exhaust severed the feedback**, in exactly the way the building's comfort reset severs it on the process side. For a fixed $d_{r\mathrm{A}}$ the scavenging inlet is a constant.

The baseline run confirms it: over twenty annual samples in `v10.6_baseline.json`, $T_{\mathrm{ri}} = 31.299\,239\,^{\circ}\mathrm{C}$ and $w_{\mathrm{ri}} = 21.289\,\mathrm{g/kg}$ to six figures at every sample, while T_{ret}, T_{ro}, T_{pd} and UA_{cool} all drift with the depleting formation.

What survives is the *water* fixed point of Section 10: T_{wi} still closes on itself through the well, the accumulator and the coil, and the scavenging inlet enters that loop as the coil's air-side boundary condition. The driver carries $(T_{\mathrm{ri}}, w_{\mathrm{ri}})$ in the same iteration vector as T_{wi}, under-relaxed by one half,

$$x \leftarrow \tfrac{1}{2}x + \tfrac{1}{2}x^{\mathrm{new}}, \qquad x \in \{T_{\mathrm{wi}},\, T_{\mathrm{ri}},\, w_{\mathrm{ri}}\}, \tag{97}$$

so their errors merely halve each sweep towards a target that never moves. Keeping them in the vector costs almost nothing and preserves the machinery for the day the topology changes again — exhaust-air heat recovery on the scavenging stream, for instance, would restore the coupling immediately.

13.7 Verification: the mass balance

The node conserves dry air identically and by construction. The exhaust branch carries $d_{r\mathrm{A}}^{\mathrm{use}}\dot{m}_{\mathrm{reg}}$, the outdoor branch $d_{r\mathrm{B}}^{\mathrm{use}}\dot{m}_{\mathrm{reg}}$, and

$$d_{r\mathrm{A}}^{\mathrm{use}} + d_{r\mathrm{B}}^{\mathrm{use}} = 1 \quad \Longrightarrow \quad (d_{r\mathrm{A}}^{\mathrm{use}} + d_{r\mathrm{B}}^{\mathrm{use}})\,\dot{m}_{\mathrm{reg}} = \dot{m}_{\mathrm{reg}}, \tag{98}$$

which the code checks with the assertion quoted in Section 13.3. Two further balances close outside the node: the wheel exhaust leaves at $\dot{m}_{\mathrm{reg}}$, matching what the node admits, and the building exhaust drawn by the scavenging stream, $d_{r\mathrm{A}}^{\mathrm{use}}\dot{m}_{\mathrm{reg}} \le \dot{m}_{\mathrm{vent}}$, is bounded by Eq. (94). The assertion is a run-time check, not a printed report; it fires as an exception rather than as a diagnostic line.

14 The air-to-air cooler

14.1 Role and configuration

The wheel returns the process air dry but hot: adsorption is exothermic, and the latent load it removed reappears as sensible heat (Eq. (88)). At the design point the wheel delivers 42.6 °C into a room held at 37 °C. Something must take that penalty back out before the air is supplied.

That something is `air_air_cooler`: an air-to-air, dry-surface, ε–NTU exchanger. The hot side is the process stream $(T_{\text{pd}}, w_{\text{pd}})$ at $\dot{m}_{\text{pro}}$; the cold side is a scavenging stream drawn from outdoors at T_{oa} with flow $\dot{m}_{\text{cool}} = 1.5\,\dot{m}_{\text{pro}} = 50.4\,\text{kg/s}$ (`cool_pro_ratio`, Section T of the configuration).

Two things follow, and they set everything else in this section. First, it is a *recuperator*: it moves heat from one air stream to another and consumes no refrigeration, no water and no geothermal energy — only fan power. It cannot manufacture cold; it can only pass heat down a gradient it is given. Second, being dry, it is sensible only:

$$\boxed{w_{\text{ps}} = w_{\text{pd}}} \tag{99}$$

where $(T_{\text{ps}}, w_{\text{ps}})$ denotes the supply state delivered to the building. (The notebook overloads the name `T_pi`, using it both for the wheel's process inlet and for the cooler outlet; the report keeps them apart, and `T_pi_req`, `w_pi_req` in the driver are $T_{\text{ps}}^{\text{req}}$, $w_{\text{ps}}^{\text{req}}$ here.)

Equation (99) is the hinge of the whole control system: the last device before the building can change temperature and cannot change humidity. Section 15.1 draws the consequence.

14.2 Rating equations

With the moist-air specific heat of Eq. (15) on the hot side and dry-air $c_{p,a}$ on the cold side,

$$C_h = \dot{m}_{\text{pro}}\,(1.006 + 1.86\,w_{\text{pd}}) \times 1000, \qquad C_c = \dot{m}_{\text{cool}}\,c_{p,a}, \qquad C_r = \frac{C_{\min}}{C_{\max}}, \tag{100}$$

and with ε from the unmixed-crossflow relation Eq. (69), the duty and outlet are

$$Q_{\text{cool}} = \varepsilon\,C_{\min}\,(T_{\text{pd}} - T_{\text{oa}}), \qquad T_{\text{ps}} = T_{\text{pd}} - \frac{Q_{\text{cool}}}{C_h}. \tag{101}$$

At the design point $C_h = 34.9\,\text{kW/K}$ and $C_c = 50.7\,\text{kW/K}$, so $C_{\min} = C_h$ and $C_r = 0.689$: the *process* stream is the limiting one, which is why oversizing the ambient fan buys progressively less.

The code also carries a trivial branch. If $T_{\text{pd}} \le T_{\text{oa}}$ it returns $T_{\text{ps}} = T_{\text{pd}}$ with $UA = 0$ and no saturation flag: there is no gradient, so there is nothing to do and nothing to report.

14.3 Target mode: back-solving the conductance

The cooler is not rated forward. The building balance, expressed through the load interface as $T_{\text{ps}}^{\text{req}} = T_{\text{pc}} - q_s/(\dot{m}_{\text{pro}}\,c_p)$, states the supply temperature the space requires, and the model asks what conductance delivers it. Inverting Eq. (101) for the effectiveness,

$$\boxed{\varepsilon^{\text{req}} = \frac{C_h\,(T_{\text{pd}} - T_{\text{ps}}^{\text{req}})}{C_{\min}\,(T_{\text{pd}} - T_{\text{oa}})}} \tag{102}$$

The crossflow relation cannot be inverted analytically for the NTU, so the code bisects. Because $\varepsilon(\text{NTU})$ increases monotonically, bisection on $\text{NTU} \in [0,\ UA_{\max}/C_{\min}]$ is unconditionally convergent; the code takes a fixed 60 halvings — far more than needed, and cheap, since each evaluation is one exponential — and returns

$$UA^{\text{req}} = \text{NTU}^{\text{req}}\,C_{\min}. \tag{103}$$

At the end of the first baseline year this gives $\varepsilon^{\text{req}} = 0.732$, $\text{NTU} = 2.42$ and $UA_{\text{cool}} = 84.4\,\text{kW/K}$, comfortably inside the ceiling $UA_{\max} = 150\,\text{kW/K}$.

Inverting the device this way is a modelling decision with a price. It reports the conductance a real coil would need rather than the behaviour of a coil that exists, so the model never tells you what happens at part load with a *fixed* exchanger — the conductance simply follows the load. For sizing, which is what this study is for, that is the more useful answer; for operation it is not, and a fixed-UA rating mode should eventually sit beside it.

14.4 The ambient floor

A recuperator cannot cool its hot stream below the temperature of the stream it rejects to. Heat flows down the gradient, and at $\varepsilon = 1$ the outlet reaches the cold inlet and stops:

$$\boxed{T_{\text{ps}} \geq T_{\text{oa}} \text{ always}} \tag{104}$$

The code enforces this by a sentinel rather than by a test: if $T_{\text{ps}}^{\text{req}} \leq T_{\text{oa}}$ it sets $\varepsilon^{\text{req}} = 2$, a value no exchanger can reach, which routes execution to the saturated branch. Saturation therefore has two causes, and the flag `sat_cool` does not distinguish them:

1. $T_{\text{ps}}^{\text{req}} \leq T_{\text{oa}}$ — the target is below ambient and *no* dry recuperator can reach it, at any size;
2. $\varepsilon^{\text{req}} > \varepsilon_{\max}$ — the target is above ambient but needs a coil larger than the ceiling.

On either branch the block returns the best it can actually do at the ceiling,

$$T_{\text{ps}} = T_{\text{pd}} - \frac{\varepsilon_{\max}\, C_{\min}\,(T_{\text{pd}} - T_{\text{oa}})}{C_h}, \qquad UA = UA_{\max}, \tag{105}$$

rather than pretending the setpoint was met.

The design criterion. Because the supply flow is chosen from the design sensible load at a design depression ΔT_{supply}, the required supply is $T_{\text{ps}}^{\text{req}} = T_{\text{pc}} - \Delta T_{\text{supply}}$ by construction, and Eq. (104) reduces to a statement about two temperature differences:

$$\boxed{\text{a dry ambient recuperator suffices} \iff \Delta T_{\text{supply}} < T_{\text{pc}} - T_{\text{oa}}} \tag{106}$$

At the shipped configuration $T_{\text{pc}} - T_{\text{oa}} = 37 - 31 = 6\,\text{K}$ and $\Delta T_{\text{supply}} = 3\,\text{K}$, so the criterion is met with a factor of two in hand and no below-ambient stage is needed. The margin is a gift of the hatchery setpoint: a room held at 37 °C is warmer than the Gulf design day, which is what makes an air-to-air recuperator viable at all. It is also fragile — raise the ambient design temperature to 35 °C, or double the supply depression to shrink the fan, and the criterion fails, at which point a below-ambient stage (a dew-point evaporative cooler, or a chiller) becomes mandatory rather than optional.

What the baseline actually shows. `sat_cool` does fire in the opening days of a run, for a different reason: while the formation is undepleted the coil delivers its hottest regeneration air, the wheel returns $T_{\text{pd}} \approx 51\,°\text{C}$, and cooling that to 34.1 °C against 31 °C air needs more than 150 kW/K. In a one-year run the flag is raised for about the first 10 steps (2.5 days) and clears thereafter, with UA_{cool} falling from the ceiling to 98 kW/K within the first month. The binding constraint in early life is the conductance ceiling, not the ambient floor — and the early-life excess of wheel duty is the same surplus that saturates the humidity controller (Section 15.3).

14.5 Is there a condensation check? No

There is none. Equation (99) is applied unconditionally: `air_air_cooler` computes only temperatures, and the driver assigns `w_pi = ap['w_pd']` with no dew-point test anywhere in the function, in the driver, or in the dashboard. A search of the notebook for a dew-point variable returns nothing. If the block were ever driven below the stream's dew point it would report a dry outlet at an impossible state, silently.

Whether that matters is a separate question, and here the answer is reassuring. The supply carries $w_{\mathrm{pd}} = 17.97\,\mathrm{g/kg}$ at the end of the first baseline year, a dew point of $23.2\,°\mathrm{C}$, while the outlet sits at $34.1\,°\mathrm{C}$: a margin of nearly 11 K. More strongly, Eq. (104) guarantees $T_{\mathrm{ps}} \geq T_{\mathrm{oa}} = 31\,°\mathrm{C}$, so *within this topology* the cooler cannot reach the dew point of air the wheel has just dried. The omission is safe by construction, not by luck.

It stops being safe the moment the cold side stops being ambient air. Add the below-ambient stage that Eq. (106) would demand at a hotter design day, or feed the cooler with chilled water from the absorption machine, and the floor drops below the dew point; the block would then need a wet-coil treatment (bypass factor or fin-efficiency), and $w_{\mathrm{ps}} < w_{\mathrm{pd}}$ would become a second, uncontrolled dehumidification path. This should be written before any below-ambient stage is added, not after.

15 Control and the coupled solution

15.1 The principle: control where there is an actuator

The building imposes a two-component setpoint, $(T_{\mathrm{pc}}, w_{\mathrm{pc}})$, and the plant must satisfy two balances. Which of them can be *controlled* and which can only be *checked* is not a modelling preference; it is decided by where the hardware sits.

Balance	Actuator downstream of the wheel?	Treatment
Sensible	**Yes** — the cooler	Control: back-solve UA_{cool}
Latent	**No** — $w_{\mathrm{ps}} = w_{\mathrm{pd}}$, fixed at the wheel	Control from *upstream*: $d_{r\mathrm{A}}$

Between the wheel outlet and the room there is exactly one device, and by Eq. (99) it moves temperature only. Temperature is therefore controlled locally, at the last possible moment; humidity has no local knob at all and must be reached backwards, through the scavenging stream, the wheel and in the last resort the well. The hierarchy is also an economic ordering — fan power first, a damper second, the geothermal resource last — and that ordering is the operational expression of what the project is trying to conserve.

15.2 The sensible channel

Nothing is left to do here beyond Section 14.3: the load interface returns $T_{\mathrm{ps}}^{\mathrm{req}}$, the cooler is inverted for UA^{req}, the floor test is applied and `sat_cool` is set if the target is unreachable. One scalar inversion, one bisection, no outer iteration — and, importantly, it happens *after* the coupled solve, because the cooler influences nothing upstream of itself.

15.3 The latent channel

The humidity setpoint is closed on the scavenging vent fraction. The model solves

$$\text{find } d_{r\mathrm{A}} \in [d_{r\mathrm{A,min}},\, 1] \quad \text{such that} \quad w_{\mathrm{pd}}(d_{r\mathrm{A}}) = w_{\mathrm{ps}}^{\mathrm{req}}, \tag{107}$$

with $d_{r\mathrm{A,min}} = 0.05$ and a residual tolerance $1 \times 10^{-4}\,\mathrm{kg/kg}$ (`w_ctrl_tol`).

The mechanism. Opening d_{rA} replaces outdoor air with building exhaust in the scavenging stream. At the design point the room, held at 37 °C and 50 %, is drier than the outdoors both absolutely (19.9 against 21.4 g/kg) and in relative terms once heated, so more exhaust means drier regeneration air, a wider F_2 gap, and a drier process outlet: $\partial w_{\text{pd}}/\partial d_{rA} < 0$. Note carefully that this sign is a property of the *design weather*, not a theorem. On a cool dry morning the outdoor air is the drier reservoir and the sign reverses. The code assumes the design sign in its first probe step and would take one wasted iteration to recover if it were wrong.

The scheme as coded. It is a bounded secant, not the bisection its predecessor used:

1. Solve the coupled inner problem at the warm-started d_{rA}; set $e_1 = w_{\text{pd}} - w_{\text{ps}}^{\text{req}}$ and seed $(r_0, e_0) = (d_{rA}, e_1)$.
2. Take the first probe as $r_1 = \text{clip}(d_{rA} \pm 0.1,\ d_{rA,\min},\ 1)$, the sign being $+$ if the air is too wet and $-$ if it is too dry.
3. Repeat at most 12 times (`d_rA_ctrl_max_it`), stopping as soon as $|e_1| <$ `w_ctrl_tol`: solve the inner problem at r_1, form e_{new}, and take the secant step

$$r_{\text{next}} = r_1 - e_{\text{new}}\,\frac{r_1 - r_0}{e_{\text{new}} - e_0}, \tag{108}$$

   guarded by $|e_{\text{new}} - e_0| > 1 \times 10^{-12}$ (below which the step is abandoned and $r_{\text{next}} = r_1$), then shift $(r_0, e_0) \leftarrow (r_1, e_{\text{new}})$ and $r_1 \leftarrow \text{clip}(r_{\text{next}})$.
4. Adopt $d_{rA} = r_0$ — the last position actually *evaluated*, never the untested extrapolation r_1 — and re-solve the inner problem there so that every reported state is mutually consistent.

Warm-starting from the previous time step matters: the loads move slowly, so the previous solution is usually already inside tolerance.

Saturation at both ends. Two flags mark the boundaries of the actuator's authority:

$$\texttt{sat\_hum}: \quad d_{rA} \geq 1 - 1 \times 10^{-6} \ \text{ and } \ w_{\text{pd}} > w_{\text{ps}}^{\text{req}} + \texttt{w\_ctrl\_tol}, \tag{109}$$

$$\texttt{sat\_dry}: \quad d_{rA} \leq d_{rA,\min} + 1 \times 10^{-6} \ \text{ and } \ w_{\text{pd}} < w_{\text{ps}}^{\text{req}} - \texttt{w\_ctrl\_tol}. \tag{110}$$

The first is the classical failure: the damper is wide open, the air is still too humid, the setpoint is unreachable at the current regeneration temperature, and the only remaining escalation is to spend geothermal energy. The second was added in v10.6 and is the failure that actually occurs: the damper is pinned at its floor, the wheel is *over*-drying, and the controller has no authority left to stop it. Before v10.6 only `sat_hum` existed, so v10.5 ran the whole twenty years over-dry while the dashboard reported the humidity target met. A one-sided flag on a two-sided actuator is not a diagnostic; it is a blind spot.

Equation (109) carries a further weakness worth recording. The controller's variable is bounded by 1, but its *effect* is bounded by $d_{rA}^{\text{cap}} = 0.425$ (Eq. (94)). Between 0.425 and 1 the residual is flat: the secant denominator collapses below its guard, the update freezes, and the iterate only reaches exactly 1 by accident. The wet-end flag can therefore be missed. The honest statement is that the damper's authority ends at $\dot{m}_{\text{vent}}/\dot{m}_{\text{reg}}$, and the bound in Eq. (107) should be d_{rA}^{cap} rather than 1.

15.4 What is coupled to what

Sub-problem	Structure	Where it is solved
Water loop	closed on T_{wi}	inner fixed point
Scavenging inlet	feed-through since v10.5 (Sec. 13.6)	carried in the same vector
Process inlet	feed-through (comfort reset)	once, before the time loop
Latent control	scalar root find on $d_{r\mathrm{A}}$	outer loop
Sensible control	scalar inversion for UA_{cool}	after convergence
Formation, accumulator	stateful, marched	once per step, at the end

The water loop and the scavenging stream share the coil, so they are solved *simultaneously* in one iteration over $(T_{\mathrm{wi}}, T_{\mathrm{ri}}, w_{\mathrm{ri}})$, never as nested loops — a full air solve inside every water iteration would multiply the cost for no gain. Convergence is measured on

$$\max\left(|\Delta T_{\mathrm{wi}}|\,,\ |\Delta T_{\mathrm{ri}}|\,,\ 1\times10^{3}\,|\Delta w_{\mathrm{ri}}|\right) < \texttt{fp\_tol} = 1\times10^{-4}, \tag{111}$$

with at most `fp_max` = 40 sweeps. The factor 1000 puts a humidity ratio in kg/kg on the same numerical scale as a temperature in kelvin; without it the humidity residual would be met automatically and would contribute nothing to the test.

Why the time loop lives in the driver. Every block exposes `init` / `set_flow` / `eval` / `advance` and none of them owns a clock. A feed-forward chain could be run block by block over the whole horizon, but feedback and memory forbid it: the water loop's reinjection temperature is an output of the same iteration that consumes it, and the formation and the accumulator carry state that depends on every earlier step. Both must be marched together, by something that sees all of them — the driver. The discipline that follows is strict and is respected in the code: `dbhe_advance` and `acc_advance` are called *once* per step, with the converged state, after the controller has finished. Advancing inside the iteration would commit history that the iteration was still free to revise.

15.5 Observed convergence

Instrumenting the baseline case, one well evaluation per inner sweep:

Quantity	Value
Well evaluations, first step (cold start)	67
Well evaluations, subsequent steps	24–25
Inner solves per step (probe + outer + final)	14
Inner sweeps per solve, warm-started	1–2
Recorded `fp_it` (final consistent solve)	1

The inner fixed point is well behaved: warm-started from the previous step, the residual of Eq. (111) is usually already met on the first sweep, which is why the recorded `fp_it` is 1 throughout the baseline. That number should be read for what it is — the cost of the *final* solve, not the cost of the step.

The outer loop is the opposite story, and it is the diagnostic that matters. At the shipped configuration the controller is pinned at $d_{r\mathrm{A,min}}$ with a residual of several g/kg, so the stopping test $|e_1| <$ `w_ctrl_tol` is never met, the clipped secant returns the same floor value every time, and the loop runs its full 12 iterations at every one of the 29 220 time steps. Roughly 12 of the 24 well evaluations per step are spent proving, over and over, that a saturated actuator is still saturated. A cheap fix exists — break out when two successive clipped iterates coincide — and it would halve the run time without changing a single result.

15.6 Open item: the humidity actuator is the wrong one

Planned for the next version: move humidity control from $d_{r\mathrm{A}}$ to the coil-flow fraction.

The baseline run pins $d_{r\mathrm{A}}$ at its floor of 0.05 at every step and raises `sat_dry` throughout: the wheel removes 2.34 g/kg in year 1 and still 1.85 g/kg in year 20, while the building asks for 0.27 g/kg, and in the opening days it removes nearly 5 g/kg. The controller is saturated at its dry end from the first step to the last, which means the plant is not being controlled at all — it is being run flat out and reported after the fact.

The diagnosis is that $d_{r\mathrm{A}}$ is too weak an actuator for this duty. It changes the *quality* of the regeneration air, and by Eq. (94) it cannot change it by much, whereas the surplus to be removed is a large fraction of the wheel's whole output. Meanwhile a large share of the geothermal duty — the coil heat that produces T_{ro} — is spent removing moisture the building does not want, and the sensible penalty of that unwanted drying then has to be taken back out by the cooler, at the cost of a bigger UA_{cool}.

The next version therefore moves the latent control to the *coil-flow fraction*: throttling the geothermal water sent to the coil lowers T_{ro} directly and therefore lowers the wheel's drying authority continuously, from full duty down to none. It is a strong, monotone actuator that acts on the resource itself, so under-drying and over-drying both become reachable, and the geothermal energy not spent is geothermal energy saved — which, unlike a damper position, shows up as slower formation depletion. When that change is made, $d_{r\mathrm{A}}$ should be retained as a secondary trim and Eq. (107) rebounded on $d_{r\mathrm{A}}^{\mathrm{cap}}$.

16 The demand side

Everything upstream of this section — the well, the water circuit, the coil, the wheel, the cooler — exists to deliver one air stream to one space. This section defines the only thing the plant is allowed to know about that space.

The model's demand side is deliberately *generic*. The building is not a hatchery, a data hall or an office as far as the plant is concerned: it is a pair of numbers per time step, a sensible load and a moisture load, plus one closure that says how much air is available to carry them. A hatchery is one way of producing that pair; Section 17 works it out in full, but it is an *instance*, not the definition. If a second application ever replaces it, nothing in Sections 1–5 of this report changes.

16.1 Two balances, three unknowns, one closure

Hypotheses, stated before the algebra. The space is treated as a single steady, well-mixed control volume at constant pressure p_{atm}, served by one supply stream of dry-air mass flow $\dot{m}_{\mathrm{sup}}$ and returning at the room condition. "Well mixed" means the extract state *is* the room state $(T_{\mathrm{pc}}, w_{\mathrm{pc}})$: no stratification, no local gradients, no distribution effectiveness. "Steady" means the room's thermal and moisture capacitance is ignored, so the balance has no time derivative and each time step is independent of the last.

Dry-air conservation. With no infiltration *through the supply path*, the dry-air flow leaving equals the flow entering, so a single $\dot{m}_{\rm sup}$ appears in both balances below. This is why humidity ratio (per kilogram of *dry* air) is the right moisture variable: it makes the water balance linear in a conserved flow.

Water balance. Let $\dot{m}_w$ be the net rate at which water vapour is released into the space by everything that is not the supply air — occupants, process, infiltration, evaporating product. Steady state requires that the supply air carry it away:

$$\dot{m}_{\rm sup}\, w_{\rm pi} + \dot{m}_w = \dot{m}_{\rm sup}\, w_{\rm pc} \qquad \Longleftrightarrow \qquad \boxed{\dot{m}_w = \dot{m}_{\rm sup}\,(w_{\rm pc} - w_{\rm pi})} \tag{112}$$

Energy balance. Likewise, with q_s the net sensible gain of the space and c_p the moist-air specific heat per kilogram of dry air,

$$\dot{m}_{\rm sup}\, c_p\, T_{\rm pi} + q_s = \dot{m}_{\rm sup}\, c_p\, T_{\rm pc} \qquad \Longleftrightarrow \qquad \boxed{q_s = \dot{m}_{\rm sup}\, c_p\,(T_{\rm pc} - T_{\rm pi})} \tag{113}$$

The code evaluates c_p at the *room* humidity by default, $c_p = (1.006 + 1.86\, w_{\rm pc}) \times 10^3$ J/(kg K), and lets the caller override it. Evaluating at the room rather than the supply state is a choice of linearisation point; at the design case of Section 17.7 the two differ by less than a part in 10^3.

Sign convention — fix it once and never revisit it. Both loads are *positive when the plant must remove something*:

Quantity	Positive means	Consequence for the supply state
$q_s > 0$	cooling required	$T_{\rm pi} < T_{\rm pc}$: supply colder than the room
$\dot{m}_w > 0$	dehumidification required	$w_{\rm pi} < w_{\rm pc}$: supply drier than the room

Negative values are perfectly legal and mean heating and humidification: the same two equations then return a supply state *above* the room in temperature or humidity. Nothing in the interface branches on the sign, which is what makes the plant capable of running in all four quadrants without a mode switch.

The counting problem. Equations (112) and (113) are two equations in *three* unknowns: $\dot{m}_{\rm sup}$, $T_{\rm pi}$ and $w_{\rm pi}$. The loads are data; the supply is not determined. Exactly one further statement — a *closure* — is required, no more and no less. The code enforces this literally: supplying both closures, or neither, raises an error rather than silently preferring one.

The two admissible closures.

Constant volume. Prescribe the flow $\dot{m}_{\rm sup}$. The supply temperature and humidity then follow from Eqs. (112)–(113) and float with the load. This is the fixed-fan machine: the air handler moves a fixed mass and the conditioning does all the work.

Variable volume. Prescribe the supply depression $\Delta T_{\rm sup} = T_{\rm pc} - T_{\rm pi}$. The flow then follows the sensible load, $\dot{m}_{\rm sup} = \max(|q_s|/(c_p|\Delta T_{\rm sup}|),\, \dot{m}_{\rm min})$, and the supply temperature is held. This is the VAV machine: the coil holds a discharge setpoint and the fan tracks the load.

Two details of the variable-volume branch are worth reading off the code. First, the absolute values: the depression is treated as a *magnitude*, so a heating load ($q_s < 0$) returns a positive flow and, through Eq. (113), a supply *above* room temperature. Second, the floor $\dot{m}_{\rm min}$: a minimum flow (typically a mandatory fresh-air rate) can be imposed, and when it binds the interface returns a `vent_limited` flag rather than quietly delivering less air than the application requires.

The room condition line. Divide Eq. (112) by Eq. (113). The flow — the one thing the closure supplies — cancels identically:

$$\frac{w_{\text{pc}} - w_{\text{pi}}}{T_{\text{pc}} - T_{\text{pi}}} = \frac{c_p\, \dot{m}_w}{q_s} \;\equiv\; S. \tag{114}$$

This is the classical *room condition line*: on a psychrometric chart, the locus of admissible supply states is a straight line through the room point $(T_{\text{pc}}, w_{\text{pc}})$ whose slope S is set by the load *ratio* alone.

> **What is fixed by what.** The load ratio fixes the *direction* in which the supply state must lie from the room point. The closure fixes *how far along that line* it sits — a large flow puts the supply near the room, a small flow pushes it far away. Direction is physics; distance is a design choice. Every disagreement about "the required supply air" is one of these two and it is worth naming which.

Why the sensible heat ratio cannot classify the load. It is tempting to compress the load pair into the single number reported by the interface,

$$\text{SHR} = \frac{q_s}{q_s + q_l}, \qquad q_l = \dot{m}_w\, h_{\text{fg}}, \tag{115}$$

and to read the operating quadrant off it. **This does not work, and the failure is not subtle.** The ratio is invariant under a simultaneous sign flip of both loads: $q_s = +100$, $q_l = +25$ and $q_s = -100$, $q_l = -25$ both give SHR $= 0.8$. Two *negative* loads — a space that needs heating and humidification, the deep-winter quadrant — return a perfectly respectable positive SHR indistinguishable from summer cooling. A ratio destroys exactly the information (the signs) that selects the quadrant. The interface therefore classifies on the signs of q_s and $\dot{m}_w$ *separately* and reports SHR as a diagnostic only.

What the steady, well-mixed hypothesis costs. Three things. There is no thermal mass, so a load spike is passed instantly to the plant instead of being buffered by the structure — the model cannot exploit pre-cooling or night flush, and it over-states peak duty on a short excursion. There is no stratification, so a tall space that could be conditioned only in the occupied layer is charged for its whole volume. And there is no room control dynamics: the space is *assumed* to be held at $(T_{\text{pc}}, \text{RH}_{\text{pc}})$, so the model answers "what must be supplied" and never "what would the room drift to if the plant fell short". The saturation flags elsewhere in the model exist precisely because that second question has been assumed away here.

16.2 Why the latent load is a moisture rate and not a power

The interface carries the latent load as $\dot{m}_w$ in kg/s, not as a power in kW. This looks like a units preference. It is not; it removes a silent error.

The double conversion. A load generator that measures evaporation — which is what physically happens — knows a *rate*. To report kilowatts it must multiply by some latent heat, $q_l = \dot{m}_w h_{\text{fg}}^{(1)}$. The interface, needing a rate for Eq. (112), must then divide by some latent heat, $\dot{m}_w^* = q_l / h_{\text{fg}}^{(2)}$. The physical quantity therefore passes through h_{fg} *twice*, and only if the two values agree exactly does the round trip return what was sent:

$$\dot{m}_w^* = \dot{m}_w\, \frac{h_{\text{fg}}^{(1)}}{h_{\text{fg}}^{(2)}}. \tag{116}$$

The error is not hypothetical. Reasonable values of h_{fg} near room temperature span roughly 2430 kJ/kg to 2500 kJ/kg depending on whether one quotes it at 25 °C, at 0 °C, or as the enthalpy datum embedded in the moist-air enthalpy correlation. The present notebook itself carries two: the load reference $h_{\mathrm{fg}} = 2.5\,\mathrm{MJ/kg}$ used by the interface and every load generator, and the datum 2501 kJ/kg inside the moist-air enthalpy function (and inside the legacy `building_loads` helper). They differ by 1 kJ/kg here, harmlessly — but nothing in a kilowatt-valued interface would have detected it had they differed by fifty.

Size of the damage. Propagating Eq. (116) through Eq. (112), the induced error falls entirely on the required supply *depression*:

$$\frac{\delta(w_{\mathrm{pc}} - w_{\mathrm{pi}})}{w_{\mathrm{pc}} - w_{\mathrm{pi}}} = \frac{\delta h_{\mathrm{fg}}}{h_{\mathrm{fg}}} \approx \frac{50}{2500} = 2\,\%. \tag{117}$$

Two per cent of a humidity difference is far below the resolution of any humidity sensor and far below the scatter of the wheel model that must deliver it. It will never be caught by inspection of a result, and it will never be caught by a unit check either, because kilowatts are kilowatts. **An error that is both systematic and invisible is worse than a large one.** The cure is structural, not numerical: convert once, at the boundary, and keep the conserved quantity — mass — in the interior.

How the code implements it. A single module-level constant `LOAD_HFG` is the only conversion the demand side may use. The generic generator `prescribed_loads` exists precisely to serve callers who genuinely have kilowatts: it accepts q_l in kW together with an *optional explicit* h_{fg}, converts there and then, and hands a rate onwards. The kilowatt figure the interface reports back, $q_l = \dot{m}_w\,$`LOAD_HFG`, is a derived display quantity and is never used to compute anything.

16.3 The guards

The two balances are linear and always return *a* solution. The question is whether that solution is a state that air can be in. Two failures used to pass silently and are now trapped.

Guard 1: negative required supply humidity. Equation (112) gives $w_{\mathrm{pi}} = w_{\mathrm{pc}} - \dot{m}_w/\dot{m}_{\mathrm{sup}}$, which goes negative when

$$\dot{m}_{\mathrm{sup}} < \frac{\dot{m}_w}{w_{\mathrm{pc}}}. \tag{118}$$

Physically: *even bone-dry supply air, at that flow, cannot carry the moisture away.* The space would have to be served air with negative water content. Equation (118) is therefore an absolute lower bound on the supply flow imposed by the latent load alone, independent of any equipment, and it is worth evaluating before sizing anything: no desiccant, however good, buys a way around it.

Guard 2: supersaturated required supply state. Even a positive w_{pi} can be unreachable. The supply temperature is fixed independently by Eq. (113), so the pair $(T_{\mathrm{pi}}, w_{\mathrm{pi}})$ may fall *below the saturation line.* The interface inverts the humidity ratio to a vapour pressure and tests it against the saturation pressure at the required supply temperature:

$$\mathrm{RH}_{\mathrm{pi}} = \frac{1}{p_{\mathrm{sat}}(T_{\mathrm{pi}})} \cdot \frac{w_{\mathrm{pi}}\, p_{\mathrm{atm}}}{0.622 + w_{\mathrm{pi}}} > 1 \quad \Longrightarrow \quad \text{no component can deliver it.} \tag{119}$$

This is not an equipment shortfall, it is an inconsistency between the load pair and the closure: the requested state does not exist. The usual cause is a small supply flow with a large sensible load and a modest latent load — the supply is driven very cold while still being asked to stay humid. The remedies are to raise the flow (moving the supply state back toward the room along the line of Eq. (114)) or to accept reheat, which means deliberately leaving the room condition line.

Reading the two guards together. Geometrically they fence the room condition line at both ends. The admissible supply state must lie on the line of Eq. (114), above $w = 0$ and below the saturation curve $w = w_{\mathrm{sat}}(T)$. The closure slides the state along that line; the guards say when it has slid off the end of the physically available segment.

A deliberate asymmetry in severity. Malformed *inputs* raise exceptions and stop the run: both closures given or neither, a zero depression, a non-positive flow. Physically impossible *outputs* raise warnings and let the run continue. That is a considered choice: a design sweep that crashes on its first infeasible corner tells you nothing about the shape of the feasible region, whereas one that flags and continues maps it. The cost is that the flags must actually be read — a warning nobody looks at is worse than an exception.

17 A worked demand case: the poultry hatchery

The hatchery is now one implementation of the generator contract of Section 16: a function of ambient state returning the pair $(q_s, \dot{m}_w)$. It is retained because it is the project's motivating application and because it exercises the interface hard — an internal heat source, an internal moisture source, a room *warmer* than the Gulf ambient, and a mandatory ventilation rate that the occupants (embryos) set and the designer cannot negotiate.

17.1 Internal terms

Embryonic metabolic heat. Developing embryos respire and the heat release rises steeply toward hatch. Written per egg,

$$Q_{\mathrm{eggs}} = N_{\mathrm{eggs}}\, q_{\mathrm{egg}}, \qquad q_{\mathrm{egg}} = 0.13\,\mathrm{W} \text{ per egg}, \tag{120}$$

a *sensible source*: it flows into the space. Note how this inverts the usual comfort-cooling intuition, in which the envelope is the load and the interior is passive.

Egg moisture loss. A setter deliberately drives controlled moisture loss from the eggs — of order 12 % of egg weight over incubation, and essential to hatchability. That water evaporates into the room air:

$$\dot{m}_{\mathrm{evap}} = N_{\mathrm{eggs}}\, r_{\mathrm{evap}}, \qquad r_{\mathrm{evap}} = 4.6 \times 10^{-9}\,\mathrm{kg/s} \text{ per egg } (\approx 0.40\,\mathrm{g} \text{ per egg per day}). \tag{121}$$

The application therefore *wants* the eggs to lose water, which is why the room humidity cannot be allowed to drift up: too humid means insufficient weight loss and a poor hatch. The humidity setpoint here is a product specification, not a comfort preference.

17.2 Envelope terms in conductance form

Sensible: two genuine conductances. Both external sensible paths are linear in a temperature difference, so both can be written as a conductance times a driving potential:

$$Q_{\text{sens,ext}} = \underbrace{H_{\text{tr}}}_{\text{material}} (T_{\text{sol}} - T_{\text{pc}}) + \underbrace{\dot{m}_{\text{inf}}\, c_{p,\text{a}}}_{\text{air mass}} (T_{\text{oa}} - T_{\text{pc}}) , \qquad H_{\text{tr}} = \sum_i U_i A_i. \tag{122}$$

H_{tr} [W/K] is a property of materials and geometry; $\dot{m}_{\text{inf}} c_{p,\text{a}}$ [W/K] is a property of the airtightness. The infiltration here is *uncontrolled leakage* and is distinct from the deliberate ventilation, which is handled in the process loop (Section 17.4).

Sol-air temperature. For the opaque envelope the driving outdoor temperature is not the air temperature: absorbed solar radiation raises the surface above it. The standard device folds this into an equivalent outdoor temperature,

$$T_{\text{sol}} = T_{\text{oa}} + \frac{\alpha_s I_{\text{sol}}}{h_o} - \frac{\varepsilon\,\Delta R}{h_o} \quad \longrightarrow \quad T_{\text{sol}} = T_{\text{oa}} + \Delta T_{\text{solar}}, \qquad \Delta T_{\text{solar}} = 4\,\text{K}, \tag{123}$$

with α_s the solar absorptance, I_{sol} the incident irradiance, h_o the outside film coefficient and the last term the long-wave sky correction. The model currently uses the constant offset on the right; under hourly weather forcing it becomes the full expression on the left. For a single-storey hatchery, whose roof is most of its envelope, this term is not cosmetic.

17.3 The latent asymmetry, which is the physically central point

The external latent load has *one* term and no counterpart to H_{tr}:

$$\dot{m}_{w,\text{ext}} = \dot{m}_{\text{inf}} \left(w_{\text{oa}} - w_{\text{pc}} \right) . \tag{124}$$

The asymmetry, stated plainly. A wall conducts heat but not vapour. There is no material vapour conductance analogous to a U-value: vapour diffusion through permeable construction exists but is second order and is neglected here. Hence the *sensible* external load has two knobs — insulation (H_{tr}) and tightness ($\dot{m}_{\text{inf}}$) — while the *latent* external load has only the second.

In a humid climate, insulation does nothing for humidity. Airflow control is the entire latent game. When a reference speaks of a "latent conductance" it means the air-mass term $\dot{m}\, h_{\text{fg}}$, never a material property.

The sign of $w_{\text{oa}} - w_{\text{pc}}$ therefore decides whether the outside air is a latent load or a latent resource. For the setter setpoint (37 °C, RH = 0.50) the room humidity ratio is $w_{\text{pc}} = 19.86$ g/kg, whose vapour pressure corresponds to a dew point of 24.9 °C: outdoor air is a dehumidification load exactly when its dew point exceeds that. At the design ambient (31 °C, RH = 0.75) it does — $w_{\text{oa}} = 21.36$ g/kg — but only by 1.5 g/kg. Gulf summer dew points straddle the criterion, which is why the computed latent demand is small and sign-changeable rather than overwhelming.

17.4 The assembled demand, and where the ventilation went

Collecting Eqs. (120)–(124) gives the hatchery generator exactly as coded:

$$q_s = N_{\text{eggs}} q_{\text{egg}} + H_{\text{tr}} \left(T_{\text{sol}} - T_{\text{pc}} \right) + \dot{m}_{\text{inf}} c_{p,\text{a}} \left(T_{\text{oa}} - T_{\text{pc}} \right) , \tag{125}$$

$$\dot{m}_w = N_{\text{eggs}} r_{\text{evap}} + \dot{m}_{\text{inf}} \left(w_{\text{oa}} - w_{\text{pc}} \right) . \tag{126}$$

(The code computes the latent term as a power using `LOAD_HFG` and the generator immediately divides it back out, so Eq. (126) is what the interface actually receives — an instance of the round trip discussed in Section 16.2, here exact because both conversions use the same constant.)

The accounting point that must be right. The mandatory fresh-air ventilation does *not* appear in Eqs. (125)–(126), and this is not an omission. In this architecture the fresh air enters the *process loop* at damper $d_{p\mathrm{A}}$ and is conditioned by the wheel and the cooler *before* it reaches the space. Its load is borne by the equipment and enters the plant through the process-inlet mixing state, not through the space balance. Counting it here as well would be double counting. Equations (125)–(126) are the loads on the *space alone*: eggs, envelope, infiltration.

Where it does appear. Combining the process-inlet mixing with the required supply humidity, the moisture the wheel must actually remove is

$$\dot{m}_{\mathrm{H_2O}}^{\mathrm{wheel}} = \dot{m}_{\mathrm{pro}} \left(w_{\mathrm{pi}}^{\mathrm{mix}} - w_{\mathrm{pi}}^{\mathrm{req}} \right) = \underbrace{\dot{m}_{\mathrm{vent}} \left(w_{\mathrm{oa}} - w_{\mathrm{pc}} \right)}_{\text{forced ventilation}} + \underbrace{\dot{m}_{\mathrm{evap}} + \dot{m}_{\mathrm{inf}} \left(w_{\mathrm{oa}} - w_{\mathrm{pc}} \right)}_{\text{space sources}}, \tag{127}$$

using the identity $\dot{m}_{\mathrm{pro}}\, d_{p\mathrm{A}} = \dot{m}_{\mathrm{vent}}$ guaranteed by the sizing of Section 17.6. At the design point the three terms are 15.0 g/s, 4.6 g/s and 4.4 g/s: a total wheel duty of 24.0 g/s against a *space* latent load of only 9.0 g/s. **Nearly two thirds of the dehumidification duty is imported by the ventilation air the eggs require.** The plant is sized by a flow that cannot be reduced multiplied by a humidity gap set by the weather.

17.5 Sizing: everything follows from N_{eggs}

Prescribing loads, flows and geometry independently allows them to describe different-sized buildings at the same time — an inconsistency that occurred during development. The cure is a single declared capacity from which everything descends.

Internal loads, linear in capacity. Equations (120) and (121).

Geometry, from packing density and a fixed ceiling height.

$$A_{\mathrm{floor}} = \frac{N_{\mathrm{eggs}}}{n_{\mathrm{egg/m^2}}}, \qquad V_{\mathrm{room}} = A_{\mathrm{floor}} H_{\mathrm{room}}, \qquad A_{\mathrm{walls}} = 4\sqrt{A_{\mathrm{floor}}}\, H_{\mathrm{room}}, \tag{128}$$

the last assuming a square footprint, so the perimeter is $4\sqrt{A_{\mathrm{floor}}}$.

Envelope, infiltration and forced ventilation.

$$H_{\mathrm{tr}} = U_{\mathrm{env}} \left(2A_{\mathrm{floor}} + A_{\mathrm{walls}} \right), \qquad \dot{m}_{\mathrm{inf}} = \frac{\rho_{\mathrm{da}} V_{\mathrm{room}}\, \mathrm{ACH_{inf}}}{3600}, \qquad \dot{m}_{\mathrm{vent}} = \frac{\rho_{\mathrm{da}} N_{\mathrm{eggs}} v_{\mathrm{egg}}}{3600}, \tag{129}$$

the factor 2 counting roof and floor.

Scaling laws. How does the plant scale with capacity? Not as the naive "surface $\propto$ volume$^{2/3}$", because **the ceiling height is fixed**: the building spreads sideways rather than growing isotropically. From Eqs. (128)–(129), everything is linear in N_{eggs} except the walls, which go as $\sqrt{N_{\text{eggs}}}$:

$$H_{\text{tr}} = \underbrace{\frac{2U_{\text{env}}}{n_{\text{egg/m}^2}} N_{\text{eggs}}}_{\text{roof+floor, } \propto N_{\text{eggs}}} + \underbrace{4U_{\text{env}} H_{\text{room}} \sqrt{\frac{N_{\text{eggs}}}{n_{\text{egg/m}^2}}}}_{\text{walls, } \propto \sqrt{N_{\text{eggs}}}} . \tag{130}$$

Numerically, from 150 000 to 10^6 eggs (a factor 6.67), H_{tr} rises from 942 W/K to 5495 W/K, a factor 5.84: an effective exponent of $N_{\text{eggs}}^{0.93}$, nearly but not exactly linear.

> **Consequence: the load *balance* is scale-invariant.** Since essentially every term scales with N_{eggs}, so do q_s, $\dot m_w$ and therefore $\dot m_{\text{pro}}$ — and the *ratios* (sensible-to-latent, and $d_{p\text{A}} = \dot m_{\text{vent}}/\dot m_{\text{pro}}$) do not move. Making the hatchery bigger does not change whether it is latent- or sensible-dominated; that is set by the setpoint and the climate. What does *not* scale is the **well**: one borehole delivers what it delivers, so raising N_{eggs} drives the plant into the coil's large-flow limit. The binding constraint is the well-to-demand ratio, not the building.

17.6 Airflow causality — which way the arrow points

The wrong way round. An earlier version prescribed the fresh-air damper and derived the supply flow from it, $\dot m_{\text{pro}} = \dot m_{\text{vent}}/d_{p\text{A}}$. That makes the fan flow a $1/d_{p\text{A}}$ hyperbola: at $d_{p\text{A}} = 0.1$ it returns 100 kg/s of supply air. This is an artifact of inverted causality, not a physical result. A damper position is not a sizing input.

The right way round. The physical logic runs in four steps.

1. The **ventilation is forced by the application**: embryos consume oxygen and produce carbon dioxide, so $\dot m_{\text{vent}}$ (Eq. (129)) is the one flow in the plant that is not a design choice.
2. The **supply flow follows the design sensible load** carried at a prescribed depression — the variable-volume closure of Section 16.1, applied once at design time:

$$\dot m_{\text{pro}}^{\text{load}} = \frac{q_s^{\text{design}}}{c_{p,\text{a}}\,\Delta T_{\text{sup}}^{\text{design}}}, \qquad \Delta T_{\text{sup}}^{\text{design}} = 3\,\text{K}. \tag{131}$$

3. The **ventilation is a floor**, and the damper is *derived*:

$$\boxed{\dot m_{\text{pro}} = \max\Big(\dot m_{\text{pro}}^{\text{load}},\, \dot m_{\text{vent}}\Big), \qquad d_{p\text{A}} = \min\left(\frac{\dot m_{\text{vent}}}{\dot m_{\text{pro}}},\, 1\right).} \tag{132}$$

   The damper opens exactly far enough to admit the required fresh air and no further.
4. If $\dot m_{\text{pro}}^{\text{load}} \le \dot m_{\text{vent}}$ the ventilation dominates: $d_{p\text{A}} \to 1$, recirculation vanishes and the plant runs 100 % once-through. The model raises a `vent_dominated` flag; this is the same condition the interface reports as `vent_limited`.

On the depression. ΔT_{sup} is the standard HVAC sizing handle: small depression means a large fan and gentle, uniform conditions; large depression means a small fan and a cold jet. **For a setter it must be small** — eggs cannot be chilled unevenly — so hatcheries run high recirculation at small temperature difference. The remaining hardware then follows the streams it serves: $\dot{m}_{\text{reg}} = 0.70\,\dot{m}_{\text{pro}}$, $\dot{m}_{\text{cool}} = 1.5\,\dot{m}_{\text{pro}}$ and $UA_{\text{coil}} = 3000\,\text{W/K}$ per kg/s of $\dot{m}_{\text{reg}}$. Freezing those as constants while scaling the plant would be the same class of error as the hyperbola above.

17.7 The chain evaluated at the baseline

Table 8 runs the whole descent at the notebook's baseline: $N_{\text{eggs}} = 10^6$, room ($37\,°\text{C},\ \text{RH} = 0.50$), ambient ($31\,°\text{C},\ \text{RH} = 0.75$).

Table 8: The sizing chain and the interface solve at the baseline configuration. Every input is read from the configuration cell; every output is the value the notebook prints.

Step	Expression	Value
Geometry and sizing		
Egg sensible heat	$10^6 \times 0.13\,\text{W}$	130.0 kW
Egg moisture	$10^6 \times 4.6 \times 10^{-9}\,\text{kg/s}$	4.60 g/s
Floor area	$10^6/200$	$5000\,\text{m}^2$
Room volume	5000×3.5	$17\,500\,\text{m}^3$
Wall area	$4\sqrt{5000} \times 3.5$	$989.9\,\text{m}^2$
Transmission	$0.5\,(2 \times 5000 + 989.9)$	5495 W/K
Infiltration	$1.2 \times 17500 \times 0.5/3600$	2.917 kg/s
Forced ventilation	$1.2 \times (10^6 \times 0.030)/3600$	10.00 kg/s
Design sensible load, Eq. (125), $T_{\text{sol}} = 35\,°\text{C}$		
eggs		+130.0 kW
envelope	$5495 \times (35 - 37)$	−11.0 kW
infiltration	$2.917 \times 1006 \times (31 - 37)$	−17.6 kW
total		+101.4 kW
Design latent load, Eq. (126), $w_{\text{oa}} - w_{\text{pc}} = +1.50\,\text{g/kg}$		
eggs		+4.60 g/s
infiltration	$2.917 \times 1.50 \times 10^{-3}$	+4.38 g/s
total		+8.98 g/s (= 22.5 kW)
Airflow, Eqs. (131)–(132)		
Load-driven flow	$101405/(1006 \times 3.0)$	33.60 kg/s
$\dot{m}_{\text{pro}}$	$\max(33.60,\ 10.00)$	33.60 kg/s (load wins)
$d_{p\text{A}}$	$10.00/33.60$	0.298 (derived)
$\dot{m}_{\text{reg}},\ \dot{m}_{\text{cool}}$	$0.70\,\dot{m}_{\text{pro}},\ 1.5\,\dot{m}_{\text{pro}}$	23.52, 50.40 kg/s
UA_{coil}	3000×23.52	70.6 kW/K
Interface solve, constant-volume closure $\dot{m}_{\text{sup}} = \dot{m}_{\text{pro}}$, $c_p = 1042.9\,\text{J/(kg K)}$		
T_{pi}	$37 - 101405/(33.60 \times 1042.9)$	34.11 °C
w_{pi}	$19.86 - 8.98/33.60$	19.59 g/kg
$\Delta T_{\text{sup}},\ \Delta w_{\text{sup}}$		2.89 K, 0.267 g/kg
RH_{pi}, SHR		0.58, 0.82

Three things to read off the table. First, **the eggs are more than the whole load**: at 31 °C ambient the room at 37 °C is warmer than outside, so the envelope and the infiltration are both *negative* — they help — and the 130 kW of metabolic heat is 128 % of the 101.4 kW net. The building is an internally driven load in a mild-by-comparison climate, and q_{egg} is by a wide margin the highest-leverage placeholder in the model.

Second, **the achieved depression is not the design depression**: 2.89 K against 3.0 K. The flow is sized with the dry-air $c_{p,\text{a}} = 1006\,\text{J/(kg K)}$ while the interface solves with the moist-air value 1042.9 J/(kg K) at w_{pc}. The 3.7 % discrepancy is harmless but real, and it is exactly the kind of double-definition the moisture-rate argument of Section 16.2 was written to avoid.

Third, **the required conditioning is tiny in humidity terms**: 0.267 g/kg of depression. The room condition line is very steep and the plant's difficulty lies almost entirely in the ventilation air it must dry on the way in (Eq. (127)), not in the space itself.

17.8 The interface contract: what a second demand case must supply

The point of the preceding section is that none of it was privileged. To attach any other application to this plant, the following — and only the following — must be provided.

1. **A load generator.** A function of the ambient state (and of whatever internal state the application has) returning the pair $(q_s, \dot{m}_w)$: sensible load in W, moisture load in kg/s, both positive when the plant must remove. If the application only knows kilowatts, it must convert once, explicitly, at its own boundary.
2. **A room setpoint** $(T_{\text{pc}}, w_{\text{pc}})$, with w_{pc} obtained from the setpoint relative humidity through the psychrometric relations, because both balances are written about the room state.
3. **Exactly one closure**: a supply flow (constant volume) or a supply depression (variable volume), optionally with a minimum flow $\dot{m}_{\text{min}}$ representing a mandatory ventilation rate. Not both, not neither.
4. **No double counting.** Any stream the plant conditions upstream of the space — ventilation air admitted at $d_{p\text{A}}$ being the case in point — must be excluded from the generator. The generator returns loads on the *space*, not on the plant.
5. **Attention to the flags.** `vent_limited`, and the two guards of Section 16.3, are warnings by design; a caller that ignores them will receive a supply state that no equipment can produce and will not be told twice.

One honest caveat. The interface is generic at *run* time, but the plant's *sizing* constants — $\dot{m}_{\text{pro}}$, $\dot{m}_{\text{reg}}$, $\dot{m}_{\text{cool}}$, UA_{coil} — still descend from the hatchery chain of Section 17.5. A second application inherits the interface for free; to inherit the hardware it must also declare its own design sensible load and design depression and re-run Eqs. (131)–(132). Generalising the demand side was the first half of that job; generalising the sizing chain is the second, and it is not yet done.

18 Specification of the components not yet implemented

> **Read this before any equation below.** *Nothing in this section is implemented in the current model version.* The four components specified here — adiabatic humidifier, indirect evaporative cooler, single-effect LiBr–water absorption chiller, wet cooling tower — exist in the master architecture and in the mode table, but no line of code evaluates them. Everywhere else in this report the equations were read off the notebook and the numbers harvested from it, so the text cannot disagree with the model. Here the

direction is reversed: this is the *design document*, written so that implementing it is transcription rather than invention. Consequently every numerical value quoted below is a *typical literature value*, offered to make the equations concrete and to give the implementer a sanity range. It is not a code constant, it has been harvested from nothing, and it must be checked against a real machine or a real correlation before it sizes hardware. Each value is flagged where it appears.

Three of the four are cheap: one algebraic block each. The fourth — the chiller, which cannot be implemented without the tower and the chilled-water loop — is the expensive one, and it is also the one that decides whether the well can serve the cooling corner at all. Section 18.5 closes with the implementation order and with the one architectural defect this exercise has exposed. Throughout, air states are carried as (T, w) pairs and enthalpies use the notebook's own correlation, $h = 1.006\,T + w(2501 + 1.86\,T)$ in kJ/kg of dry air, so the new blocks compose with the existing ones without a units seam. Each block returns a dictionary of outlet states and duty, matching the pattern of `regen_coil` and `desiccant_wheel`.

A prerequisite that is not a component. Section 8.7 recorded that the notebook contains *no* wet-bulb routine and no dew-point routine, because nothing in the present solution path needs one. That ceases to be true the moment evaporative equipment enters. The implementation must therefore first add two scalar solvers,

$$h(T_{\mathrm{wb}},\, w_s(T_{\mathrm{wb}})) - h(T, w) = (w_s(T_{\mathrm{wb}}) - w)\, c_{p,\mathrm{w}}\, T_{\mathrm{wb}}, \qquad p_{\mathrm{sat}}(T_{\mathrm{dp}}) = \frac{w\, p_{\mathrm{atm}}}{0.622 + w}, \tag{133}$$

with $w_s(T) = 0.622\, p_{\mathrm{sat}}(T)/(p_{\mathrm{atm}} - p_{\mathrm{sat}}(T))$ and $c_{p,\mathrm{w}} = 4.186\,\mathrm{kJ/(kg\,K)}$. The left is the adiabatic-saturation (thermodynamic wet-bulb) definition; the right inverts the Magnus correlation (9) in closed form. Both are monotone and safe for bisection. The makeup term on the right of the first shifts T_{wb} by at most 0.1 K to 0.2 K at the humidities of interest, which is why "constant wet bulb" and "constant enthalpy" are used interchangeably below.

18.1 Adiabatic humidifier (unlocks M4)

Status: specification only. Not implemented in the current model version.

18.1.1 Hypothesis and equations

Air passes over a wetted surface — rigid cellulose media, or an atomising spray — fed with recirculated water that has equilibrated at the air's own wet bulb. No heat crosses the casing and no work is done, so the energy to evaporate the water can come from one place only: the sensible heat of the air itself. The process is adiabatic saturation, and on the chart it walks the air *up the constant wet-bulb line* toward saturation.

With inlet $(T_{\mathrm{hi}}, w_{\mathrm{hi}})$ and its wet bulb $T_{\mathrm{hi}}^{\mathrm{wb}}$ from (133), the device is characterised by one number, the *saturation effectiveness* — the fraction of the available wet-bulb depression the media actually delivers:

$$\boxed{\varepsilon_{\mathrm{sat}} \equiv \frac{T_{\mathrm{hi}} - T_{\mathrm{ho}}}{T_{\mathrm{hi}} - T_{\mathrm{hi}}^{\mathrm{wb}}}} \qquad \Longrightarrow \qquad T_{\mathrm{ho}} = T_{\mathrm{hi}} - \varepsilon_{\mathrm{sat}} \left(T_{\mathrm{hi}} - T_{\mathrm{hi}}^{\mathrm{wb}}\right). \tag{134}$$

The outlet humidity ratio is *not* a second free parameter: it is fixed by the adiabatic constraint $h_{\mathrm{ho}} = h_{\mathrm{hi}}$, inverted with the notebook's enthalpy correlation,

$$w_{\mathrm{ho}} = \frac{h_{\mathrm{hi}} - 1.006\,T_{\mathrm{ho}}}{2501 + 1.86\,T_{\mathrm{ho}}}, \qquad h_{\mathrm{hi}} = 1.006\,T_{\mathrm{hi}} + w_{\mathrm{hi}}(2501 + 1.86\,T_{\mathrm{hi}}). \tag{135}$$

The water consumed is the moisture actually added,

$$\dot{m}_{\text{w,hum}} = \dot{m}_{\text{pro}} \left(w_{\text{ho}} - w_{\text{hi}} \right), \tag{136}$$

plus a sump bleed of 10 % to 25 % of (136) for recirculating media (a *typical literature figure*, not a code constant), and zero for a once-through atomiser. The block returns $\{T_{\text{ho}}, w_{\text{ho}}, \dot{m}_{\text{w,hum}}, \varepsilon_{\text{sat}}\}$ and, being adiabatic, no duty. Assert $w_{\text{ho}} \leq w_s(T_{\text{ho}})$ on exit: with $\varepsilon_{\text{sat}} < 1$ it cannot fail, and if it ever does the enthalpy inversion was called with an inconsistent state.

Typical values. Rigid cellulose 200 mm to 300 mm deep at 2.5 m/s face velocity gives $\varepsilon_{\text{sat}} = 0.85$ to 0.95; atomising and ultrasonic units run 0.70 to 0.90. Use 0.90 as a placeholder. *Literature ranges quoted for concreteness; not harvested from the model, because the model does not contain this component.*

18.1.2 What it means physically, and why M4 needs no well heat

Equation (135) says the enthalpy the air gains as vapour it loses as temperature. As a duty balance across the device,

$$q_s^{\text{del}} = \dot{m}_{\text{pro}}\, c_{p,\text{a}} \left(T_{\text{hi}} - T_{\text{ho}} \right), \qquad q_l^{\text{del}} = -\, \dot{m}_{\text{pro}}\, h_{\text{fg}} \left(w_{\text{ho}} - w_{\text{hi}} \right), \qquad q_s^{\text{del}} + q_l^{\text{del}} \approx 0. \tag{137}$$

This is the whole point of mode M4. *Adiabatic saturation is direct evaporative cooling*: the humidifier is not a component that humidifies while something else cools, it is one device discharging both requirements of the cooling-plus-humidification quadrant simultaneously, and it does so by moving heat *within* the air stream rather than importing any. No geothermal water is drawn, no wheel turns, no chiller runs. That is why the mode table records "water required: none" for M4 — the one mode in the architecture that is free of the reservoir entirely, and therefore the one mode that survives unchanged for the whole plant life.

The constraint that comes with the gift. Equation (137) is also a restriction. Because the process is pinned to a constant-enthalpy line, the ratio of sensible to latent delivery is not a design variable: it is whatever the psychrometrics of that line happen to be. The device has *one* degree of freedom — ε_{sat}, or a bypass damper — and one degree of freedom can satisfy q_s or q_l, not both. Modulation slides the outlet along the same ray from the inlet state; it does not rotate that ray. M4 therefore serves the sub-region of its quadrant whose load ratio lies near that ray, and loads far off it fall through to M5, where a chilled-water coil supplies the missing sensible capacity and the humidifier trims the latent. That handoff is where the ordering defect of Section 18.5 lives.

18.2 Indirect evaporative cooler and the M-cycle (unlocks M2)

Status: specification only. Not implemented in the current model version.

18.2.1 Hypothesis

Two streams are separated by an impermeable plate. The *primary* (product) stream flows on the dry side and is cooled sensibly; the *secondary* (working) stream flows on the wet side, evaporates water into itself, and is vented. Because the plate is impermeable the product gains no moisture,

$$w_{1\text{o}} = w_{1\text{i}}, \tag{138}$$

which is the property that makes this, and not a direct evaporative cooler, the correct partner for the desiccant wheel. The wet side is taken as saturated at exit.

18.2.2 Effectiveness, in two flavours

In a *conventional* IEC the working air enters the wet channel at the inlet state, so its wet bulb is the inlet wet bulb and that is the coldest sink available:

$$\boxed{\varepsilon_{\mathrm{wb}} \equiv \frac{T_{1\mathrm{i}} - T_{1\mathrm{o}}}{T_{1\mathrm{i}} - T_{1\mathrm{i}}^{\mathrm{wb}}}} \qquad T_{1\mathrm{o}} = T_{1\mathrm{i}} - \varepsilon_{\mathrm{wb}} \left(T_{1\mathrm{i}} - T_{1\mathrm{i}}^{\mathrm{wb}}\right). \tag{139}$$

In the *M-cycle* (Maisotsenko) regenerative arrangement the working air is first pre-cooled sensibly along a dry channel and only then turned through perforations into the adjacent wet channel. Each successive element of working air therefore enters the wet side already colder, with a depressed wet bulb, and the asymptotic sink is not the inlet wet bulb but the inlet *dew point*. The effectiveness must then be referred to the dew-point depression:

$$\boxed{\varepsilon_{\mathrm{dp}} \equiv \frac{T_{1\mathrm{i}} - T_{1\mathrm{o}}}{T_{1\mathrm{i}} - T_{1\mathrm{i}}^{\mathrm{dp}}}} \qquad T_{1\mathrm{o}} = T_{1\mathrm{i}} - \varepsilon_{\mathrm{dp}} \left(T_{1\mathrm{i}} - T_{1\mathrm{i}}^{\mathrm{dp}}\right). \tag{140}$$

The two are not interchangeable, and confusing them is the classic error in this literature: since $T^{\mathrm{dp}} < T^{\mathrm{wb}} < T$, a machine with $\varepsilon_{\mathrm{dp}} = 0.6$ can exhibit an *apparent* $\varepsilon_{\mathrm{wb}}$ above unity. That violates nothing; it says only that the wet bulb was never the correct reference. The block should accept both and record which was used.

18.2.3 Outlets, duty and water

Let $\dot{m}_{\mathrm{in}}$ be the air delivered, split by the *working-air fraction* ϕ:

$$\dot{m}_{\mathrm{sec}} = \phi\, \dot{m}_{\mathrm{in}}, \qquad \dot{m}_{\mathrm{pri}} = (1 - \phi)\, \dot{m}_{\mathrm{in}}. \tag{141}$$

The primary outlet is (138) with (139) or (140), and the duty removed from the product stream is

$$Q_{\mathrm{IEC}} = \dot{m}_{\mathrm{pri}}\, c_{p,\mathrm{ma}} \left(T_{1\mathrm{i}} - T_{1\mathrm{o}}\right), \qquad c_{p,\mathrm{ma}} = 1.006 + 1.86\, w_{1\mathrm{i}}. \tag{142}$$

(The existing coil block uses the dry-air value `CP_A` in the same place; at $w \approx 10\,\mathrm{g/kg}$ the difference is under $2\,\%$. Whichever is chosen, choose it once and use it in both.) The secondary stream carries that heat away and leaves saturated; its state follows from an energy balance closed against the saturation line — one monotone root-find in $T_{2\mathrm{o}}$:

$$\dot{m}_{\mathrm{sec}} \left[h(T_{2\mathrm{o}}, w_s(T_{2\mathrm{o}})) - h_{2\mathrm{i}}\right] = Q_{\mathrm{IEC}} + \dot{m}_{\mathrm{w,IEC}}\, c_{p,\mathrm{w}}\, T_{\mathrm{sump}}, \qquad w_{2\mathrm{o}} = w_s(T_{2\mathrm{o}}), \tag{143}$$

with water consumption

$$\dot{m}_{\mathrm{w,IEC}} = \dot{m}_{\mathrm{sec}} \left(w_{2\mathrm{o}} - w_{2\mathrm{i}}\right). \tag{144}$$

Dropping the makeup-enthalpy term costs a few tenths of a kelvin in $T_{2\mathrm{o}}$ and makes (143) explicit; retaining it costs one iteration and is cleaner. The block returns both outlet states, the duty Q_{IEC}, the water rate $\dot{m}_{\mathrm{w,IEC}}$ and — this is the one easy to forget — the *reduced* product flow $\dot{m}_{\mathrm{pri}}$, which the downstream mass balance must honour.

Typical values. Conventional IEC $\varepsilon_{\mathrm{wb}} = 0.55$ to 0.75; M-cycle $\varepsilon_{\mathrm{dp}} = 0.55$ to 0.75, i.e. an apparent wet-bulb effectiveness of roughly 1.0 to 1.2; working-air fraction $\phi = 0.30$ to 0.40. *Literature ranges, not code constants.*

18.2.4 Why the M-cycle is the indicated partner for the wheel

The wheel leaves the process air *hot and dry* — the central consequence recorded in Section 12.10: latent load has been converted into sensible load. Three things follow, and together they select the M-cycle.

1. **The moisture margin must be protected.** After the wheel w already sits below the supply target, and regeneration heat was spent to put it there. A direct evaporative cooler hands that margin straight back as vapour; the IEC cools at constant w by construction, Eq. (138), and spends none of it.
2. **Dry air has a deep dew point.** The depression $T_{1\mathrm{i}} - T_{1\mathrm{i}}^{\mathrm{dp}}$ that (140) multiplies grows rapidly as w falls. The wheel does not merely leave a load for the M-cycle to remove; it *enlarges the M-cycle's driving potential* in the act of creating that load. The two are complementary, not merely sequential.
3. **Sub-wet-bulb is what makes the target reachable.** A conventional IEC bottoms out at T^{wb}, which for post-wheel air may still sit above the supply setpoint. The M-cycle's floor is T^{dp}, several kelvin lower, and that margin is what turns M2 from marginal into comfortable.

The cost of ϕ. The working-air fraction is the design lever and it cuts both ways. Raising ϕ adds wet-channel surface and enthalpy sink per unit of product, so both effectivenesses rise and $T_{1\mathrm{o}}$ falls — but the product flow $(1-\phi)\dot{m}_{\mathrm{in}}$ falls with it, and the delivered capacity $Q_{\mathrm{IEC}} \propto (1-\phi)\,\varepsilon(\phi)$ is a rising factor times a falling one. It has an interior maximum, typically near $\phi \approx 0.3$ to 0.4, so the design question is not "how cold can the product get" but "at what ϕ does the product *duty* peak". This is the coil's temperature-versus-throughput tension (Section 11) in a different currency.

18.3 Single-effect LiBr–water absorption chiller (unlocks M3 and M5)

Status: specification only. Not implemented in the current model version, and it cannot be implemented alone — the chilled-water loop and the tower of Section 18.4 belong to the same step.

18.3.1 Hypothesis: a characteristic equation, not a cycle model

A full cycle model carries solution concentrations, a solution heat exchanger and crystallisation margins, and needs property data for aqueous lithium bromide. None of that is warranted here, because the plant-level question is only how much cooling the machine delivers from a given hot-water temperature at a given rejection temperature. The *characteristic equation* method answers exactly that with three external temperatures and two regression constants.

Its basis is the Dühring diagram: for a given pair, the boiling temperature of the solution is very nearly linear in the refrigerant saturation temperature at the same pressure. Following that line through generator, condenser, absorber and evaporator collapses the cycle into a single driving potential, the *characteristic temperature difference*

$$\boxed{\Delta\Delta T = T_{\mathrm{gen}} - A\,T_{\mathrm{ac}} + (A-1)\,T_{\mathrm{evap}}} \tag{145}$$

with T_{gen} the generator driving temperature, T_{ac} the common condenser/absorber (rejection) temperature, T_{evap} the evaporator temperature and A the Dühring slope, $A \approx 1.15$ for single-effect LiBr–water — a *typical literature value*, not a code constant. Because $A > 1$, a kelvin lost at the rejection side costs more than a kelvin gained at the generator side; Section 18.4 turns that into an architectural argument. The evaporator duty is linear in the potential:

$$\boxed{Q_{\mathrm{evap}} = a\,\Delta\Delta T + b} \qquad (a > 0,\ b < 0). \tag{146}$$

The constants a and b belong to one machine, obtained by regressing manufacturer part-load data, and they absorb every internal approach the model does not resolve. Their signs carry meaning: a is the machine's gain, and the negative intercept b encodes the internal losses that must be overcome before any refrigerant is boiled, so (146) already contains its own shut-off,

$$\Delta\Delta T_{\min} = -\,b/a, \qquad Q_{\text{evap}} \equiv 0 \;\; \text{for } \Delta\Delta T \le \Delta\Delta T_{\min}. \tag{147}$$

For a nominal 100 kW single-effect machine, $a \approx 3.5\,\text{kW/K}$ and $b \approx -30\,\text{kW}$ reproduce the name-plate at $\Delta\Delta T \approx 37\,\text{K}$. *Illustrative literature-scale figures for the reader's arithmetic — not code constants and not a product selection.*

18.3.2 Generator side and the water-temperature drop

Define the thermal COP and take the generator duty from it:

$$\text{COP} = \frac{Q_{\text{evap}}}{Q_{\text{gen}}}, \qquad Q_{\text{gen}} = \frac{Q_{\text{evap}}}{\text{COP}}. \tag{148}$$

Single-effect machines run COP = 0.65 to 0.75; 0.70 is the usual placeholder and the value behind the resource tables of the topology report. *Literature value.* The geothermal stream feeding the generator then drops by

$$\Delta T_{\text{gen,w}} = \frac{Q_{\text{gen}}}{\dot m_{\text{gen}}\, c_{p,\text{w}}}, \qquad T_{\text{w,gen,out}} = T_{\text{w,gen,in}} - \Delta T_{\text{gen,w}}, \tag{149}$$

and $T_{\text{w,gen,out}}$ returns to the hydraulic mixing node exactly as the coil's T_{wh} does. Design drops are 8 K to 10 K with a maximum near 18 K — *literature values*, and the ones behind the flow-squeeze table of the topology report.

One consistency point not to be skipped: T_{gen} in (145) is an *external* temperature, because a and b were regressed against external temperatures. Use the mean of the generator water inlet and outlet,

$$T_{\text{gen}} = \tfrac{1}{2}\left(T_{\text{w,gen,in}} + T_{\text{w,gen,out}}\right), \tag{150}$$

and do *not* add a separate internal approach — that approach already lives in b. Since (150) depends on Q_{gen}, which depends on Q_{evap}, which depends on (145), the generator side is itself a small fixed point converging in two or three passes.

18.3.3 Heat rejected

Steady-state balance on the machine, solution-pump work neglected (well under 1 % of Q_{gen}):

$$Q_{\text{rej}} = Q_{\text{gen}} + Q_{\text{evap}} = Q_{\text{evap}}\left(1 + \frac{1}{\text{COP}}\right) \approx 2.4\, Q_{\text{evap}} \;\; \text{at COP} = 0.70. \tag{151}$$

This is the number that sizes the tower, and it is worth pausing on: an absorption machine rejects roughly *two and a half times* its useful cooling, against about 1.25 for a vapour-compression chiller. The tower is correspondingly large, and so is its water consumption. That is the price of being driven by heat the well supplies for nothing.

18.3.4 The two limits, and why this component binds the architecture

Equation (146) is smooth and will return a duty at any temperature handed to it. The physical machine will not. Two bounds must be imposed *outside* the characteristic equation, because neither is contained in it:

Lower bound, $T_{\mathrm{gen}} \gtrsim 78\,^{\circ}\mathrm{C}$. Below this the generator cannot raise the solution above its boiling point at condenser pressure and no refrigerant vapour is produced. The machine does not degrade gracefully; it stops. Implement as a hard gate returning $Q_{\mathrm{evap}} = 0$ with an explicit warning, in the spirit of the wheel's regime diagnostic (Section 12.11) — a silent zero is worse than none.

Upper bound, $T_{\mathrm{gen}} \lesssim 110\,^{\circ}\mathrm{C}$. Above this the concentration required at the generator exit approaches the crystallisation line, and the corrosion rate of carbon steel in hot concentrated LiBr rises steeply. The penalty is not lost performance but a blocked machine and a maintenance event. Implement as a clamp with a warning, never as an extrapolation.

Both thresholds are *typical literature values* for hot-water-fired single-effect machines; different manufacturers quote 75 °C to 80 °C and 110 °C to 120 °C. They are not code constants, and the implementation should carry them as named configuration parameters so a real data sheet can replace them in one place.

The architectural consequence, stated plainly. Of every component in the master architecture, *this one alone requires the high-temperature tier.* The desiccant wheel regenerates at roughly 66 °C; the winter heating duty needs only the supply-air temperature; the humidifier and the IEC need no geothermal water at all. Only the absorption generator asks for 78 °C or better, and only modes M3 and M5 contain it. As the reservoir depletes and the return temperature drifts down, the plant therefore does not lose capability uniformly: it crosses the 78 °C gate first, and at that moment M3 and M5 switch off while every other mode continues untouched. *The serviceable region of the load plane shrinks from the cooling corner first.* That is the strongest system-level result the architecture yields, and it is entirely a consequence of the lower bound above — which is why the bound must be a declared parameter and not a magic number buried in a branch.

18.4 Wet cooling tower (serves the chiller)

Status: specification only. Not implemented in the current model version.

18.4.1 Hypothesis: Merkel, and why the potential is enthalpy

Water falls through fill against an upward air stream and part of it evaporates. Heat and mass transfer occur together, so a temperature difference is not the correct driving potential — a tower can cool water *below* the ambient dry bulb, which no temperature-driven exchanger can do. Merkel's simplification resolves this: assume the interface air is saturated at the local bulk water temperature and that the Lewis number is unity, and the coupled problem collapses to one transfer equation driven by an *enthalpy* difference. The tower then admits the same ε–NTU treatment as the regeneration coil, with enthalpy in place of temperature.

18.4.2 Equations

Define the saturated-air enthalpy function, built from the notebook's own p_{sat} and enthalpy correlations,

$$h_s(T) \equiv h(T, w_s(T)) = 1.006\,T + \frac{0.622\,p_{\mathrm{sat}}(T)}{p_{\mathrm{atm}} - p_{\mathrm{sat}}(T)}\,(2501 + 1.86\,T)\,, \tag{152}$$

and its secant slope over the operating range, which plays the role of a specific heat on the water side:

$$C_s = \frac{h_s(T_{\mathrm{t,i}}) - h_s(T_{\mathrm{t,o}})}{T_{\mathrm{t,i}} - T_{\mathrm{t,o}}}. \tag{153}$$

The water stream's *equivalent* capacity rate — its capacity expressed in enthalpy units so it can be compared with the air's — is $\dot{m}_{\mathrm{t}} c_{p,\mathrm{w}}/C_s$, while the air's is simply $\dot{m}_{\mathrm{a,t}}$, the potential already being an enthalpy. Hence

$$C_r = \frac{\dot{m}_{\mathrm{a,t}}\,C_s}{\dot{m}_{\mathrm{t}}\,c_{p,\mathrm{w}}}, \qquad \mathrm{NTU_t} = \frac{UA_{\mathrm{t}}}{\dot{m}_{\mathrm{a,t}}}, \tag{154}$$

with UA_{t} a mass-transfer conductance carrying the units of a mass flow. The air-side effectiveness uses the counterflow relation already coded for the coil,

$$\varepsilon_{\mathrm{t}} = \frac{1 - \exp[-\mathrm{NTU_t}(1 - C_r)]}{1 - C_r \exp[-\mathrm{NTU_t}(1 - C_r)]}, \qquad \varepsilon_{\mathrm{t}} = \frac{\mathrm{NTU_t}}{1 + \mathrm{NTU_t}} \ \text{ if } C_r = 1, \tag{155}$$

and the duty is the effectiveness times the maximum enthalpy potential — the gap between saturation at the hot water inlet and the actual inlet air:

$$Q_{\mathrm{t}} = \varepsilon_{\mathrm{t}}\,\dot{m}_{\mathrm{a,t}}\,[h_s(T_{\mathrm{t,i}}) - h_{\mathrm{a,i}}]\,. \tag{156}$$

Since $h_{\mathrm{a,i}} = h_s(T^{\mathrm{wb}}_{\mathrm{a,i}})$ to within the makeup term, the potential in (156) is measured from the *ambient wet bulb* — the analytical statement of the fact that a tower chases the wet bulb and not the dry bulb. Outlet temperature, range and approach follow:

$$T_{\mathrm{t,o}} = T_{\mathrm{t,i}} - \frac{Q_{\mathrm{t}}}{\dot{m}_{\mathrm{t}} c_{p,\mathrm{w}}}, \qquad \mathrm{range} = T_{\mathrm{t,i}} - T_{\mathrm{t,o}}, \qquad \mathrm{approach} = T_{\mathrm{t,o}} - T^{\mathrm{wb}}_{\mathrm{a,i}} > 0. \tag{157}$$

Equations (153)–(157) are implicit in $T_{\mathrm{t,o}}$ through C_s, so the block iterates internally; two or three passes from $T_{\mathrm{t,o}} = T^{\mathrm{wb}}_{\mathrm{a,i}} + 4\,\mathrm{K}$ suffice.

Water leaves the tower three ways — evaporation, from the air-side moisture pickup with the outlet air taken as saturated at its own enthalpy (the usual Merkel closure); blowdown, which removes dissolved solids; and drift, which is mechanical carryover:

$$\dot{m}_{\mathrm{evap}} = \dot{m}_{\mathrm{a,t}}\,(w_{\mathrm{a,o}} - w_{\mathrm{a,i}})\,, \qquad \dot{m}_{\mathrm{blow}} = \frac{\dot{m}_{\mathrm{evap}}}{N_c - 1}, \qquad \dot{m}_{\mathrm{makeup}} = \dot{m}_{\mathrm{evap}} + \dot{m}_{\mathrm{blow}} + \dot{m}_{\mathrm{drift}}. \tag{158}$$

A useful implementation check: $\dot{m}_{\mathrm{evap}}$ should come out near 1 % of the circulating flow for every 6 K of range, which follows from $c_{p,\mathrm{w}}/h_{\mathrm{fg}}$ and the fact that roughly three-quarters of a wet tower's duty is latent.

Typical values. Design approach 3 K to 5 K; design range 5 K to 8 K; $\mathrm{NTU_t} = 1.5$ to 2.5; cycles of concentration $N_c = 3$ to 5; drift 0.005 % to 0.02 % of circulating flow. *All literature ranges for mechanical-draught counterflow towers; none is a code constant.*

18.4.3 Why the tower cannot be solved after the chiller

Here is the statement this subsection exists to make. The tower's approach sets $T_{\mathrm{t,o}}$; the chiller rejects into that water, so its condenser/absorber temperature is

$$T_{\mathrm{ac}} = T_{\mathrm{t,o}} + \delta_{\mathrm{ac}}, \tag{159}$$

with δ_{ac} the rejection-side approach inside the machine (3 K to 6 K, *literature*). But T_{ac} is exactly the term the Dühring slope multiplies in (145), so

$$\frac{\partial Q_{\mathrm{evap}}}{\partial T_{\mathrm{t,o}}} = -\,a\,A \approx -4\,\mathrm{kW/K} \qquad \text{against} \qquad \frac{\partial Q_{\mathrm{evap}}}{\partial T_{\mathrm{gen}}} = a \approx 3.5\,\mathrm{kW/K} \tag{160}$$

at the illustrative constants above. *A kelvin surrendered at the cooling tower costs more evaporator duty than a kelvin gained at the well returns.* And the loop closes: Q_{evap} sets Q_{rej} by (151), which sets the tower duty, which sets $T_{\mathrm{t,o}}$, which sets T_{ac}, which sets Q_{evap}. Solving the tower after the chiller — fixing T_{ac} at a nominal number and sizing the tower to whatever falls out — is not a mild approximation. It discards the dominant feedback and reports capacity the plant does not have, on exactly the humid design days when the wet bulb is high and the error is largest.

The two must therefore be closed simultaneously. The cheapest correct implementation is a one-dimensional residual in $T_{\mathrm{t,o}}$,

$$R(T_{\mathrm{t,o}}) = T_{\mathrm{t,o}} - \left[T_{\mathrm{t,i}} - \frac{Q_{\mathrm{t}}}{\dot m_{\mathrm{t}} c_{p,\mathrm{w}}}\right] = 0, \qquad T_{\mathrm{t,i}} = T_{\mathrm{t,o}} + \frac{Q_{\mathrm{rej}}(T_{\mathrm{t,o}})}{\dot m_{\mathrm{t}} c_{p,\mathrm{w}}}, \tag{161}$$

driven by bisection on the physically bounded interval $T_{\mathrm{a,i}}^{\mathrm{wb}} < T_{\mathrm{t,o}} < T_{\mathrm{t,i}}$. It is monotone, it cannot leave the interval, and it costs perhaps a dozen evaluations of two algebraic blocks — negligible beside the DBHE march.

18.5 Implementation order, and one unresolved defect

Status: none of the following is implemented.

Table 9 gives the sequence, ordered by unlocked capability per unit of effort. It is not arbitrary: item 0 is a prerequisite for 1, 2 and 4, and items 3–5 cannot be separated, because Section 18.4 showed that chiller and tower share a fixed point.

Table 9: Implementation order for the pending components. "Unlocks" refers to the modes of the master architecture. Nothing in this table is implemented.

#	Item	Unlocks	Note
0	Wet-bulb and dew-point solvers	—	Eq. (133); not a component
1	Adiabatic humidifier	M4 (and M8)	One block; needs no geothermal water
2	Indirect evaporative cooler	M2	One block plus a saturated-outlet solve
3	Chilled-water loop	—	Hydraulic; prerequisite for 4
4	Absorption chiller	M3, M5	Must be solved with 5
5	Wet cooling tower	M3, M5	Must be solved with 4

Item 1 is the highest-value single change in the backlog: a dozen lines, no new loop, and it opens the one operating mode that is independent of the reservoir and therefore immune to depletion. Items 4 and 5 are the largest change, and they open the two modes that are the *first* to be lost to depletion. That asymmetry is worth keeping in view when the work is scheduled.

18.5.1 The unresolved defect: process ordering in M5

Mode M5 activates the absorption chiller with a dry chilled-water coil *and* the humidifier, to serve deep cooling-plus-humidification loads that M4 cannot reach. In the architecture as drawn, the humidifier sits on prong E, *upstream* of the chilled-water coil in fork 2. The air is therefore humidified before it is cooled. This is the wrong order, for three reasons:

1. **It may saturate.** Adiabatic saturation drives the air toward RH = 1 along a constant-enthalpy line. Presenting a nearly saturated stream to a cold coil face guarantees condensation on the fins.
2. **It undoes itself.** Whatever condenses on the coil is moisture the humidifier has just evaporated. The plant pays evaporative duty to add water and chiller duty to take the same water back out — latent work performed twice for a net result of zero.
3. **It wastes the free cooling.** The humidifier's own sensible cooling, Eq. (137), is delivered upstream of the coil, where the coil simply sees a colder inlet and throttles back. Downstream of the coil the same cooling would *add* to delivered capacity instead of substituting for it.

Cool first, then humidify is the correct sequence: the coil brings the air down at nearly constant w, and the humidifier then walks it up the constant-wet-bulb line to the supply state, its own evaporative cooling counting toward the sensible duty rather than against it. **The architecture does not currently provide that path** — prong E has no branch that reaches the humidifier after fork 2.

Three remedies exist, and the choice is a genuine open question rather than an oversight to be quietly patched:

- *Reposition.* Add a duct branch placing the humidifier downstream of the chilled-water coil. Physically correct, and the only option that captures the humidifier's cooling as useful duty. Costs a new prong and one more damper in a valve network that is already the most intricate part of the plant.
- *Duplicate.* Install humidification capacity in both positions and select by mode. Correct in every mode, but it buys generality with hardware idle most of the year.
- *Detune.* Keep the present order and cap ε_{sat} so the coil face never sees $\text{RH} > \text{RH}_{\text{max}}$. A pure control workaround: it removes the condensation risk but keeps defects 2 and 3, and it forfeits latent capacity precisely in the loads that forced M5 in the first place.

Whichever is adopted, the humidifier block of Section 18.1 is unchanged — only its position in the stream sequence moves. That is the advantage of specifying components as pure functions of their inlet states, and it is why this defect is a topology question and not a modelling one.

Part III

Verification, and what changed when

19 Verification protocol

Each model version is frozen as a regression fixture recording every tracked state at annual resolution. Fixtures are named by version and are *never overwritten*: a mutable reference would always pass and prove nothing. A refactor that should not change physics is verified by reproducing the previous fixture to 1×10^{-9}; a change that should move results is verified by explaining every value that moved.

The harness loads the notebook's own code rather than re-implementing it, so if the two ever disagree the harness is wrong by definition. It is validated in both directions: an untouched notebook must pass, and a notebook perturbed by 0.45% in formation conductivity must fail.

A recurring failure mode. Four defects found so far shared one shape — a guard that degraded quietly instead of complaining:

1. a missing property library substituted constant water properties, misstating viscosity by up to 90% at the hot end;
2. a global warning filter suppressed the notebook's own diagnostics, so a formation-model safety check could never fire;
3. the desiccant wheel could reverse sign and humidify the process air while the dashboard still reported the humidity target as met;
4. the scavenging mixing node conserved energy but not moisture, using two different damper positions for the two balances.

None raised an error. Each produced plausible output. The protocol now requires that any guard whose failure would otherwise be invisible must announce itself.

19.1 What writing Part II turned up

Part II was written against the source rather than from memory, and the act of having to state each model precisely surfaced nine discrepancies that reading the code casually had not. Three of them are the same failure mode again — a quiet degradation — which is the strongest evidence yet that the pattern is structural rather than coincidental. None is fixed at v10.6; all are recorded here so that the report does not describe a plant more careful than the code.

Finding	**Kind**	**Consequence**
Wheel Newton inversion has a 60-iteration cap and no failure branch	quiet	On non-convergence it returns the last iterate with no flag, no exception and no warning. The caller cannot tell a converged state from a failed one.
Regeneration-outlet balances carry no physical bound	quiet	At a scavenging-to-process flow ratio of 0.30 with the shipped self-check inlets they return $T_{rd} = -3.2\,°\mathrm{C}$: arithmetically consistent, physically impossible, unflagged.
Humidity controller brackets d_{rA} on $[\cdot, 1]$ though availability caps it at 0.425	quiet	Above the cap the residual is flat, the secant denominator collapses into its 1×10^{-12} guard, and the wet-end saturation flag can be missed. Re-bracketing on the cap fixes it.
Coil uses dry-air $c_p = 1006\,\mathrm{J/(kg\,K)}$	bias	The $1.86\,w$ term is used by the cooler and the load routines but dropped here, understating the air-side capacity rate by about 3.7 % at $w = 20\,\mathrm{g/kg}$. The same mismatch makes the achieved supply depression 2.89 K against a design 3.0 K.
Pump work is charged on the well stream but spread over the loop stream	bias	Identical only when no flow bypasses the well. Under any bypass schedule the temperature rise attributed to pumping is wrong.

Finding	Kind	Consequence
Two values of h_{fg} coexist	bias	2501 kJ/kg in the enthalpy routines, 2500 kJ/kg in the load interface. The disagreement is 1 kJ/kg and harmless — but it is precisely the ambiguity the interface was built to prevent.
Well-flow floor applied in one place and not the other	latent	`dbhe_set_flow` floors the well fraction at 0.05; the mixing junction uses the raw fraction. Harmless at the baseline, wrong under any schedule that goes below the floor.
Cooler performs no dew-point check	latent	The outlet humidity is returned unchanged unconditionally. Safe only because the ambient floor currently sits above the supply dew point; not safe by construction.
Configuration comment for d_{rA} is stale	doc	It still describes the pre-v10.5 meaning — the opposite of what the scavenging kernel now does. A comment that contradicts its own code is worse than no comment.

The first three belong in the next maintenance version alongside the humidity actuator change. The middle three are small but should be fixed together, because each is an instance of one rule not being applied uniformly: use the moist-air specific heat everywhere, attribute pump work to the stream that receives it, define h_{fg} once.

20 Version history

Year-20 values at the default operating point (4500 m, 10 m^3/h, insulated tubing), taken directly from the frozen fixtures:

v	T_{ret} [°C]	T_{wa} [°C]	T_{ro} [°C]	w_{ri} [g/kg]	Δw [g/kg]	UA_{cool} [kW/K]	d_{rA} [–]
10.2	72.83	70.37	50.83	30.86	0.618	66.9	0.230
10.3	72.83	70.37	50.83	30.86	0.618	66.9	0.230
10.4	76.74	74.05	58.18	38.58	0.634	90.7	0.050
10.5	70.60	68.27	46.64	21.29	1.853	70.4	0.050
10.6	70.60	68.27	46.64	21.29	1.853	70.4	0.050

v10.2 Wheel-regime diagnostics; warning suppression fixed. Announced the desiccant-wheel sign reversal, and stopped `warnings.filterwarnings(ignore)` from swallowing the notebook's own diagnostics.

v10.3 Generic load interface. The plant sees the building only through a two-equation, three-unknown interface with an explicit closure. Verified bit-identical to v10.2.

v10.4 Scavenging mixing-node mass balance. The moisture balance used the global damper constant while the enthalpy balance used the controlled one; the weights summed to 1.07 and moisture was not conserved.

v10.5 Scavenging fed from building exhaust. The loop had been recirculating the wheel's own desorbed moisture. Moisture removal nearly tripled.

v10.6 Two-sided humidity limit flag. The saturation flag could only see under-drying. Reporting only; verified bit-identical to v10.5.

On the two largest movements. The mass-balance correction at v10.4 moved results in the direction *opposite* to the one predicted. Fixing the balance improves the wheel at any given damper position, so the controller needs less venting and drives the damper to its floor; at high recirculation the loop accumulates more of the wheel's own moisture, and the regeneration inlet ends up wetter rather than drier. The open-loop estimate and the closed-loop result disagreed in sign. At v10.5, removing recirculation entirely then reduced the regeneration inlet humidity by roughly 17 g/kg and nearly tripled moisture removal.

Current open item. The humidity controller is saturated at its dry end: the supply is drier than the space requires and the damper has no authority left to reduce drying. Approximately 41% of the regeneration coil duty is presently spent removing moisture the building does not want. The remedy is to move humidity control from the scavenging damper to v_G, the share of water flow reaching the regeneration coil, which throttles the coil duty directly and returns that heat to the loop.